

InfraSense: Proposed Solution vs. Existing Systems

A Comparative Technical Analysis

Comparative Analysis Report

*High-Sensitivity Microbarometer Infrasonics Sensor — Proposed
Solution vs. Existing Systems*

Problem Statement ID: SIH26144

Organization: National Technical Research Organisation (NTRO)

Team Name: [Your Team Name]

1. Introduction

This report presents a focused comparison between the proposed InfraSense high-sensitivity microbarometer infrasound sensor and the infrasound-sensing solutions that currently exist — spanning professional/imported instruments, documented hobbyist designs, and basic commodity MEMS sensors. The objective is to clearly position what the proposed solution does differently, and why.

2. Existing Systems — Overview

2.1 Professional / Imported Electromagnetic Microbarometers

The CTBTO's International Monitoring System (IMS) uses professional-grade stations (e.g., MB2000, MB2005, MB3) based on an aneroid capsule read out via an electromagnetic (LVDT) transducer. Commercial vendors such as Hyperion, Chaparral, and CET offer similar instruments using piezoelectric, differential-capacitive, and LVDT sensing respectively. These are highly sensitive and field-proven, but expensive and import-dependent.

2.2 DIY / Hobbyist Strain-Gauge Microbarometers

Documented amateur designs use a metal diaphragm instrumented with bonded foil strain gauges in a Wheatstone-bridge configuration, with a calibrated capillary leak venting the reference cavity. This is more sensitive than a bare MEMS sensor and cheaper than professional instruments, but requires precise manual fabrication and has no built-in wind-noise suppression, temperature compensation, or documented calibration.

2.3 Basic MEMS-Only Sensors

Commodity MEMS barometric sensors (e.g., BMP388/BMP390-class) are inexpensive and widely available, but used alone — without a reference chamber, differential technique, or wind filtering — they are typically limited to strong, nearby events and are vulnerable to wind-induced noise and baseline drift outdoors.

3. Side-by-Side Comparison

Parameter	Imported Professional (CTBTO-grade)	DIY Strain-Gauge (Hobbyist)	Basic MEMS-Only Sensor	Proposed InfraSense
Sensing method	Electromagnetic (LVDT) / capacitive / piezoelectric	Strain gauge on metal diaphragm	Bare MEMS capacitive die	Differential MEMS + reference chamber
Approx. cost / unit	Several lakh ₹ (imported)	Low-moderate (DIY parts)	Very low (single sensor)	~₹6,500–8,000 (prototype); ₹3,500–5,000 at scale
Wind-noise handling	Dedicated hose/pipe arrays at station level	None built-in	None	Porous-hose spatial-averaging filter
Long-period equalization	Precision engineered	Capillary leak (manual tuning)	Not addressed	Calibrated capillary-leak reference path
Temperature compensation	Factory-calibrated	Partial (extra gauges)	Minimal	Dedicated sensing + firmware correction
Calibration methodology	Certified lab facilities	Manual, ad hoc	None	Documented multi-point + reference comparison
Data acquisition / software	Proprietary station systems	Arduino-class, manual	Basic logging only	Open-market digitizer + Python/ObsPy waveform GUI
Fabrication complexity	Very high (factory-grade)	High (precision manual bonding)	Very low	Moderate — achievable with standard workshop/lab tools
Import dependency	High	Low	Low	None — fully indigenous
Deployment density feasible	Very low (cost-limited)	Low (build complexity)	High but unreliable	High — low cost + practical build

4. Analysis by Dimension

4.1 Sensitivity & Accuracy

Imported professional sensors remain the most sensitive option, benefiting from decades of refinement and factory calibration. DIY strain-gauge designs achieve moderate sensitivity sufficient for regional-scale events. Basic MEMS-only sensors are the weakest, lacking any reference-chamber or differential technique. InfraSense targets the middle-to-upper range of this spectrum by combining a differential MEMS element with a purpose-built reference chamber — a combination that neither the DIY nor basic-MEMS approach implements together.

4.2 Cost & Accessibility

This is where InfraSense differentiates most sharply. Imported instruments cost several lakh rupees per unit, making dense deployment financially impractical. InfraSense is estimated at ₹6,500–8,000 per prototype unit, falling to ₹3,500–5,000 at production scale — a reduction exceeding 99% relative to imported systems, while still incorporating engineering safeguards (wind filtering, calibration, temperature compensation) that basic MEMS and most DIY designs omit.

4.3 Wind-Noise Rejection

Wind turbulence is one of the dominant sources of false or degraded readings in any outdoor infrasound sensor. Professional stations address this with dedicated hose/pipe arrays. Neither typical DIY builds nor basic MEMS sensors include this feature. InfraSense adopts the same principle as professional stations — a distributed porous-hose wind filter — scaled to a single-unit, low-cost implementation.

4.4 Calibration & Long-Term Stability

Professional systems are calibrated at certified facilities; DIY builds rely on manual, ad hoc calibration; basic MEMS sensors are typically used uncalibrated. InfraSense introduces a documented, repeatable multi-point calibration and periodic zero-check methodology — directly satisfying the NTRO requirement for a defined calibration methodology, and going beyond what either the DIY or basic-MEMS category typically provides.

4.5 Data Acquisition & Software

Professional stations rely on proprietary acquisition systems. DIY builds typically use manual, one-off Arduino-based logging. InfraSense uses an open-market digitizer combined with Python/ObsPy-based real-time waveform display and analysis — satisfying the NTRO requirement for open-market data-acquisition and analysis software while remaining fully reproducible and extensible.

5. Positioning Summary

InfraSense is positioned as the practical middle path between two extremes: the very expensive, import-dependent professional station, and the inexpensive but unreliable basic-MEMS or manually-built DIY sensor. It borrows the differential-reference and capillary-leak principle proven in both professional and DIY designs, adds systematic wind-noise suppression and documented calibration — largely absent from DIY and basic-MEMS approaches — and remains buildable at a small fraction of imported system cost.

5.1 Key Differentiators

- **Cost:** Over 99% cheaper than imported professional-grade sensors.
- **Completeness:** Only the proposed solution combines differential sensing, wind-noise filtering, temperature compensation, and documented calibration in a single low-cost design.
- **Indigenuity:** Fully sourceable and buildable within India, with no dependency on imported sensing technology.
- **Reproducibility:** Built on open-market components and open-source software (Python/ObsPy), unlike proprietary professional systems.
- **Practicality:** Unlike DIY strain-gauge designs, does not require precision manual strain-gauge bonding — uses a ready-made differential MEMS sensing element instead.

5.2 Honest Limitations

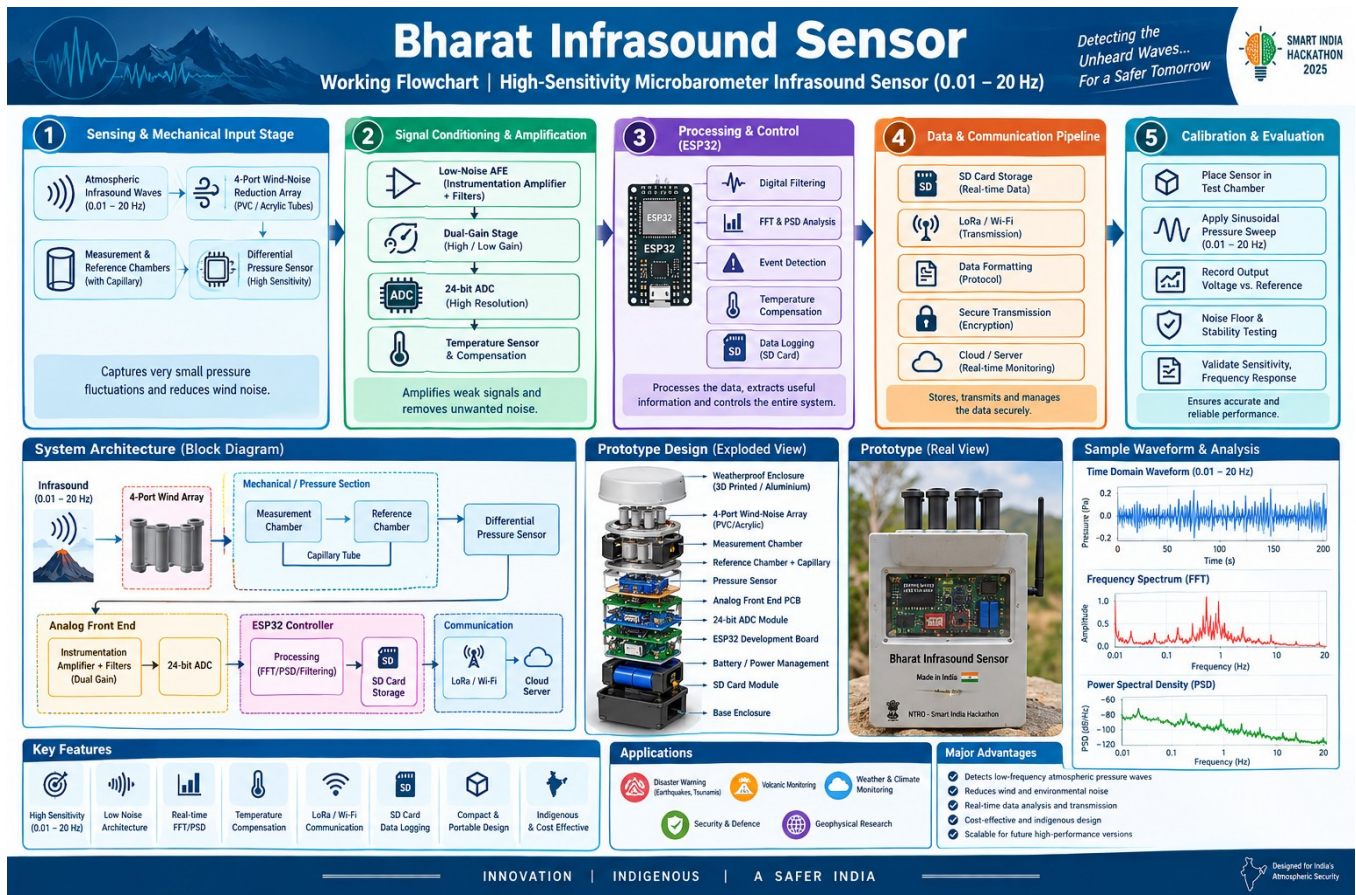
- Raw sensitivity of the chosen MEMS sensing element is not expected to match professional capacitive/optical instruments — this is a deliberate cost/practicality trade-off for the prototype stage.
- Long-term field reliability (multi-year drift, extreme-weather survivability) of professional stations is backed by decades of deployment history that a new design cannot yet claim.
- All InfraSense performance figures in this report are design targets to be confirmed through the calibration and validation protocol, not yet field-proven results.

6. Conclusion

Compared against imported professional systems, DIY strain-gauge builds, and basic MEMS-only sensors, the proposed InfraSense solution offers the most balanced combination of low cost, engineering completeness, and practical buildability. It does not claim to exceed the raw sensitivity of certified professional instruments, but it is the only option in this comparison that combines wind-noise suppression, temperature compensation, and documented calibration methodology at a cost low enough to enable the dense, indigenous deployment that India's infrasound-monitoring needs currently lack.

BHARAT INFRASOUND SENSOR

Design & Development of a High-Sensitivity Microbarometer Infrasound Sensor



Project Documentary | Prototype Architecture, Working Principle, Signal Processing, Production Plan & Validation

Proposed frequency band: 0.01–20 Hz | Controller: ESP32 | High-resolution ADC: 24-bit | Local storage: SD card | Wireless: Wi-Fi / optional LoRa

Prepared as an SIH-ready technical document. The waveform/FFT/PSD plots supplied with the project are treated as reference or illustrative plots unless generated from the final calibrated prototype.

1. Executive Summary

The proposed system is a compact, modular microbarometer architecture for detecting very low-frequency atmospheric pressure fluctuations. The design combines wind-noise reduction, differential pressure sensing, low-noise analog conditioning, high-resolution conversion, ESP32 processing and local/wireless data handling. The main objective is to build a student-level proof-of-concept that can be calibrated and progressively improved toward the 0.01–20 Hz target band.

- Detect low-frequency atmospheric pressure variations in the infrasonic band.
- Reduce wind-induced disturbances before the pressure reaches the sensing element.
- Measure differential pressure using a two-port sensing mechanism with a reference pressure path.
- Amplify and filter weak analog signals using a low-noise AFE.
- Convert the conditioned signal with a high-resolution external ADC.
- Use ESP32 for acquisition, filtering, FFT/PSD analysis, event detection and data logging.
- Store data locally on SD and provide Wi-Fi-based remote monitoring.
- Validate sensitivity, noise floor, frequency response, temperature stability and repeatability through controlled testing.

Design philosophy

Build → Measure → Calibrate → Improve. The documentary avoids treating the target 0.01–20 Hz performance as already proven. Published research shows that high-sensitivity infrasound instruments require careful mechanical filtering, low-noise electronics and traceable calibration.

Target	Proposed implementation
Frequency band	0.01–20 Hz target; actual validated band to be measured
Pressure input	4-port wind-noise inlet + measurement/reference chambers
Sensing	Differential pressure sensor / diaphragm mechanism
Electronics	Low-noise AFE + external 24-bit ADC
Processing	ESP32 + digital filtering + FFT + PSD
Data	SD logging + Wi-Fi; LoRa optional

2. Problem Statement & Engineering Requirements

Infrasound consists of atmospheric pressure fluctuations below the normal human hearing range. A practical microbarometer must resolve small pressure variations while rejecting large unwanted components such as very slow meteorological pressure changes and wind-generated noise.

Core engineering requirements

- High sensitivity to small dynamic pressure fluctuations.
- Wide and well-characterized response around the intended 0.01–20 Hz band.
- Low electronic and environmental noise.
- Stable response across temperature changes.
- Mechanical filtering / pneumatic equalization to control out-of-band pressure variations.
- Reliable digitization and time-stamped data logging.
- Repeatable calibration against a known pressure reference.
- Weather-resistant packaging for controlled outdoor trials.

Why differential sensing is useful

A differential pressure transducer compares pressure at two ports. A measurement port can receive the atmospheric signal while a reference path is connected through a controlled pneumatic element. This arrangement can reduce sensitivity to slow pressure drift and can create a designed frequency response. Marcillo et al. describe a closely related low-cost architecture using a micromachined differential pressure transducer, a backing volume and capillary tube. ■cite■turn0search0■

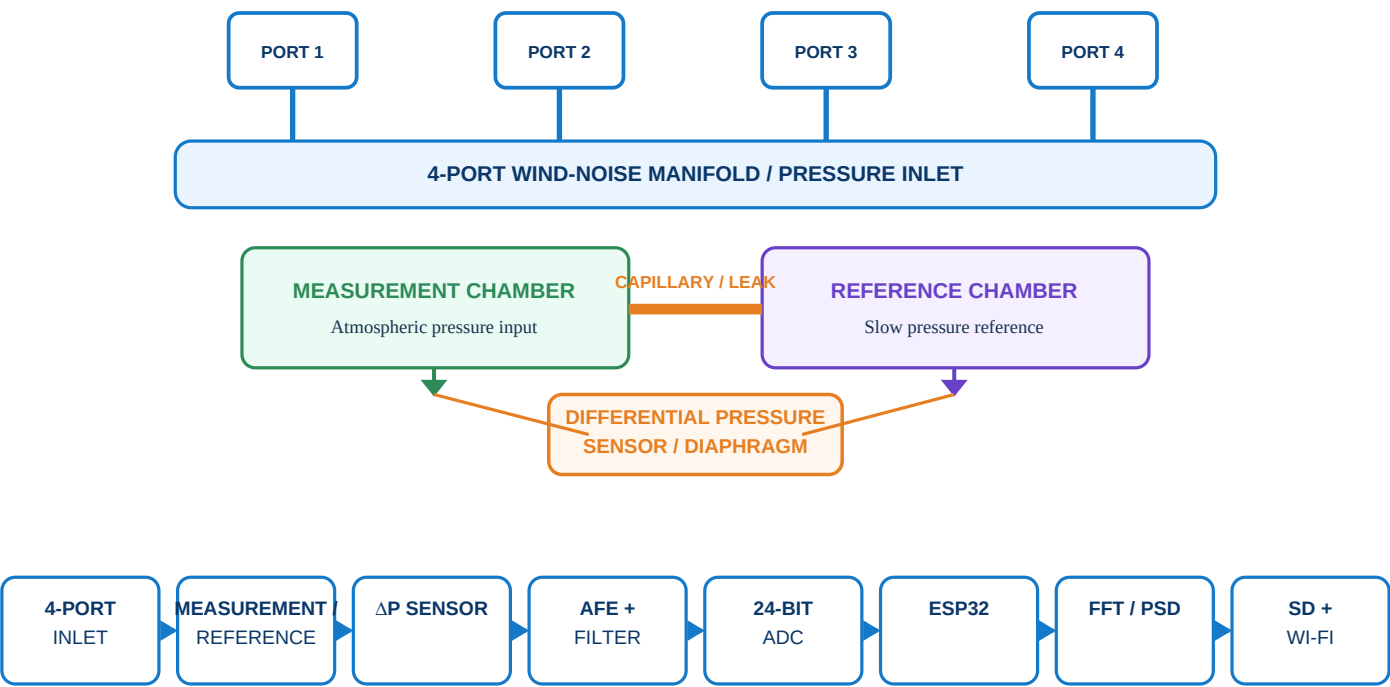
Important performance statement

The project should report **measured** sensitivity, noise floor and frequency response only after calibration. Published values from other instruments are research references, not specifications of this prototype.



3. Proposed System Architecture

The complete proposed signal chain is organized into five stages so that the mechanical, analog, digital and validation functions remain modular.



Architecture note: The 4-port array is a wind-noise-reduction inlet, while the differential diaphragm/pressure sensor is the sensing element. They are complementary, not the same component. Pipe-array wind-noise systems are used in operational infrasound arrays to suppress incoherent wind-generated pressure perturbations while passing coherent long-wavelength signals with limited distortion. ■cite■turn0search2■

4. Mechanical & Pressure-Sensing Subsystem

The mechanical subsystem determines how atmospheric pressure reaches the differential sensor. Its design must balance wind-noise rejection, pneumatic time constant, leakage, mechanical robustness and the desired low-frequency response.

A. 4-Port Wind-Noise Reduction Array

- Use four distributed inlet ports connected to a common manifold or pressure collection volume.
- Use identical tubing paths where possible to maintain similar pneumatic resistance.
- Place the inlet ports away from direct obstruction and avoid sharp, unnecessary bends.
- Use a mechanically rigid manifold and sealed joints to prevent leakage.
- Characterize the wind-noise response experimentally rather than assuming a fixed attenuation value.

B. Measurement & Reference Chambers

- Measurement chamber receives the pressure signal through the wind-noise inlet.
- Reference chamber provides a slower pressure reference path.
- A capillary/restricted passage between the reference volume and atmosphere sets the pneumatic time constant.
- Chamber volume and capillary dimensions directly affect the frequency response and must be selected and validated experimentally.

C. Dual-Vent / Differential Diaphragm Concept

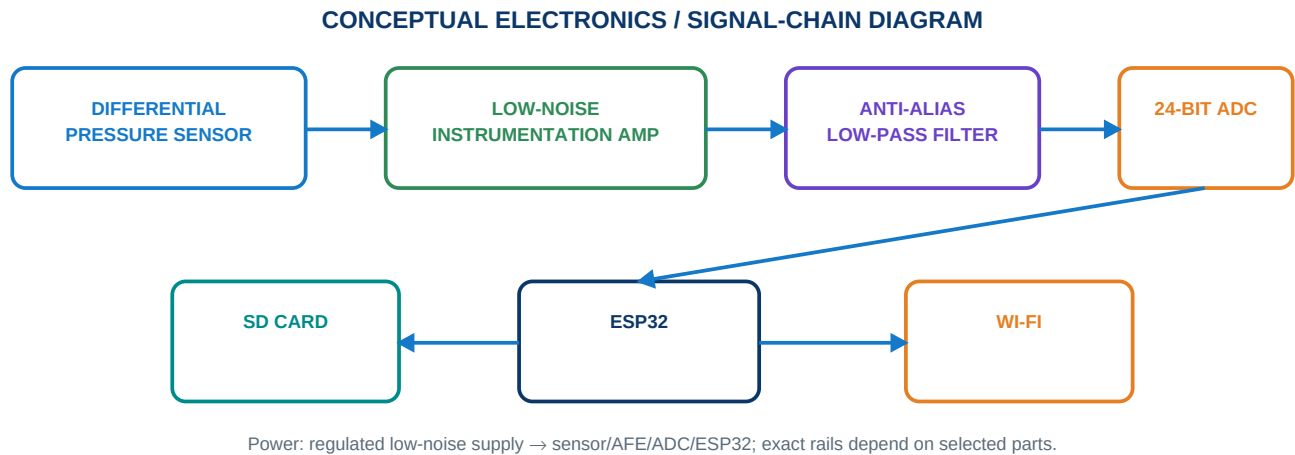
- Two pressure ports act on opposite sides of the diaphragm or differential sensing element.
- The diaphragm responds to pressure difference $\Delta P = P_{\text{measurement}} - P_{\text{reference}}$.
- The sensor converts diaphragm stress/deflection into an electrical output.
- The mechanical filter and sensing element should be treated as one calibrated pressure-input chain.

Research basis

Marcillo et al. report a micromachined differential pressure transducer with two ports, a flexible silicon diaphragm, and a capillary/backing-volume mechanical filter. Their paper also emphasizes calibration and characterization of the complete sensor chain. [■cite■turn0search0■](#)

5. Electronics: Low-Noise AFE & ADC

The analog front end is responsible for preserving small pressure-induced signals before digitization. The design shown below is conceptual; final component values should be selected after the chosen sensor, ADC input range, supply rails and measured noise are known.



AFE functional blocks

- Instrumentation/low-noise amplifier: raises the small differential sensor output.
- Dual-gain option: provides a high-gain path for weak signals and a lower-gain path for stronger events.
- Anti-alias low-pass filter: limits unwanted high-frequency energy before ADC sampling.
- Stable reference and clean power: reduce gain drift and electrical noise.
- 24-bit external ADC: provides higher-resolution conversion than relying on the ESP32 ADC alone.

Design caution

A nominal “24-bit” ADC does not automatically deliver 24 effective bits. The final performance depends on ADC noise, reference stability, analog layout, grounding, gain, sampling strategy and the sensor’s own noise floor.

6. ESP32 Processing, Storage & Communication

The ESP32 acts as the digital controller. It should receive clean, time-stamped samples from the external ADC and perform lightweight signal processing before storage or transmission.



Firmware functions

- Initialize ADC, sensor and temperature channels.
- Acquire samples at a rate appropriate for the intended 0.01–20 Hz analysis band.
- Apply digital filtering and remove obvious DC/bias components where appropriate.
- Calculate FFT-based frequency information.
- Estimate PSD using a documented method and fixed processing parameters.
- Attach timestamps and temperature metadata to each record.
- Write raw or processed data to SD card for traceability.
- Send selected summaries or waveform segments through Wi-Fi when required.

Data record structure

Timestamp | Pressure sample | Temperature | Sensor status | Gain mode | Processing window | Event flag

Local storage should remain the primary evidence source during testing; wireless transmission is an additional convenience layer.

7. Wind-Noise Reduction & Frequency Response

Wind can produce pressure fluctuations at the sensor inlet that are unrelated to the desired infrasonic signal. Operational infrasound systems therefore use wind-noise reduction systems such as porous hoses or pipe arrays. Research describes these systems as suppressing incoherent wind-generated fluctuations while allowing coherent long-wavelength signals to pass. [■cite■turn0search2■](#)

Prototype approach

- Use the 4-port inlet as the first-stage environmental interface.
- Keep the four pressure paths mechanically symmetric.
- Use the measurement/reference chamber arrangement to create a controlled pneumatic response.
- Measure response with and without the wind-noise inlet to quantify its effect.
- Use the final combined inlet + chamber + sensor assembly during calibration because the WNRS and sensor form one measurement system.

Frequency-response concept

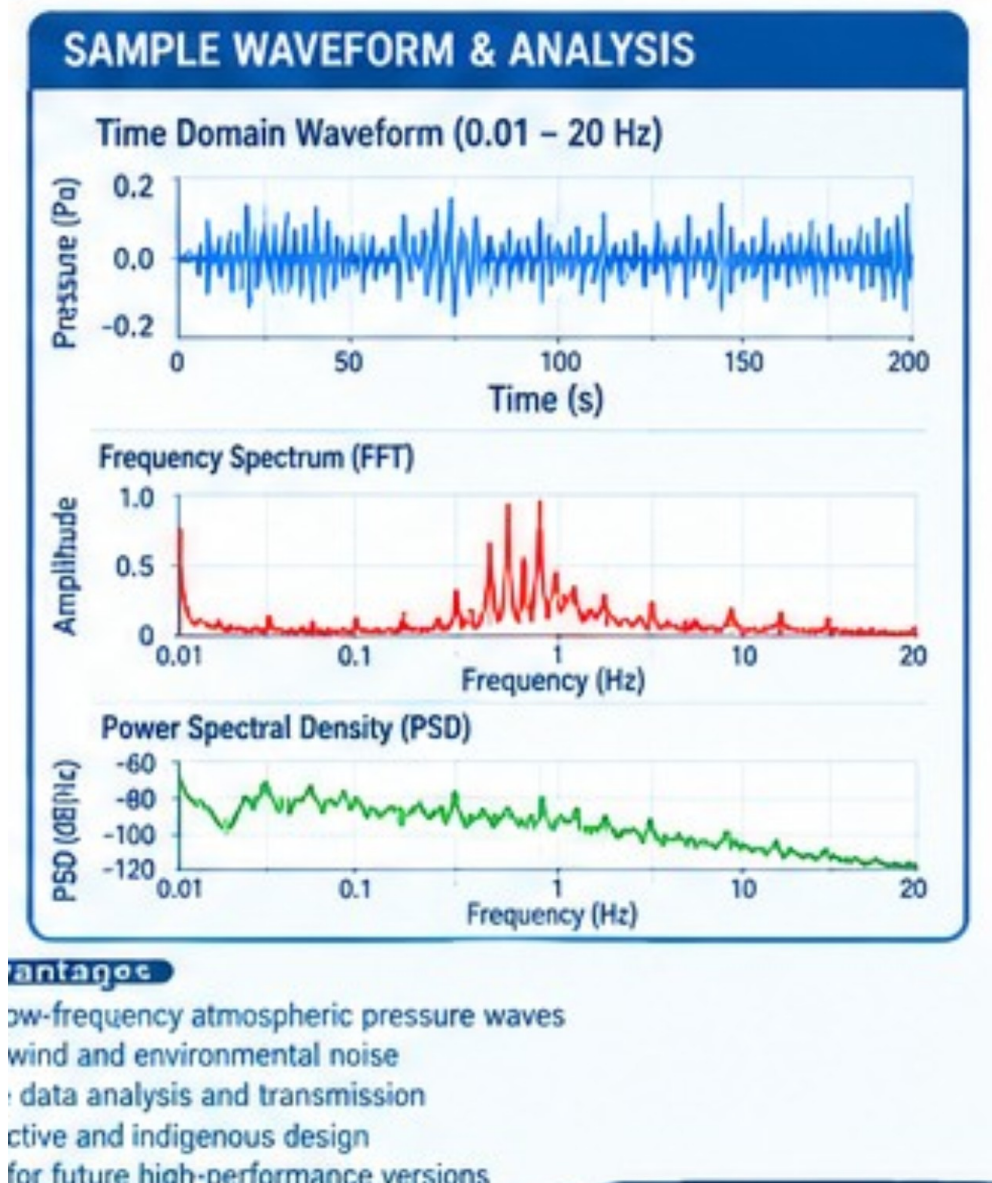


Published benchmark

Marcillo et al. reported a low-cost sensor with a flat response from 0.01 to at least 40 Hz, while using a pneumatic filter to reject very-low-frequency pressure changes below the target band. This is a benchmark for design reasoning, not a guaranteed performance value for the proposed prototype. [■cite■turn0search0■](#)

8. Waveform, FFT & PSD Analysis

The supplied project material contains representative plots for time-domain pressure, frequency spectrum (FFT) and Power Spectral Density (PSD). These plots are useful for explaining the analysis pipeline, but they should be labelled as reference/illustrative until generated from the calibrated prototype.



Interpretation

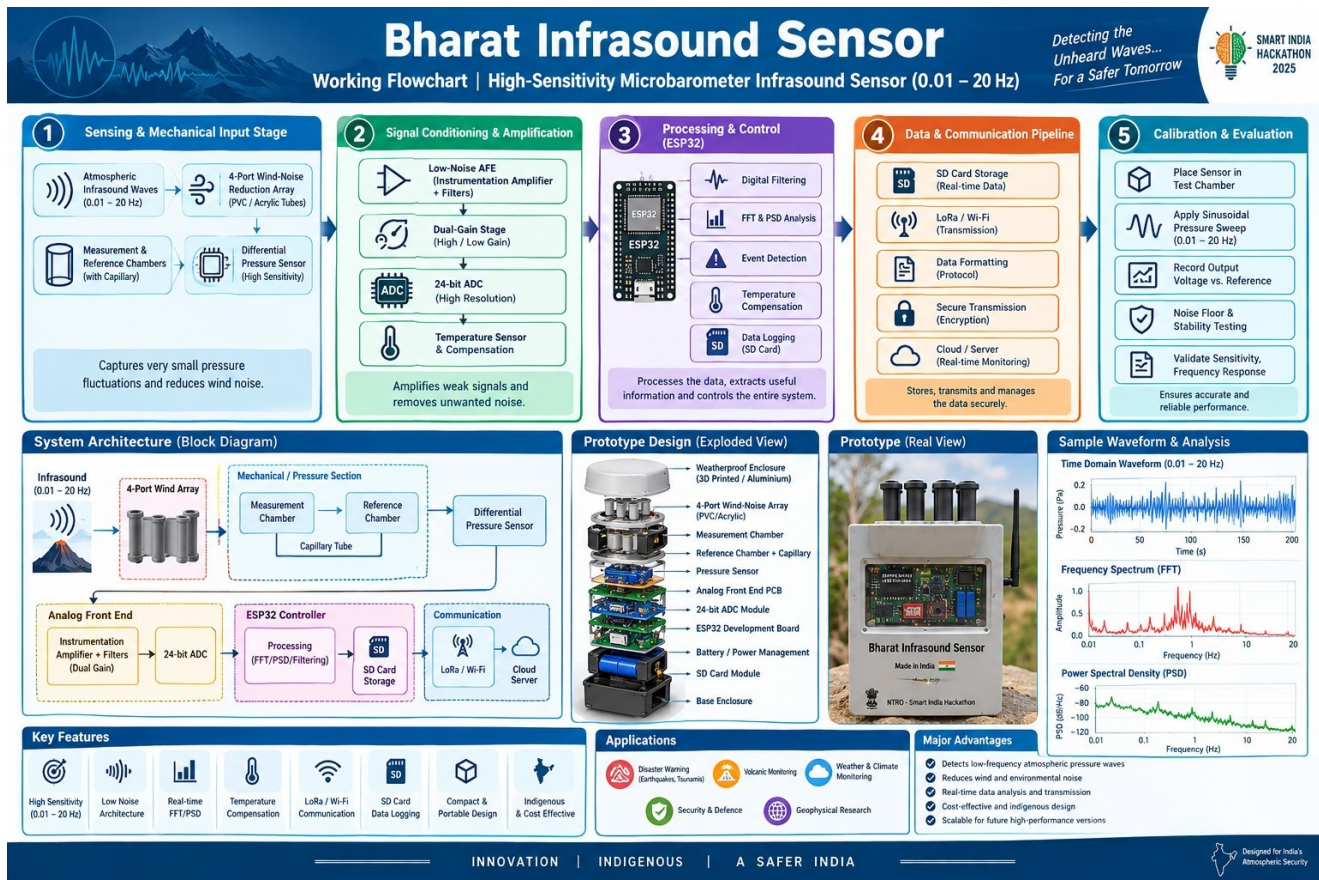
- Time-domain waveform: shows pressure variation versus time.
- FFT: shows which frequency components are present in the measured record.
- PSD: shows how signal/noise power is distributed across frequency.
- A low-frequency peak may indicate a coherent infrasonic source, but source identification requires additional evidence and array/event analysis.
- For research-grade comparison, PSD should be computed with a documented window, segment length, overlap, sampling rate and instrument-response correction. Bowman & Lees used overlapping FFTs and instrument-response correction to characterize ambient infrasound noise. [■cite■turn0search1■](#)

Microbarom reference

Global infrasound studies report a prominent microbarom band around 0.2 Hz. This makes the low-frequency region especially useful when validating a microbarometer, while also requiring careful separation of ambient background noise from events of interest. [■cite■turn0search1■turn0search6■](#)

9. Prototype Design & Physical Integration

The prototype should be built as a stack of serviceable modules. The mechanical pressure path remains isolated from the noisy digital electronics as far as practical, while all pressure connections remain accessible for leak testing and calibration.



Prototype layer structure

- Top: weather-resistant cover and 4-port wind-noise inlet.
- Pressure layer: measurement/reference chambers, capillary path and differential pressure sensor.
- Analog layer: AFE, gain selection and anti-alias filtering.
- Digital layer: 24-bit ADC and ESP32 controller.
- Storage/communication layer: SD card and Wi-Fi; optional LoRa module.
- Bottom: regulated power system and mechanically stable enclosure base.

Assembly principle

Keep pressure tubing short, sealed and mechanically supported. Keep analog sensor wiring away from switching regulators and high-current digital paths. Provide test points for sensor output, AFE output and ADC input so each stage can be debugged independently.

10. Prototype Production Plan

The following sequence converts the proposed architecture into a buildable student prototype while keeping testing checkpoints between stages.



Stage-by-stage production

- 1. Mechanical fabrication — make the enclosure, four-port manifold, measurement chamber and reference chamber. Verify dimensions and airtight joints.
- 2. Pressure-path assembly — install tubing, capillary and fittings. Perform a simple leak/stability check before connecting electronics.
- 3. Sensor + PCB integration — mount the differential pressure sensor and connect the AFE/ADC using short, clean signal paths.
- 4. Firmware — acquire raw ADC samples first; then add filtering, FFT, PSD and SD logging.
- 5. Calibration — apply known low-frequency pressure variations and compare output against a reference instrument.
- 6. Enclosure + field test — close the weatherproof housing only after bench tests pass; then compare wind-noise behaviour under controlled outdoor conditions.

Suggested prototype BOM

Subsystem	Representative item
Mechanical	PVC/acrylic/aluminium enclosure, 4 inlet tubes, chambers, capillary, fittings
Sensing	Differential pressure sensor / diaphragm sensing element
Analog	Low-noise instrumentation amplifier, filter, gain stage, reference
Digital	24-bit ADC module, ESP32 development board, temperature sensor
Data	microSD module/card, Wi-Fi through ESP32; optional LoRa
Power	Regulated supply, protection and connectors

11. Calibration & Validation Method

Calibration is the evidence that connects the prototype output to physical pressure input. The final report should include measured plots, test conditions and uncertainty/limitations rather than only theoretical claims.

Calibration flow



Recommended validation tests

- Sensitivity test: apply several known pressure amplitudes and plot output versus input.
- Linearity test: verify whether output follows the reference input over the intended range.
- Frequency response: test multiple frequencies across the intended band; report amplitude and phase if available.
- Noise-floor test: operate with controlled input and quantify RMS noise / PSD.
- Wind-noise test: compare inlet configurations under similar wind conditions.
- Temperature test: record output while temperature changes and estimate drift.
- Long-duration stability: log continuously and inspect baseline drift and repeatability.
- Data integrity test: verify SD files, timestamps, missing samples and wireless packet loss.

Published calibration precedent

Marcillo et al. evaluated the sensor in an infrasonic test chamber using a reference-sensor approach and reported sensitivity from controlled pressure tests. The paper is useful as a model for how the student prototype can document its own calibration procedure.

■cite■turn0search0■

12. Signal Processing & Data Products

The digital processing chain should preserve raw evidence while generating compact products for real-time monitoring.

Processing sequence



Primary outputs

- Raw pressure waveform with timestamp.
- Filtered waveform for visual monitoring.
- FFT amplitude spectrum for frequency identification.
- PSD curve for noise and power characterization.
- Temperature and sensor-status channels.
- Event flags based on documented thresholds or spectral criteria.
- Calibration coefficients and instrument configuration metadata.

Recommended data folder for GitHub

- README.md — project overview, architecture and how to reproduce the analysis.
- docs/ — project documentary PDF and technical notes.
- hardware/ — block diagram, conceptual circuit and PCB files when available.
- firmware/ — ESP32 source code.
- data/ — small example datasets only; do not upload private credentials.
- analysis/ — FFT/PSD scripts and example plots.
- references/ — citation list and source notes.
- images/ — prototype photos and diagrams with source/credit information.

Important GitHub practice

Keep secrets such as Wi-Fi passwords, API keys and private credentials out of the repository. Use a configuration template such as **config.example.h** instead.

13. Existing Approach vs Proposed Prototype

Aspect	Typical research-grade system	Proposed student prototype
Pressure sensing	Specialized microbarometer / calibrated sensor	Differential pressure sensor + diaphragm/reference path
Wind-noise control	Dedicated pipe/porous-hose WNRS	Compact 4-port inlet concept for proof-of-concept testing
Analog electronics	Precision low-noise instrumentation	Low-noise AFE + gain + anti-alias filter
Digitization	Precision scientific digitizer	External 24-bit ADC
Processing	Dedicated scientific data systems	ESP32 + software DSP
Data	Research-grade network / archive	SD + Wi-Fi; optional LoRa
Validation	Traceable calibration infrastructure	Controlled chamber/reference testing as available

Risk → mitigation

Risk	Mitigation
Wind noise dominates signal	4-port inlet, chamber design, controlled wind testing
Sensor drift	Temperature monitoring, calibration and baseline tracking
Electrical noise	Low-noise AFE, clean power, grounding and PCB layout
Low-frequency response not as expected	Measure frequency response; adjust chamber/capillary geometry
ADC resolution not equal to effective resolution	Measure system noise and ENOB-related performance rather than assuming nominal bits
Prototype claims exceed evidence	Label plots as reference until measured; publish calibration results

14. Applications, Roadmap & References

Potential application areas

- Atmospheric and environmental monitoring.
- Geophysical and atmospheric research.
- Volcanic and severe-weather research support.
- Meteoroid / rocket / energetic-event monitoring research.
- Distributed sensor-array experiments.
- Educational and indigenous instrumentation development.

Development roadmap

- Prototype V1: bench-top pressure sensing and raw data logging.
- Prototype V2: wind-noise inlet + reference chamber + calibrated AFE/ADC.
- Prototype V3: controlled frequency-response testing + temperature compensation.
- Prototype V4: weatherproof enclosure + long-duration field trials.
- Future: multi-sensor array, improved synchronization, advanced event classification and deeper environmental validation.

Key references used for the technical basis

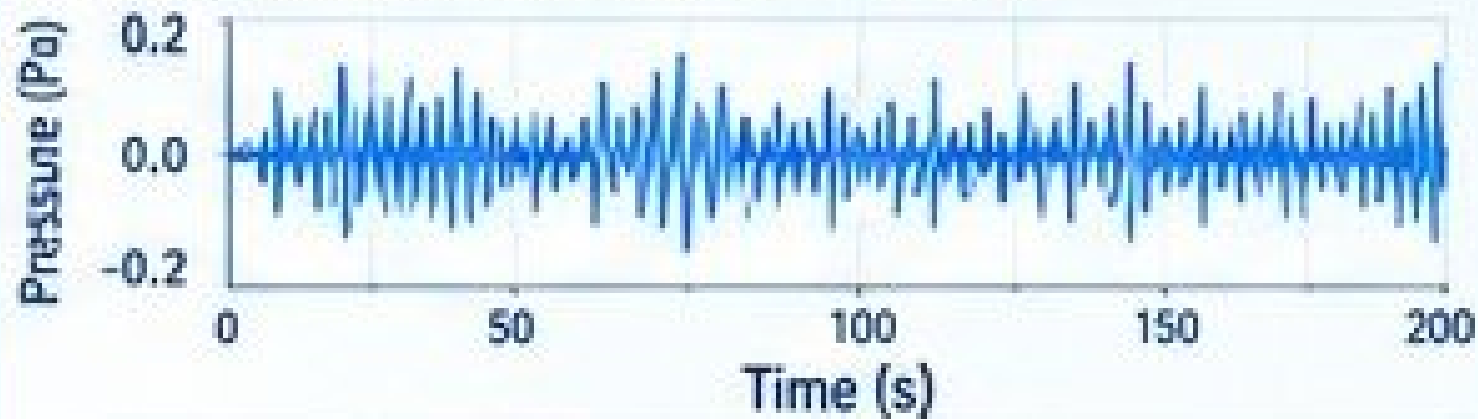
1. Marcillo, O., Johnson, J. B., & Hart, D. (2012). *Implementation, Characterization, and Evaluation of an Inexpensive Low-Power Low-Noise Infrasound Sensor Based on a Micromachined Differential Pressure Transducer and a Mechanical Filter*. *Journal of Atmospheric and Oceanic Technology*, 29(9), 1275–1284. DOI: 10.1175/JTECH-D-11-00101.1. ■cite■turn0search0■
2. Bowman, J. R., & Lees, J. M. (2005). *Ambient infrasound noise*. *Geophysical Research Letters*. DOI: 10.1029/2005GL022486. ■cite■turn0search1■
3. *Identifying suitable time periods for infrasound measurement system response estimation using across-array coherence*. *Geophysical Journal International*. Discusses pipe-array wind-noise reduction systems and complete array-element response. ■cite■turn0search2■
4. De Carlo et al. (2021). *Global Microbarom Patterns: A First Confirmation of the Theory for Source and Propagation*. *Geophysical Research Letters*, 48, e2020GL090163. ■cite■turn0search6■
5. User-supplied project materials: technical approach sheets, prototype concept visuals, FFT/PSD plots and sample waveform analysis. These are treated as project reference/illustrative materials unless explicitly identified as measured prototype data.

Final documentation statement: The proposed Bharat Infrasound Sensor is a modular, low-cost proof-of-concept architecture. Its engineering value should be demonstrated through measured sensitivity, noise floor, frequency response, calibration and repeatability—not by claiming published research-grade performance before validation.

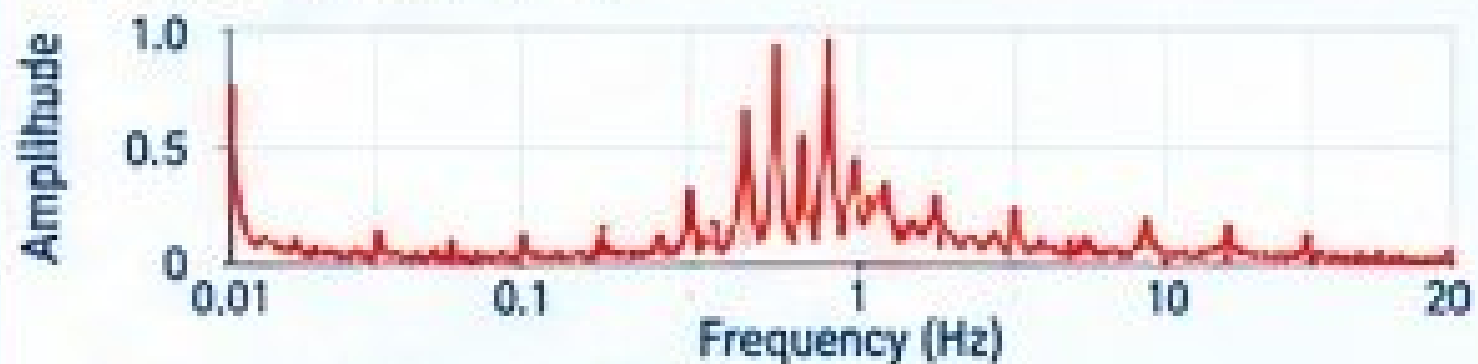
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SAMPLE WAVEFORM & ANALYSIS

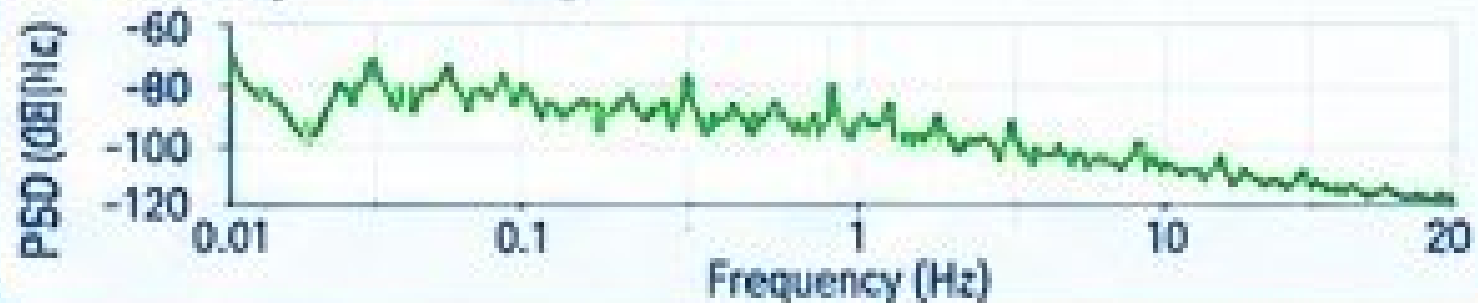
Time Domain Waveform (0.01 – 20 Hz)



Frequency Spectrum (FFT)

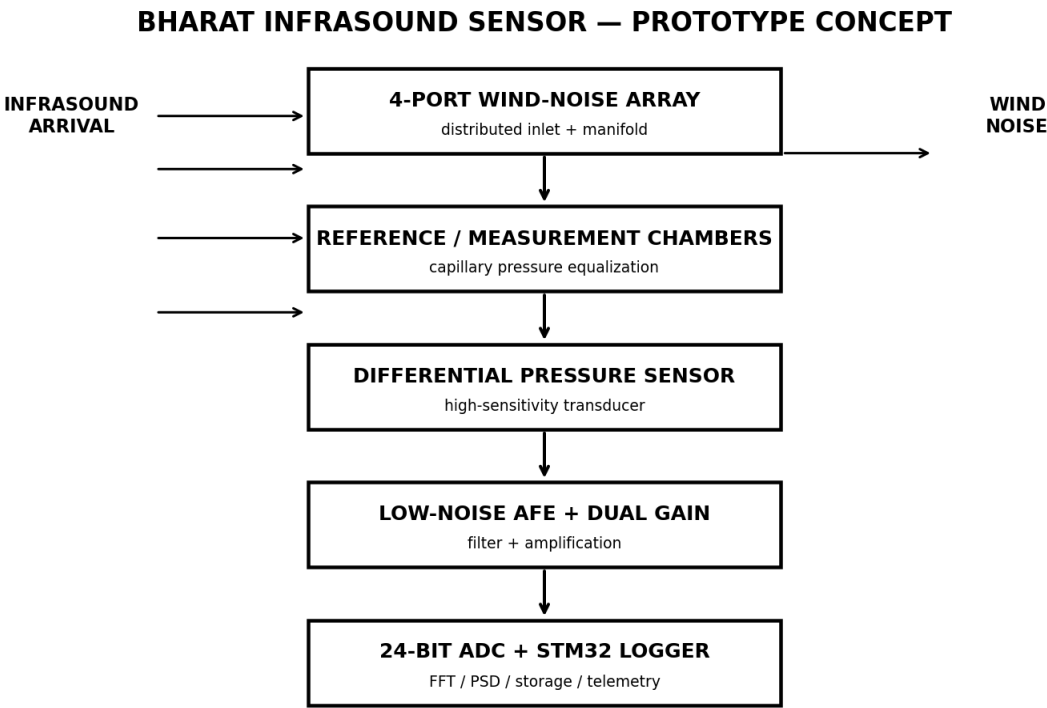


Power Spectral Density (PSD)



BHARAT INFRASOUND SENSOR

High-Sensitivity Microbarometer for Low-Frequency Atmospheric Pressure Monitoring



Smart India Hackathon — Project Documentary

Target measurement band: approximately 0.01 Hz to 20 Hz

1. PROJECT OVERVIEW

The Bharat Infrasound Sensor is a proposed compact, high-sensitivity microbarometer designed to detect very low-frequency atmospheric pressure fluctuations. The concept combines wind-noise reduction, differential pressure measurement, low-noise analog electronics, high-resolution digitization and digital signal processing.

- Detects infrasonic pressure variations below the normal human hearing range.
- Designed around a target frequency range of approximately 0.01–20 Hz.
- Uses a multi-port wind-noise interface to reduce local turbulence effects before sensing.
- Uses measurement and reference chambers with controlled pressure equalization for stable long-period response.
- Uses a differential pressure sensor followed by a low-noise analog front end and dual-gain path.
- Uses a 24-bit ADC and STM32-class controller for waveform acquisition, processing and logging.
- Supports FFT, power spectral density (PSD), event-feature extraction and remote data transmission.
- Prototype architecture is intended for atmospheric science, environmental monitoring and research applications.

PROPOSED SYSTEM ARCHITECTURE

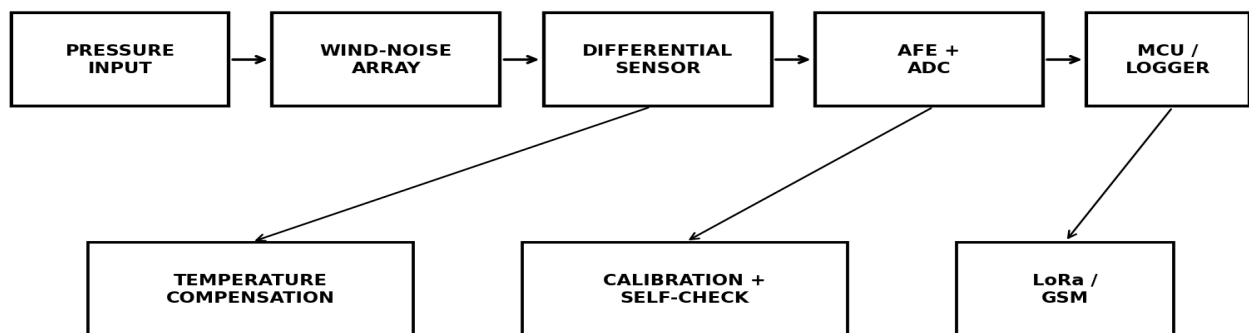


Figure 1. Proposed end-to-end sensor architecture.

2. PROBLEM STATEMENT & DESIGN NEED

Problem: Build a high-sensitivity microbarometer infrasound sensor capable of measuring very low-frequency atmospheric pressure fluctuations with low self-noise and stable long-period behaviour.

Key engineering challenges

- Infrasound signals can be extremely small compared with normal atmospheric pressure changes.
- Wind turbulence can produce pressure fluctuations that mask the desired infrasonic signal.
- Temperature changes can introduce sensor drift and electronic offset changes.
- Very low-frequency measurement requires careful pressure equalization and low-noise electronics.
- A useful prototype must preserve signal integrity from the mechanical interface through digitization and storage.
- Calibration and repeatability are essential for meaningful measurements.

Design response

- Wind-noise reduction before the pressure transducer.
- Differential sensing between measurement and reference pressure paths.
- Low-noise amplification with selectable gain.
- Temperature sensing and compensation in firmware/electronics.
- High-resolution ADC acquisition with timestamped data logging.
- Digital spectral analysis for waveform, FFT and PSD outputs.

3. WORKING PRINCIPLE

WORKING FLOW — FROM ATMOSPHERIC WAVE TO DIGITAL EVENT DATA

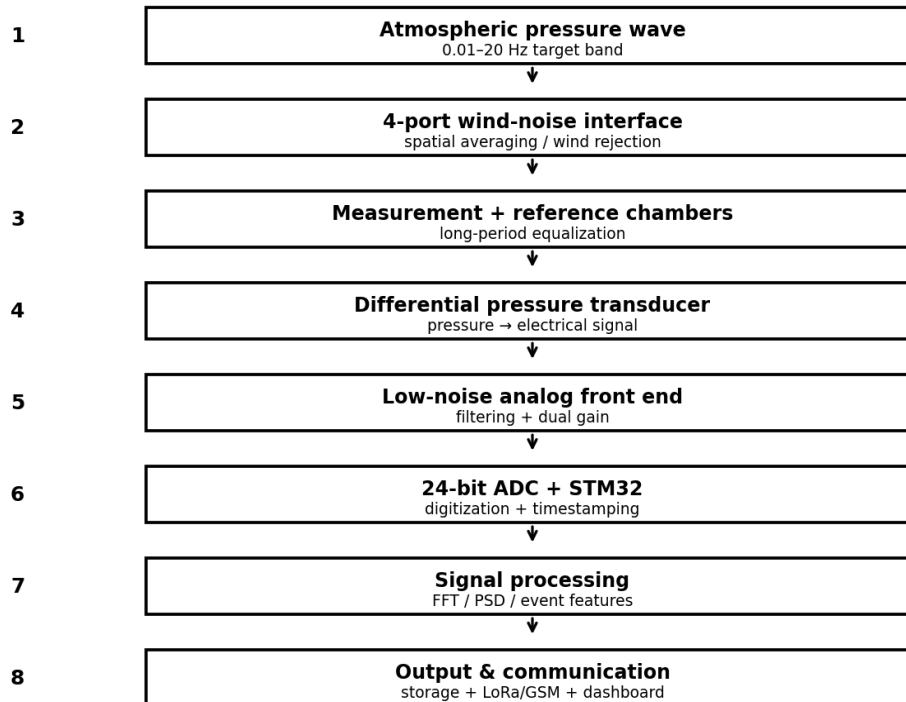


Figure 2. Working flow from atmospheric pressure disturbance to digital output.

The proposed signal chain starts with an atmospheric pressure disturbance. The multi-port inlet distributes the arriving pressure wave through a wind-noise reduction network. The resulting pressure is presented to a differential sensing element. The analog signal is filtered and amplified, converted by a high-resolution ADC, and processed by the controller. The final data can be stored locally and transmitted for further analysis.

4. PROTOTYPE DESIGN

BHARAT INFRASOUND SENSOR — PROTOTYPE CONCEPT

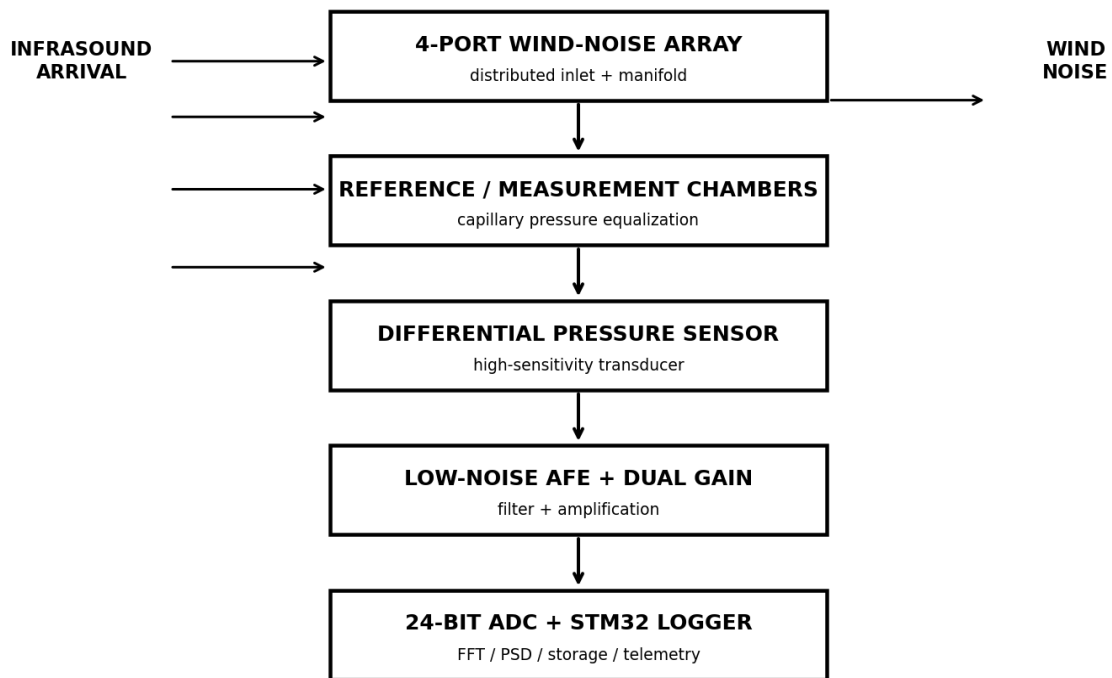


Figure 3. Conceptual prototype architecture — not a manufacturing drawing.

Main prototype modules

- **4-port wind-noise array:** distributed ports and manifold arrangement to spatially average turbulent pressure variations.
- **Measurement/reference chambers:** provide the pressure paths needed for differential measurement and controlled low-frequency response.
- **Capillary equalization path:** provides a controlled slow pressure-balancing mechanism.
- **Differential pressure sensor:** converts small pressure differences into an electrical signal.
- **Low-noise analog front end:** instrumentation-grade amplification, filtering and dual-gain operation.
- **24-bit ADC:** high-resolution digitization of the conditioned sensor output.
- **STM32-class controller:** sampling, timestamping, FFT/PSD processing and data management.
- **Communication:** LoRa/GSM option for remote data transfer.
- **Environmental enclosure:** protects electronics while keeping the pressure interface mechanically stable.

5. REPRESENTATIVE INFRASOUND WAVEFORM

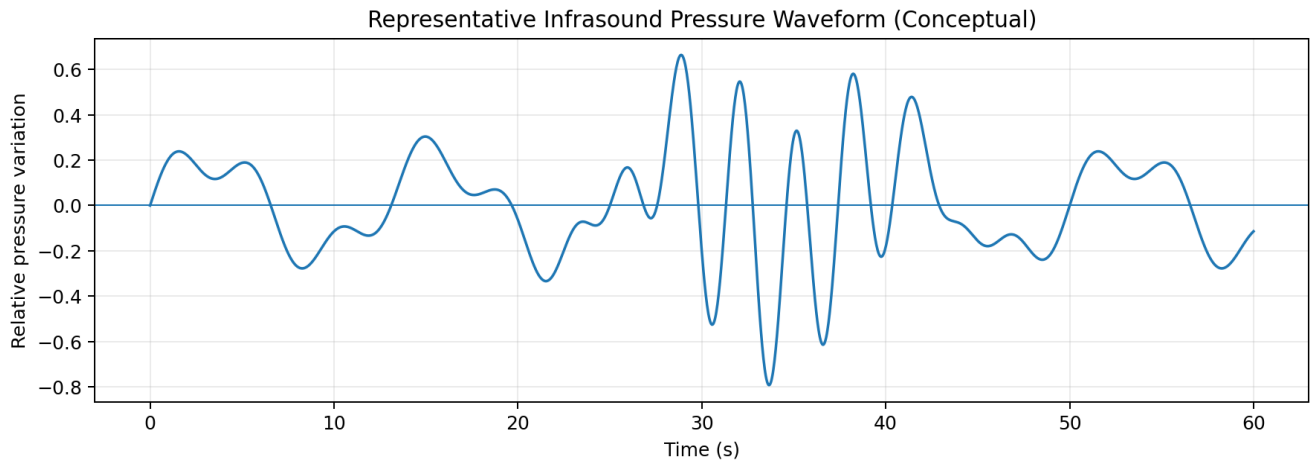


Figure 4. Conceptual waveform showing low-frequency atmospheric variation and a transient event feature. This is an illustrative signal, not measured prototype data.

What the waveform-processing stage should provide

- Continuous time-domain waveform for raw signal inspection.
- FFT spectrum to identify dominant frequency components.
- PSD estimate to compare signal energy across frequency bands.
- Event detection features such as amplitude, duration, onset and dominant frequency.
- Quality indicators for clipping, saturation, sensor drift and abnormal noise.
- Time-stamped records suitable for later correlation with other observations.

6. EXISTING APPROACH vs PROPOSED CONCEPT

Parameter	Typical conventional approach	Bharat Infrasound Sensor concept
Wind-noise control	External or basic wind protection	4-port distributed interface + manifold
Pressure measurement	Single pressure sensing path	Differential measurement using measurement/reference paths
Low-frequency stability	Depends strongly on sensor implementation	Controlled capillary equalization + chamber design
Analog conditioning	Fixed/basic amplification	Low-noise AFE with dual-gain path
Digitization	Application-dependent	High-resolution 24-bit ADC target
Temperature handling	External correction or limited compensation	Integrated temperature sensing + compensation
Processing	Raw recording or external analysis	STM32-class edge processing with FFT/PSD
Connectivity	Local data collection	Local storage + optional LoRa/GSM telemetry
Calibration	Periodic external calibration	Designed for structured calibration/self-check workflow

The comparison is architectural: final performance must be demonstrated through calibration, noise testing and prototype measurements.

7. ELECTRONICS & SIGNAL PROCESSING

- **Sensor interface:** low-bias, low-noise differential acquisition path.
- **Analog filtering:** anti-alias filtering and suppression of unwanted high-frequency content.
- **Dual gain:** selectable gain path to preserve sensitivity while reducing overload risk.
- **ADC:** 24-bit conversion target for high-resolution waveform capture.
- **Temperature channel:** records sensor/enclosure temperature for drift compensation.
- **Microcontroller:** manages sampling, buffering, timestamping and diagnostics.
- **Signal processing:** FFT, PSD, digital filtering and event-feature extraction.
- **Data logging:** stores waveform and diagnostic information for later analysis.
- **Telemetry:** LoRa/GSM option for remote monitoring where connectivity is available.

Proposed processing sequence

- Acquire synchronized pressure and temperature samples.
- Apply calibration coefficients and temperature compensation.
- Check signal quality and sensor health.
- Generate waveform, FFT and PSD products.
- Extract event features when a detection threshold is crossed.
- Store raw/processed data and transmit selected summaries.

8. CALIBRATION, TESTING & VALIDATION

- **Static pressure test:** verify zero/offset response and repeatability.
- **Dynamic pressure test:** apply controlled low-frequency pressure variations.
- **Frequency response:** evaluate sensitivity across the intended band.
- **Self-noise test:** measure sensor output with controlled quiet input conditions.
- **Temperature test:** characterize offset and sensitivity changes across expected operating temperatures.
- **Wind-noise test:** compare the sensing path with and without the wind-noise interface.
- **ADC/electronics test:** verify gain, clipping limits, sampling stability and timing.
- **End-to-end test:** compare known input signals with the final digital waveform and spectral output.

Validation outputs

- Sensitivity versus frequency plot.
- Self-noise / noise-floor plot.
- Temperature drift curve.
- Waveform and FFT examples.
- Calibration coefficient table.
- Long-duration stability record.
- Communication/data-integrity test report.

9. PROPOSED INNOVATION POINTS

- Compact wind-noise-reduction interface integrated with the sensor architecture.
- Differential pressure measurement with controlled reference-path equalization.
- Dual-gain analog front end for sensitivity and dynamic-range management.
- Temperature-aware digital compensation to reduce long-term drift.
- 24-bit acquisition architecture targeted for small pressure fluctuations.
- Edge FFT/PSD processing to reduce dependence on external computers.
- Built-in sensor-health and signal-quality checks.
- LoRa/GSM option for distributed remote monitoring.
- Modular enclosure and pressure interface intended for field-serviceability.
- India-focused prototype architecture using commonly accessible embedded-electronics building blocks.

Important engineering note

The items above describe the proposed design concept. Claims about final sensitivity, noise floor, accuracy, environmental range and detection performance should be stated as measured specifications only after prototype calibration and testing.

10. APPLICATIONS

- Atmospheric and environmental monitoring.
- Geophysical and atmospheric research.
- Volcanic and severe-weather research.
- Meteor and rocket-launch acoustic research.
- Long-range low-frequency acoustic event studies.
- Distributed scientific sensor networks and educational instrumentation.

TECHNICAL REFERENCES

- CTBTO — Infrasound Monitoring: microbarometers measure minute atmospheric pressure changes; array elements use wind-noise-reducing pipe/manifold systems.
- CTBTO Science & Technology Conference 2021 — Microbarometer for infrasound monitoring systems: published example with a 0.01–20 Hz operating range.
- CTBTO — International Monitoring System: overview of global infrasound stations, station architecture and data processing.
- CTBTO Science & Technology Conference 2013 — MB3 microbarometer: discussion of high-sensitivity, low-noise sensing and internal calibration concepts.

Source domains: ctbto.org and conferences.ctbto.org

END OF PROJECT DOCUMENTARY

PHOTONICS Research

Miniaturized and highly sensitive fiber-optic Fabry–Perot sensor for mHz infrasound detection

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Infrasound detection is important in natural disasters monitoring, military defense, underwater acoustic detection, and other domains. Fiber-optic Fabry–Perot (FP) acoustic sensors have the advantages of small structure size, long-distance detection, immunity to electromagnetic interference, and so on. The size of an FP sensor depends on the transducer diaphragm size and the back cavity volume. However, a small transducer diaphragm size means a low sensitivity. Moreover, a small back cavity volume will increase the low cut-off frequency of the sensor. Hence, it is difficult for fiber-optic FP infrasound sensors to simultaneously achieve miniaturization, high sensitivity, and extremely low detectable frequency. In this work, we proposed and demonstrated a miniaturized and highly sensitive fiber-optic FP sensor for mHz infrasound detection by exploiting a Cr-Ag-Au composite acoustic-optic transducer diaphragm and a MEMS technique-based spiral micro-flow hole. The use of the spiral micro-flow hole as the connecting hole greatly reduced the volume of the sensor and decreased the low-frequency limit, while the back cavity volume was not increased. Combined with the Cr-Ag-Au composite diaphragm, a detection sensitivity of $-123.19 \text{ dB re } 1 \text{ rad}/\mu\text{Pa}$ at 5 Hz and a minimum detectable pressure (MDP) of $1.2 \text{ mPa}/\text{Hz}^{1/2}$ at 5 Hz were achieved. The low detectable frequency can reach 0.01 Hz and the flat response range was 0.01–2500 Hz with a sensitivity fluctuation of $\pm 1.5 \text{ dB}$. Moreover, the size of the designed sensor was only $12 \text{ mm} \times \Phi 12.7 \text{ mm}$. These excellent characteristics make the sensor have great practical application prospects. © 2024 Chinese Laser Press

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1. INTRODUCTION

Infrasound waves are widely present in nature, with frequencies below the threshold of human hearing ($< 20 \text{ Hz}$) [1]. Before some natural disasters occur, infrasound waves are generated, such as seismic activity [2–5], volcanic eruptions [6], debris flows [7,8], and snow avalanches [9,10]. Hence, early warning of disasters can be achieved by detecting the infrasound waves. In the military field, infrasound waves have been used as an attack weapon since they can resonate with certain organs of the human body, and thus infrasound detection/defense is also essential. On the other hand, pipeline leakage monitoring [11,12] and underwater acoustic detection [13,14] also include the infrasound frequency band. Therefore, the development of infrasound sensors is crucial for practical applications and is moving toward miniaturization, high performance, and intelligence. Different principle-based infrasound sensors transpose the infrasound signals into different measurable physical quantities through the implementation of various energy conversion mechanisms such as capacitive, piezoelectric, and optical inter-

ference [15]. In recent years, the establishment of the International Monitoring System (IMS) infrasound network has resulted in unprecedented advancements in infrasound research [16]. The commercial electrical infrasound sensor (Martec MB2005) was applied to the IMS for the Comprehensive Nuclear-Test-Ban Treaty. However, although electrical infrasound sensors are convenient to use, they are susceptible to electromagnetic interference, difficult to achieve long-distance detection and multiplexing networking [17], which limits their application in certain specific fields, such as pipeline leakage monitoring and underwater acoustic detection [18].

Fiber-optic Fabry–Perot (FP) acoustic sensors, based on the optical FP interference principle, have the advantages of high sensitivity, small structure size, long-distance detection, immunity to electromagnetic interference, easy to multiplexing networking, and so on [19–27]. For an FP acoustic sensor, acoustic-optic transducer diaphragms of various materials were usually considered to improve the detection sensitivity of acoustic pressure response, such as metallic diaphragms [28–31],

silicon-related diaphragms [32–34], polymer diaphragms [35,36], two-dimensional materials diaphragms [37–40], and special structure diaphragms [41–44]. On the other hand, the detection sensitivity is directly proportional to the size of the diaphragm. When pursuing miniaturized sensors, the detection sensitivity will be sacrificed due to the small diaphragm size. Furthermore, the detection sensitivity of the sensor is not only constrained by the materials and sizes of transducer diaphragms but also depends on the back cavity structures, which are designed to improve the response characteristics and balance the influence of the environmental factors such as temperature. For infrasound detection, to decrease the low cut-off frequency, the back cavity volume can be increased, but this will also increase the size of the sensor. Another optional approach is mechanical filtering [45,46]. In 2021, Zhang *et al.* reported a fiber-optic FP sensor for 0.01 Hz infrasound detection by connecting the capillary glass tube through the pressure difference mechanical filtering device. The size of the capillary tube and the back cavity jointly restrict the low cut-off frequency. Nevertheless, the limitation of capillary tube length leads to an increase in the back cavity volume and the sensor size, resulting in a very large size of $>20 \text{ cm} \times \Phi 8 \text{ cm}$. In addition, the frequency response range of the sensor was only 0.01–1 Hz [46]. Therefore, for infrasound detection, it is a challenge for fiber-optic FP sensors to simultaneously achieve miniaturization, high sensitivity, and extremely low detectable frequency.

In this work, a fiber-optic FP sensor with a very small size of $12 \text{ mm} \times \Phi 12.7 \text{ mm}$ was proposed and demonstrated for highly sensitive mHz infrasound detection. A chromium-silver-gold (Cr-Ag-Au) composite metal diaphragm was selected as the acoustic-optic transducer diaphragm, which has high sensitivity and stable performance. To decrease the low cut-off frequency and reduce the sensor size, the equivalent circuit of the acoustic sensing system was analyzed, and a miniaturized spiral micro-flow hole was designed and fabricated by micro electromechanical system (MEMS) technique to be employed as the connecting hole with a long length. The low detectable frequency can reach 0.01 Hz and the flat response range was from 0.01 to 2500 Hz with a sensitivity fluctuation of $\pm 1.5 \text{ dB}$. The infrasound signals were demodulated accurately in real time by using the white light interferometry

(WLI) Fourier phase demodulation method. The achieved performances of the designed sensor were superior with respect to the reported results, in terms of acoustic pressure sensitivity, minimum detectable pressure (MDP), and low detectable frequency as well as sensor size. Such an infrasound sensor provides a promising solution for the application in natural disaster warning, underwater acoustic detection, pipeline monitoring, and other domains.

2. PRINCIPLES AND SIMULATION

A schematic diagram of the proposed fiber-optic FP acoustic sensor is shown in Fig. 1(a). The end surface of the fiber and the transducer diaphragm formed an FP interference resonator (FP cavity), which can be simplified as a two-beam interferometer due to the low reflectivity of the fiber end surface. When the acoustic wave acts on the sensitive transducer diaphragm, the diaphragm is elastically deformed, and thus the reflection interference spectrum is modulated by the acoustic wave.

When the thin diaphragm vibrates, the edge experiences minimal displacement while the center experiences maximum since the surrounding of the diaphragm is clamped. For an FP sensor, the fiber end surface must align with the diaphragm center to determine the displacement sensitivity of the diaphragm by calculating the ratio of the shape variable at the midpoint to the applied acoustic pressure. The diaphragm's tension is $T = Ph$, where P represents the tensile stress and h represents the thickness, and the sensitivity can be expressed as [47]

$$S_d = \frac{1}{k^2 Ph} \left[\frac{1}{J_0(ka)} - 1 \right], \quad (1)$$

where $k = \omega/c$, $J_0(ka)$ is the zeroth-order Bessel function, and a is the radius of the diaphragm. As a result, to improve the sensitivity, the thickness and the tensile stress of the diaphragm should be decreased, and the radius of the diaphragm should be increased. The complete theoretical model of diaphragm vibration is depicted in Appendix A.

When selecting diaphragm materials, theoretical analysis and calculations were conducted. On the other hand, for silicon-related diaphragms or polymer diaphragms, due to their low reflectivity, a layer of metal needs to be coated on them to

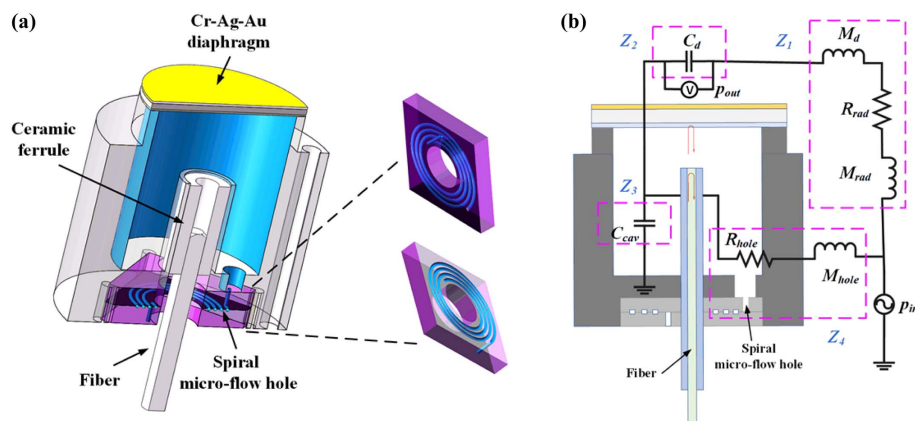


Fig. 1. (a) Schematic diagram of the fiber-optic FP acoustic sensor. (b) Electrical-mechanical-acoustic equivalent model of the sensor.

improve reflectivity, which undoubtedly increases the production cost and process difficulty. As for two-dimensional materials, such as graphene, the large-scale preparation and the consistency of sensors are difficult to guarantee. By the calculation and simulation of diaphragm vibration theory, the Cr-Ag-Au composite metal diaphragm was selected as the acoustic-optic transducer diaphragm, which has high sensitivity and stable performance. Cr was selected as the bottom layer because Cr has excellent corrosion resistance and wear resistance, and Cr was easier to grow on the photoresist during evaporation. The temperature of Ag can be kept at a constant level at a low temperature during the coating process, which can control the residual thermal stress of the diaphragm caused by temperature change. As a result, the aggregation of thermal stress can be reduced, and thus the sensitivity can be further improved. Consequently, Ag was selected as the main deformation layer. However, Ag is prone to oxidation in the air to form silver oxide, which can affect the stability of the sensor performance. Hence, a layer of Au diaphragm was evaporated on the upper layer due to the good chemical stability of Au. In addition, Au can provide a high reflectivity for optical interferometry measurement. Furthermore, Ag is a face-centered cubic crystal structure and Cr is a body-centered cubic crystal structure. The interface composed of the two different crystal structures has a strong ability to hinder dislocation motion [48].

Moreover, to improve the low-frequency response of the FP acoustic sensor, the electrical-mechanical-acoustic equivalent model was applied to the sensor design, as shown in Fig. 1(b). In the studied acoustic system, the size of each element is much smaller than the wavelength of the infrasound wave, which can be analyzed as a lumped system [49]. For a diaphragm-based

acoustic sensor, the diaphragm's performance can be easily affected by external environments such as temperature if the sensor cavity is closed, so an open cavity structure can eliminate the influence of external factors by making a connecting hole in the sensor cavity. However, the low-frequency acoustic pressure response of the sensor will be reduced due to the existence of the connecting hole. In the equivalent circuit model, the actual response of the acoustic sensor is determined by the sensitivity S_d of the diaphragm and the transfer function H . When the sensor is placed in an acoustic field with an acoustic pressure of p_{in} , only the acoustic pressure acting on the diaphragm can cause the deformation of the diaphragm, which is the effective acoustic pressure p_{out} . The ratio of p_{out} to p_{in} is defined as the transfer function H . The elements in Fig. 1(b) are the acoustic mass M_d of the diaphragm, the acoustic compliance C_d of the diaphragm, the radiation acoustic mass M_{rad} of the diaphragm, the radiation acoustic impedance R_{rad} of the diaphragm, the acoustic impedance R_{hole} of the connected hole, the acoustic mass M_{hole} of the balanced hole, and the acoustic compliance C_{cav} of the cavity. The complete electrical-mechanical-acoustic equivalent model is depicted in Appendix B. It can be seen from the transfer function expression that the low cut-off frequency f of the sensor can be expressed as follows:

$$f = \frac{1}{2\pi} \frac{C_{cav} + C_d}{C_d R_{hole} (1 + C_{cav}/C_d)}. \quad (2)$$

Figures 2(a) and 2(b) show the simulation results for the effect of connecting holes size on the transfer function. The presence of the connecting hole does not affect the sensor response in the flat area. However, in the low-frequency range, both the length and radius of the connecting hole greatly in-

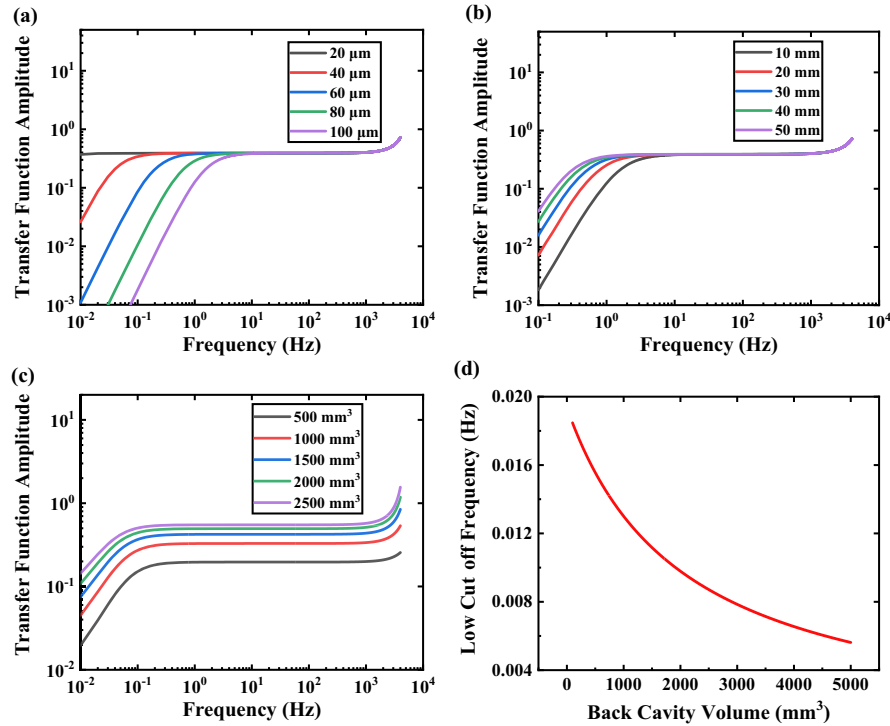


Fig. 2. (a) Effects of connecting hole radius on transfer function when connecting hole length is fixed. (b) Effects of connecting hole length on transfer function when connecting hole radius is fixed. (c) Effects of back cavity volume on transfer function. (d) Effects of back cavity volume on low cut-off frequency.

fluence the frequency response of the sensor. Specifically, shorter connecting hole length and larger connecting hole radius can result in more severe low-frequency attenuation. In addition, the effects of back cavity volume on the transfer function and low cut-off frequency are shown in Figs. 2(c) and 2(d), respectively. It can be seen that a large back cavity volume can lower the cut-off frequency, but a large back cavity volume means a large sensor size. Hence, to enhance the sensor's response sensitivity in the low-frequency band and minimize attenuation, it is recommended to increase the length of the connecting hole and decrease its radius as much as possible within the constraints of the structure and process, while ensuring that the sensor size does not increase.

3. EXPERIMENTAL SETUP AND RESULTS

A. Device Design

The MEMS technique was applied to the processing of the precision structures of the composite diaphragm. As shown in Fig. 3(a), the photoresist was used as the sacrificial layer in a silicon wafer and the electron beam evaporation process was adopted to complete the preparation of the high-performance composite metal diaphragms. The thickness was regulated to

20 nm, 500 nm, and 30 nm of Cr, Ag, and Au layers, respectively, by controlling the process parameters, and the density of the composite diaphragm was 10.856 g/cm^3 . A custom metal encapsulation was bonded to the composite diaphragm by epoxy glue and placed on a heating table for heating the epoxy glue to cure. The curing temperature was 80°C and the curing time was 60 min; then the diaphragm was transferred to the metal part by soaking in acetone. The effective sensing radius of the diaphragm was 4 mm.

To decrease the low cut-off frequency of the FP infrasound sensor, a design of a spiral micro-flow hole was proposed based on the MEMS technique, which can effectively increase the connecting hole length and reduce the sensor size, as shown in Figs. 3(b) and 3(c). First, the photoresist as a mask was coated on two silicon wafers. Second, the designed patterns of two mirror-symmetrical miniature grooves with a same diameter of about $48 \mu\text{m}$ were transferred to the photoresist in the two silicon wafers by photoetching, respectively. Third, the two grooves were etched on the two silicon wafers, respectively. The spiral shape extended the total length of each groove to 96.6 mm. Then the photoresist was cleaned off. Fourth, the starting point of one spiral groove and the end point of another spiral groove were etched into two through-holes by using the

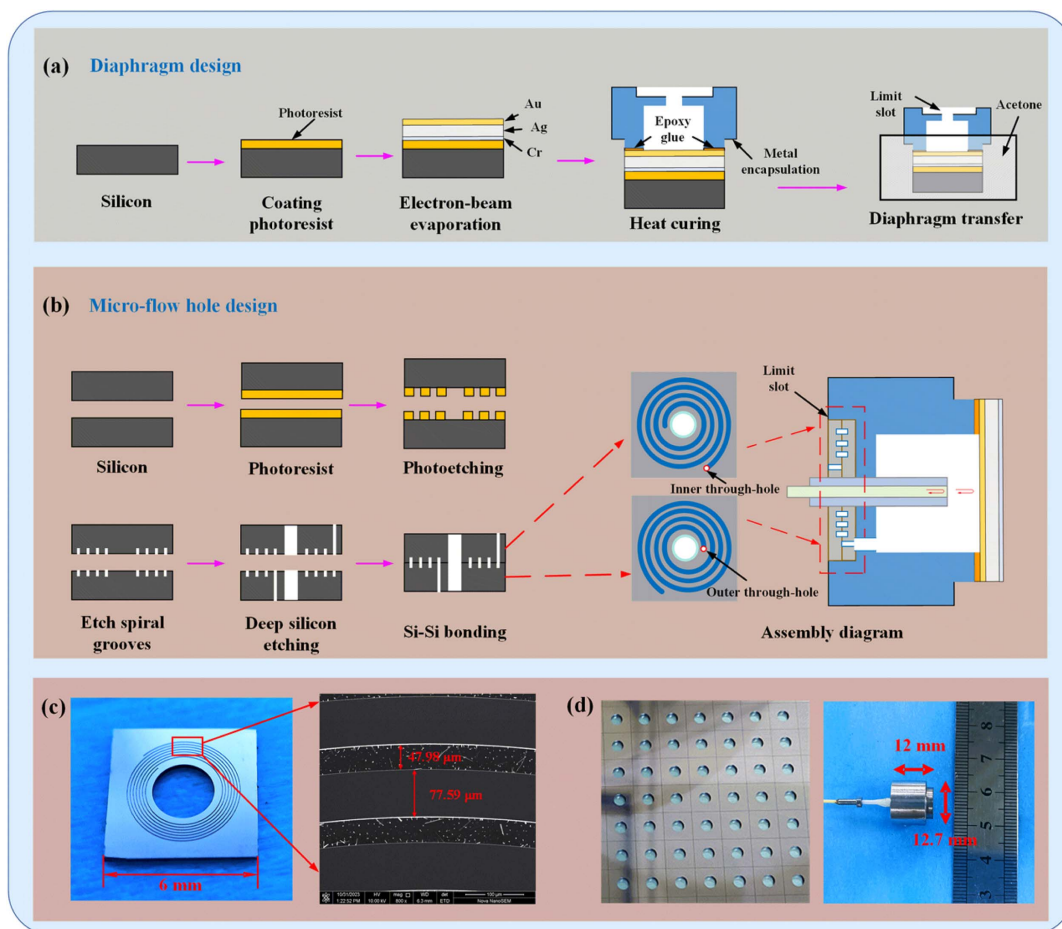


Fig. 3. (a) Fabrication process of the diaphragm. (b) Fabrication process of the spiral micro-flow hole and assembly of the sensor. (c) Left: picture of the single layer of silicon chip before bonding. Right: scanning electron microscope (SEM) image of the spiral groove structure. (d) Left: picture of the dual layers of large-area silicon wafers after bonding. Right: picture of the fiber-optic FP sensor.

deep silicon etching method, respectively. The through-holes for optical fiber penetration were simultaneously etched out. Finally, through the silicon–silicon bonding process for the two silicon wafers, the spiral micro-flow holes were formed by the mirror-symmetrical surface-etched spiral grooves inside two layers of silicon wafers. Due to the limitation of the silicon–silicon bonding process, two large-area silicon wafers were used in the above steps. Each small silicon chip with a size of $6\text{ mm} \times 6\text{ mm}$ containing a spiral micro-flow hole was separated from the large-area silicon wafer by the slicing method after bonding. Figure 3(c) shows the picture of the single layer of silicon chip before bonding and the scanning electron microscope (SEM) image of the spiral groove structure. Figure 3(d) left shows the picture of the dual layers of large-area silicon wafers after bonding. A limit slot was designed on the metal encapsulation to fix the chip, as shown in Figs. 3(a) and 3(b). The physical picture of the completed fiber-optic FP sensor is displayed in Fig. 3(d) right. The volume of the packaged sensor back cavity is 417 mm^3 .

B. Experimental Testing

According to the principle of acoustic metrology, we constructed an aeroacoustic testing system and conducted a performance test on the fiber-optic FP sensor, as shown in Fig. 4. Since the coupling cavity can suppress the low-frequency noise in the environment, we utilized the coupling cavity comparison method to measure the sensor response in the low-frequency band (0.01–100 Hz). Moreover, the free-field comparison method was employed to measure the sensor response in the high-frequency band (100 Hz to 5 kHz) due to the uneven distribution of acoustic field in the coupling cavity.

A C-band amplified spontaneous emission (ASE) light source module was implemented as the probe light source, with a wavelength range of 1526–1563 nm. Moreover, the WLI Fourier phase demodulation method was employed for acoustic signal extraction [50,51]. The ASE light was coupled into the fiber-optic FP sensor through a variable optical attenuator (VOA) and a fiber-optic circulator, and the reflected light signals were imported to a fiber Bragg grating analyzer (FBGA). The spectral signal was collected in real-time and input into the computer via the FBGA, which has a high sampling rate of 20 kHz to detect acoustic signals in a wider frequency range.

The spectral signals were then processed by a LabVIEW program that integrated a WLI demodulation algorithm to obtain the phase change information.

A low-frequency coupling cavity was used as an acoustic source for the sensor's low-frequency response measurement from 0.01 to 100 Hz. A commercial B&K 4193-L-004 microphone was exploited as the reference. Both the FP infrasound sensor and the reference microphone were simultaneously positioned at the two input ports of the low-frequency coupling cavity. To measure the high-frequency response from 100 Hz to 5 kHz, a commercial B&K 4190-L-001 microphone and a commercial speaker were used as the reference and the acoustic source, respectively. The FP infrasound sensor and the reference microphone were symmetrically placed on either side of the acoustic source axis at the free field with a distance of approximately 2.0 m from the test acoustic source. To reduce the influence of the environment, all experimental devices were placed in a fully anechoic chamber. When testing at both high frequency and low frequency, the signal generator of a B&K 3160-A-042 acoustic analyzer produced a sinusoidal signal and drove the test sound source to generate acoustic waves through a power amplifier with a gain of 18 dB. The output electrical signal of the reference microphone was collected and analyzed by the acoustic analyzer to obtain the value of test acoustic pressure. Furthermore, the detection sensitivity of the FP infrasound sensor can be obtained by the WLI Fourier phase demodulation algorithm. The photos of the sensor calibration system are shown in Appendix C.

C. Experimental Result

Figures 5(a)–5(c) show the measured frequency spectra of the sensor output signals by the WLI Fourier phase demodulation algorithm under the acoustic waves at 0.01 Hz, 0.05 Hz, and 0.1 Hz, respectively, which demonstrate the superior frequency resolution of the sensor. The demodulated 0.1 Hz acoustic signal with an excellent sine fitting, as shown in Fig. 5(d), means that the acoustic signal was accurately restored. In addition, the linear response of the sensor was also measured, as shown in Fig. 5(e). The linearity of the sensor reached 0.9999 at 5 Hz acoustic wave.

The measured acoustic pressure sensitivity of the sensor from 0.01 Hz to 5 kHz is shown in Fig. 6. The results indicate that

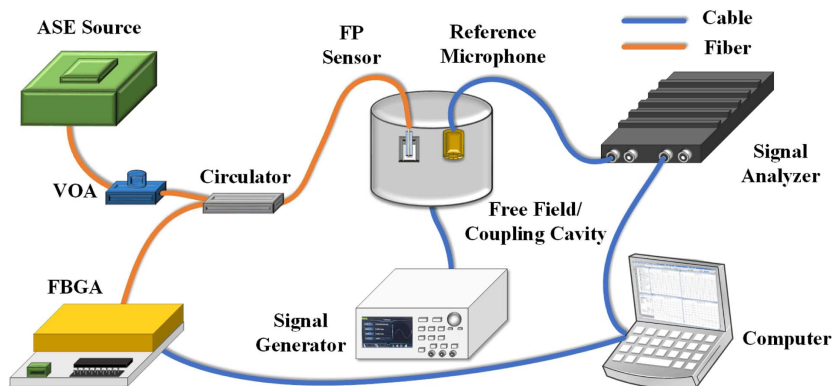


Fig. 4. Infrasound sensor calibration experimental system.

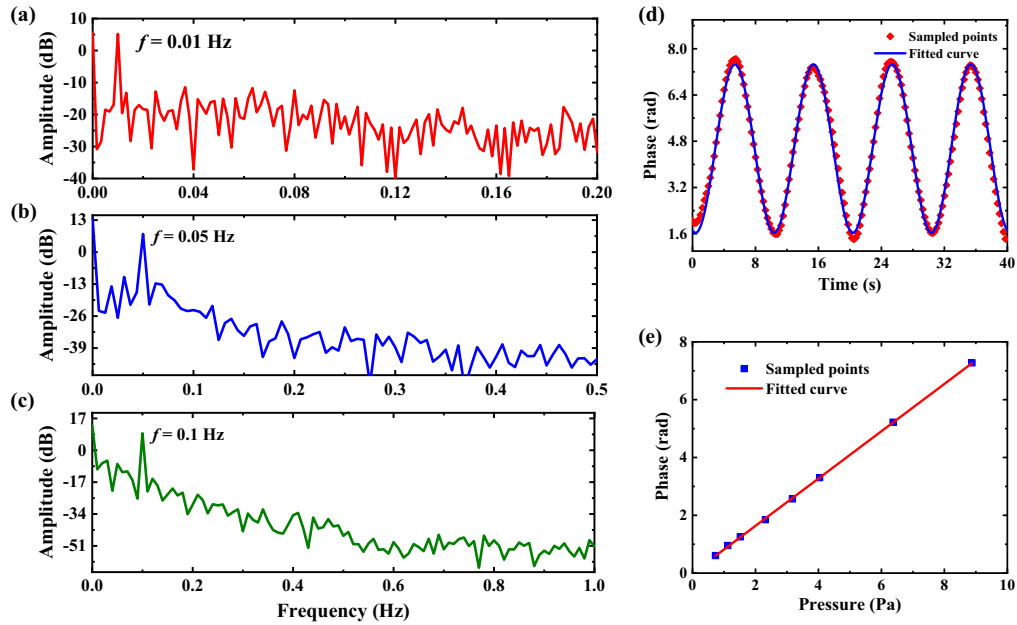


Fig. 5. Frequency spectra of the sensor output signals under the acoustic waves with different frequencies: (a) 0.01 Hz, (b) 0.05 Hz, (c) 0.1 Hz. (d) Sinusoidal fitting of time domain signal of 0.1 Hz. (e) Linearity of the sensor at 5 Hz acoustic wave.

the designed fiber-optic FP acoustic sensor can achieve 0.01 Hz acoustic detection and has a flat response from 0.01 to 2500 Hz with a sensitivity fluctuation of ± 1.5 dB, while the average acoustic pressure sensitivity reached -123.19 dB re 1 rad/ μ Pa at 5 Hz.

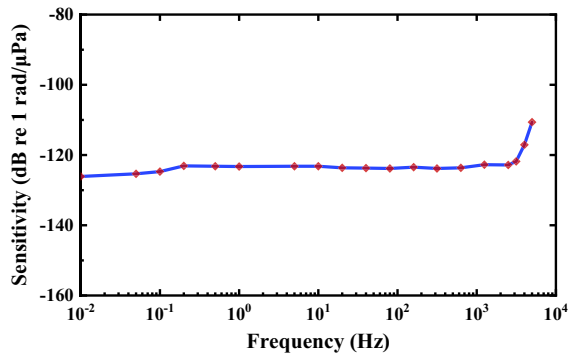


Fig. 6. Acoustic pressure sensitivity of the fiber-optic FP acoustic sensor from 0.01 Hz to 5 kHz.

The MDP is another critical index for an acoustic sensor, which is determined by both sensitivity and noise. To accurately assess the sensor noise characteristic, the sensor was placed in a fully anechoic chamber to effectively suppress the potential influence of environmental noise. The noise output signal, denoted as $n(t)$, was collected over a specific period. Subsequently, a frequency domain analysis of $n(t)$ was conducted, and the noise spectrum $N(f)$ was derived through Fourier transformation, as depicted in Fig. 7(a). Given that the total noise sampling time is T_s , the frequency spectrum abscissa point interval (or frequency resolution) $\Delta f = 1/T_s$ post-Fourier transformation. The MDP of the acoustic sensor with phase demodulation at each frequency can be calculated using Eq. (3). For instance, the MDPs were 1.2 mPa/Hz $^{1/2}$ at 5 Hz and 29.7 μ Pa/Hz $^{1/2}$ at 250 Hz. The MDPs under different frequencies are shown in Fig. 7(b):

$$\text{MDP} = \frac{N(f)}{S(f)\sqrt{\Delta f}} \left[\frac{\text{Pa}}{\text{Hz}^2} \right]. \quad (3)$$

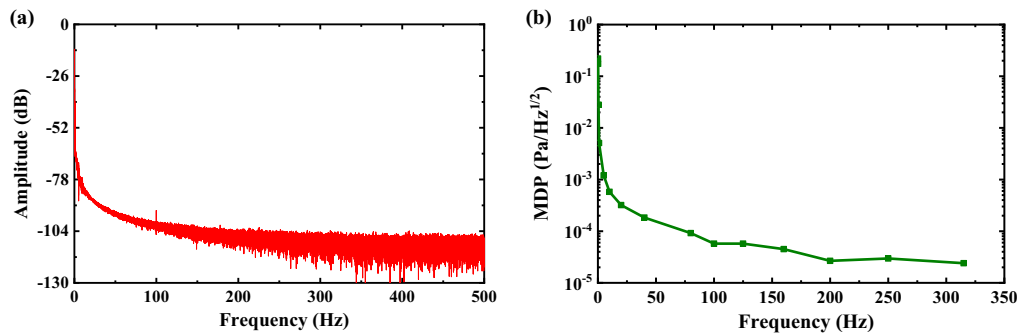


Fig. 7. Noise test results. (a) Noise spectrum, (b) minimum detectable pressure (MDP) under different frequencies.

Table 1. Comparison between Different Fiber-Optic Infrasound Sensors

Diaphragm	Sensitivity	MDP	Frequency (Hz)	Reference
Silicon nitride	-152 dB re 1 rad/ μ Pa at 100 Hz	/	0.1–250	[1]
Gold	-130.6 dB re 1 rad/ μ Pa at 100 Hz	10.2 mPa/Hz ^{1/2} at 5 Hz	0.8–250	[28]
Silicon	-154.6 dB re 1 rad/ μ Pa	/	10–2000	[44]
PPS (polyphenylene sulfide)	/	88.9 mPa/Hz ^{1/2}	0.01–1	[46]
Cr-Ag-Au	-123.19 dB re 1 rad/ μ Pa at 5 Hz	1.2 mPa/Hz ^{1/2} at 5 Hz	0.01–2500	This work

Compared with the currently reported fiber-optic infrasound sensor, the developed sensor in this work has a higher acoustic pressure sensitivity, a better MDP, and a lower detectable frequency. The comparison between different fiber-optic infrasound sensors is listed in Table 1.

4. CONCLUSION AND DISCUSSION

In conclusion, a fiber-optic FP infrasound sensor was designed and fabricated from principle to verification, which used a highly sensitive Cr-Ag-Au composite diaphragm as the acoustic-optic transducer diaphragm and a miniaturized MEMS spiral micro-flow hole with a total length of 96.6 mm as a connecting hole. The infrasound sensor achieved 0.01 Hz infrasound detection and a flat response from 0.01 to 2500 Hz with a sensitivity fluctuation of ± 1.5 dB. The MDP reached 1.2 mPa/Hz^{1/2} at 5 Hz and 29.7 μ Pa/Hz^{1/2} at 250 Hz. Most importantly, the size of the developed infrasound sensor was only 12 mm $\times$ Φ 12.7 mm. Hence, the sensor simultaneously achieved miniaturization, high sensitivity, mHz detectable frequency, and wide frequency response range. Due to the frequency limitation of the reference microphone, the frequency signal below 0.01 Hz was not detected, but it can also be responded in theory. Moreover, the all-fiber detection approach makes the sensor possess the advantages of all-optical passivity, intrinsic safety, immunity to electromagnetic interference, resistance to harsh environments, long-distance detection ability, and multiplexing ability. In practical applications, it is necessary to further optimize the strength of the diaphragm, and in the future, materials with high mechanical strength can be found as transducer diaphragms while ensuring a high sensitivity. Furthermore, the developed small-sized sensor is very convenient to integrate with other sensor devices to compose a multifunctional intelligent detection system. In addition, a sensor array can be formed due to the excellent consistency of the proposed MEMS-based fiber-optic infrasound sensor, and some corresponding algorithms can be combined to achieve the localization or orientation for the sound sources. This work provides a new idea and scheme for the military, disaster warning, pipeline leakage monitoring, and other fields.

APPENDIX A: PRINCIPLES OF THE DIAPHRAGM VIBRATION

Theoretical analysis of the forced vibration of the diaphragm is needed to select the appropriate material for the energy conversion diaphragm. According to the characteristics of the forced vibration of the diaphragm, the vibration equation can be obtained:

$$\nabla_r^2 \xi - \frac{1}{c^2} \cdot \frac{\partial^2 \eta}{\partial t^2} = -\frac{p}{c^2 \sigma}, \quad (\text{A1})$$

where η is the amount of displacement of the diaphragm, $c = \sqrt{T/\sigma}$, σ represents the areal density of the diaphragm, and T represents the tension of the diaphragm. When the diaphragm vibrates, the surrounding of the diaphragm is clamped, and the boundary condition of its vibration is $\xi(r=a) = 0$, where r is the distance from the center of the circle and a is the radius of the diaphragm. The amount of displacement of the diaphragm can be obtained by solving the above equation, and the formula is expressed as follows:

$$\xi(t, r) = \frac{p_a}{k^2 T} \left[\frac{J_0(kr)}{J_0(ka)} - 1 \right] \cdot e^{j\omega t}. \quad (\text{A2})$$

In the formula, $k = \omega/c$, $J_0(kr)$ is the zeroth-order Bessel function. The maximum value of ξ can be obtained when $J_0(ka) = 0$. The resonant frequency equation for the forced vibration of the diaphragm and the first-order resonance frequency equation when the first-order root $\delta_1 = 2.405$ can be denoted as

$$f_n = \frac{\delta_n}{2\pi a} \sqrt{\frac{T}{\sigma}}, \quad (\text{A3})$$

$$f_1 = \frac{2.405}{2\pi a} \sqrt{\frac{T}{\sigma}}. \quad (\text{A4})$$

The formula above explains that the amount of displacement on a diaphragm when acoustic pressure is applied depends on its distance from the center. The edge experiences minimal displacement while the center experiences maximum. For FP sensors, the fiber end face must align with the center to determine the displacement sensitivity of the diaphragm by calculating the ratio of the shape variable at the midpoint to the acoustic pressure applied. The diaphragm's tension is $T = Ph$, where P represents the tensile stress and h represents the thickness, and the sensitivity can be expressed as Eq. (1) in the main text.

The simulation of the influence of various parameters on the diaphragm sensitivity is shown in Fig. 8. It can be seen that in order to improve the sensitivity of the diaphragm, it is necessary to reduce the tensile stress, increase the diameter, and reduce the thickness of the diaphragm. Considering that a diaphragm with a too large or too thin size is more fragile and prone to damage during practical use, a Cr-Ag-Au composite diaphragm with a radius of 4 mm and a total thickness of 550 nm was selected.

APPENDIX B: PRINCIPLES OF THE ELECTRICAL-MECHANICAL-ACOUSTIC EQUIVALENT MODEL

When studying acoustic sensors, the parameters in acoustics can be equivalent to those in circuits, and it is possible to

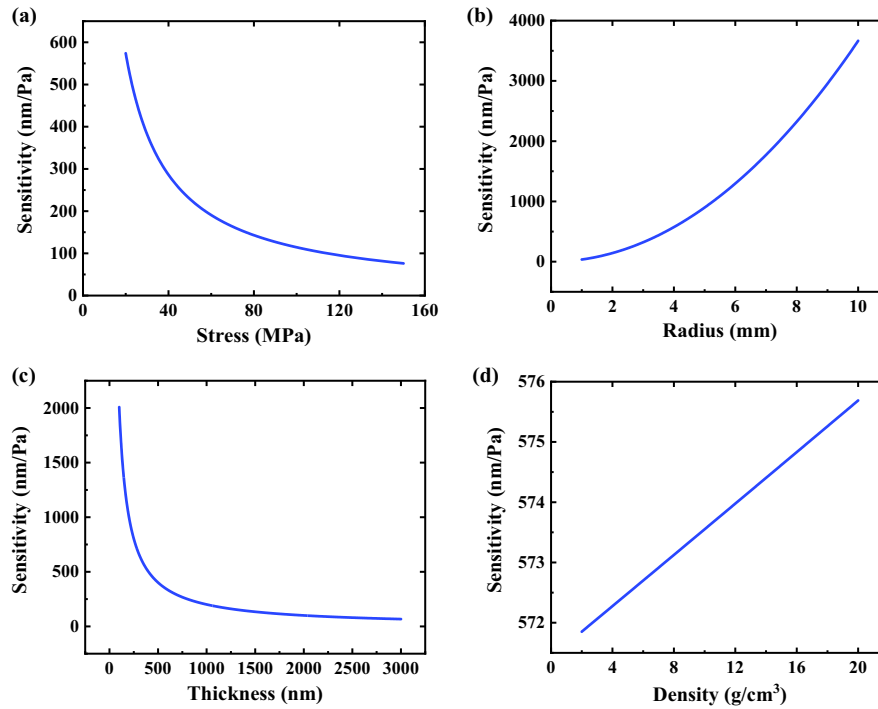


Fig. 8. Relationship between diaphragm sensitivity and various parameters. (a) Stress, (b) radius, (c) thickness, (d) density.

analyze the acoustic phenomenon by solving the equivalent circuit equation. But it should be noted that the electric acoustic equivalence model is mainly applicable to lumped systems, that is, the sizes of the acoustic objects (cavities, stubs, small holes, etc.) are much smaller than the wavelength of the acoustic wave. In the studied acoustic system, the size of each element is much smaller than the wavelength of the acoustic wave, which can be analyzed as a lumped system. For a diaphragm acoustic sensor, an open cavity structure with connecting holes can be used to eliminate sensor performance changes caused by external environments. However, the low-frequency response of the sensor will be reduced due to the connecting hole. In the equivalent circuit model, the actual response of the acoustic sensor is determined by the sensitivity S_d of the diaphragm and the transfer function H :

$$S_{\text{fact}} = H \cdot S_d. \quad (\text{B1})$$

The internal structure and the equivalent circuit system of the proposed FP acoustic sensor are shown in Figs. 1(a) and 1(b), respectively, among them $Z_1 = R_{\text{rad}} + j\omega(M_d + M_{\text{rad}})$, $Z_2 = 1/j\omega C_d$, $Z_3 = 1/j\omega C_{\text{cav}}$, $Z_4 = R_{\text{hole}} + j\omega M_{\text{hole}}$. Therefore, the transfer function H can be expressed as

$$H = \frac{p_{\text{out}}}{p_{\text{in}}} = \frac{Z_2 Z_4}{Z_3 Z_4 + (Z_1 + Z_3)(Z_3 + Z_4)}. \quad (\text{B2})$$

The lumped element definitions are shown in Table 2.

It can be seen from the transfer function expression that the sensing system has two resonance frequencies:

$$f_{\text{high}} = \frac{1}{2\pi} \sqrt{\frac{C_{\text{cav}} + C_d}{C_{\text{cav}} C_d (M_{\text{rad}} + M_d)}}, \quad (\text{B3})$$

$$f_{\text{low}} = \frac{1}{2\pi} \frac{C_{\text{cav}} + C_d}{C_d R_{\text{hole}} (1 + C_{\text{cav}}/C_d)}. \quad (\text{B4})$$

f_{high} is the resonant frequency, and f_{low} is the cut-off frequency of low frequency.

Table 2. Lumped Element Definitions

Model Parameter	Symbol	Definition ^a
Diaphragm acoustic mass	M_d	$\frac{4b\rho_f}{3\pi a^2}$
Diaphragm acoustic compliance	C_d	$\frac{\pi a^4}{8hP}$
Diaphragm radiation mass	M_{rad}	$\frac{8\rho_{\text{air}}}{3\pi^2 a}$
Diaphragm radiation resistance	R_{rad}	$\frac{\pi\rho_{\text{air}}c_{\text{air}}}{\lambda^2}$
Hole acoustic mass	R_{hole}	$\frac{8\varepsilon_{\text{air}}l_{\text{hole}}}{\pi a_{\text{hole}}^4}$
Hole radiation resistance	M_{hole}	$\frac{4\rho_{\text{air}}l_{\text{hole}}}{3\pi a_{\text{hole}}^2}$
Cavity acoustic capacity	C_{cav}	$\frac{V}{\rho_{\text{air}}c_{\text{air}}^2}$

^a ρ_f is the density of the diaphragm, ρ_{air} is the density of air, c_{air} is the acoustic velocity of air, V is the volume of the back cavity, ε_{air} is the viscosity coefficient of air, a_{hole} is the radius of the connecting hole, l_{hole} is the length of the connecting hole, and λ is the wavelength of the acoustic wave.

APPENDIX C: ACOUSTIC TEST PHOTOS OF THE FIBER-OPTIC FP SENSOR

The photos of the sensor calibration system are shown in Fig. 9, including high-frequency testing and low-frequency testing, respectively.

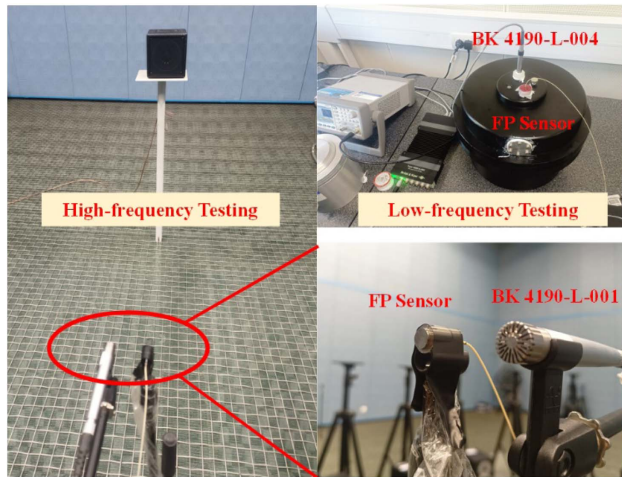


Fig. 9. Acoustic test photos of the fiber-optic FP sensor.

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Disclosures. The authors declare no competing interests.

Data Availability. The data that support the findings of this study are available from the corresponding author upon reasonable request.

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HIGH-SENSITIVITY MICROBAROMETER INFRASOUND SENSOR

Proposed Solution & Prototype Development Document

A compact, low-noise and intelligent atmospheric sensing platform for detecting very-low-frequency pressure fluctuations from approximately 0.01 Hz to 20 Hz.

Project Theme	Precision Infrasound Monitoring
Core Innovation	MEMS + optical readout + low-noise electronics + digital intelligence
Target Frequency	~0.01 Hz to 20 Hz
Deployment	Outdoor and distributed monitoring environments

Professional project-document format • Simple English • Prototype-oriented • Human-readable

1. Executive Summary

The proposed system is a high-sensitivity microbarometer designed to detect low-frequency atmospheric pressure variations that are difficult to capture with conventional pressure sensors. The design combines a sensitive mechanical transducer, differential pressure measurement, optical displacement readout, low-noise analog electronics, temperature compensation, pressure equalization and wind-noise reduction.

- Designed for infrasonic pressure fluctuations in the approximate 0.01–20 Hz range.
- Uses a precision diaphragm-based sensing mechanism to convert pressure changes into mechanical displacement.
- Uses an optical readout concept with VCSEL and photodiode to improve displacement sensitivity.
- Uses a custom low-noise analog front end and digital processing chain for clean signal acquisition.
- Uses temperature compensation to reduce sensor drift caused by environmental temperature changes.
- Uses a multi-port wind-noise reduction structure to reduce turbulence-related pressure fluctuations.
- Uses an STM32-class microcontroller for acquisition, filtering, event detection and communication.
- Supports LoRa/GSM-based remote monitoring for distributed deployments.

2. Problem Definition

Infrasound consists of very-low-frequency atmospheric pressure waves. The challenge is not only detecting a pressure variation, but extracting a very small signal from temperature drift, wind turbulence, mechanical vibration, electronic noise and long-term pressure changes.

Key Design Challenges

- Very small pressure variations require high sensitivity.
- The target band extends to very long periods, so baseline drift must be controlled.
- Outdoor wind turbulence can be much larger than the desired infrasonic signal.
- Temperature changes can alter mechanical and electronic characteristics.
- Low-frequency measurements require stable pressure equalization without suppressing the signal.
- The prototype must combine sensitive mechanics with practical electronics and field protection.

3. Proposed Solution - System Architecture

The proposed architecture follows a complete sensing chain from atmospheric pressure disturbance to a validated digital event record.

Stage	Purpose
Atmospheric Pressure Wave	External infrasonic disturbance
Wind-Noise Reduction Array	Four-port spatial averaging / cancellation concept
Microbarometer Diaphragm	Pressure-to-displacement conversion
Differential / Optical Readout	Sensitive measurement of diaphragm motion
Low-Noise Analog Front End	Amplification and signal conditioning
Temperature Compensation	Drift correction using temperature sensing
ADC + STM32	High-resolution digitization and embedded processing
Digital Filtering + TinyML	Noise rejection and event classification
Communication	LoRa / GSM remote transmission
Monitoring Dashboard	Time-series signal and event visualization

4. Core Technical Design

4.1 High-Sensitivity Pressure Sensing Unit

- Use a precision pressure-sensing chamber with a very small effective volume.
- Use a mechanically optimized diaphragm to obtain measurable displacement from small pressure changes.
- Control diaphragm stiffness, thickness, geometry and stress during design simulation.
- Use stable materials and mechanical mounting to reduce long-term drift.

4.2 Mechanical Diaphragm Transducer

- A thin diaphragm acts as the primary mechanical transducer.
- Atmospheric pressure variation produces controlled diaphragm deflection.
- MEMS fabrication can be explored using an SOI wafer and DRIE-based micromachining.
- COMSOL simulation can be used to study deformation, resonance, stress and sensitivity before fabrication.

4.3 Differential Pressure Measurement

- A differential configuration helps reject common environmental pressure changes and mechanical disturbances.
- The reference side can be coupled to a controlled reference volume through a long-period equalization path.
- The pneumatic path must be designed carefully so the required low-frequency response is preserved.

4.4 Optical Readout

- Use a VCSEL light source and photodiode receiver as a contactless displacement measurement concept.
- A Fabry-Pérot cavity can convert diaphragm displacement into an optical intensity or phase-related signal.
- Optical sensing reduces direct electrical loading of the mechanical element.
- The optical path should be mechanically isolated from vibration and protected from contamination.

5. Low-Noise Electronics

Analog Front End

- Use a low-noise preamplifier close to the sensing element.
- Use controlled gain stages to avoid saturation during strong atmospheric disturbances.
- Use anti-alias filtering before ADC conversion.
- Separate analog and digital power/ground paths where practical.
- Use shielding and careful PCB layout to reduce electromagnetic interference.

Digital Acquisition

- Use a high-resolution ADC suitable for low-frequency precision measurement.
- Use an STM32-class microcontroller for real-time acquisition and processing.
- Store raw and processed measurements for later validation.
- Add timestamping so events can be compared across multiple sensor nodes.

6. Environmental Compensation & Noise Control

Temperature Compensation

- Measure temperature near the mechanical and electronic sensing region.
- Characterize sensor output at different temperatures during calibration.

- Apply a temperature-dependent correction model in firmware.
- Continuously monitor temperature so compensation remains active during field operation.

Long-Period Pressure Equalization

- Provide a controlled pneumatic path for slow static pressure equalization.
- Use a high-flow-resistance path to avoid rapidly short-circuiting the target infrasonic signal.
- Optimize tubing length, restrictor geometry and chamber volume during prototype testing.
- Validate the frequency response experimentally rather than relying only on theoretical calculations.

Wind-Noise Reduction

- Use a four-port spatial inlet structure around the pressure sensor.
- Route each port through matched pneumatic paths before combining the signals.
- Use spatial averaging to reduce localized turbulent fluctuations.
- Place the sensing enclosure away from sharp edges and strong direct airflow.
- Use field calibration to compare wind-noise performance with and without the inlet array.

7. Intelligent Signal Processing

- Apply high-pass, low-pass and band-limited digital filters according to the target frequency band.
- Use adaptive noise estimation to identify changing background conditions.
- Use beamforming / spatial processing when multiple sensor channels are available.
- Use TinyML for lightweight event classification at the edge.
- Classify events using signal features such as amplitude, duration, spectral content and multi-channel correlation.
- Transmit only important event summaries when communication bandwidth is limited, while retaining raw data locally.

8. Proposed Prototype – Physical Construction

Mechanical Stack

- Weather-resistant outer enclosure.
- Four-port wind-noise reduction inlet.
- Internal acoustic / pneumatic chamber.
- Precision diaphragm assembly.
- Optical emitter-receiver module.
- Reference volume and pressure equalization path.
- Vibration-isolated internal mounting.

Electronics Stack

- Low-noise analog front-end PCB.
- Temperature sensor.
- High-resolution ADC.
- STM32 microcontroller board or custom control PCB.
- Local storage for data logging.
- LoRa / GSM communication module.
- Regulated power supply with battery/solar-ready option.

9. Innovative Features

Innovation	Implementation Idea
Hybrid Optical-MEMS Readout	Combines precision mechanical sensing with contactless optical displacement measurement.
Four-Port Wind-Noise Array	Uses spatially distributed pneumatic inputs to reduce turbulence-related noise.
Adaptive Digital Filtering	Changes filtering behaviour according to measured background noise.
Edge TinyML Classification	Performs lightweight event recognition locally before transmission.
Temperature-Aware Calibration	Uses continuous temperature data to correct sensor drift.
Distributed Sensor Networking	Multiple nodes can share timestamped events for spatial confirmation.
Energy-Aware Communication	Transmits event summaries while keeping detailed data locally available.
Simulation-to-Prototype Workflow	Uses multiphysics simulation before fabrication to reduce design iterations.

10. Design & Development Workflow

01. Define target sensitivity, bandwidth and environmental requirements.
02. Model the diaphragm, chamber and pressure path in COMSOL or equivalent simulation software.
03. Select materials and mechanical dimensions based on sensitivity and stability.
04. Develop the optical readout and validate displacement response.
05. Design the low-noise analog front end and ADC interface.
06. Build the first laboratory prototype.
07. Perform static, dynamic, temperature and vibration characterization.
08. Develop wind-noise reduction hardware and compare measured noise floors.
09. Implement digital filtering, calibration and TinyML event classification.
10. Integrate communication and remote monitoring.
11. Perform controlled field trials and compare results with reference instruments where available.
12. Freeze the validated design and prepare the final enclosure and documentation.

11. Testing & Validation Plan

Test	Validation Method
Sensitivity Test	Apply known small pressure variations and measure output response.
Frequency Response	Evaluate amplitude and phase response across the target band.
Noise Floor Test	Measure baseline noise in a controlled low-disturbance environment.
Temperature Test	Characterize output drift over the intended operating temperature range.
Wind Test	Compare signal noise under different airflow conditions.
Long-Duration Test	Monitor baseline stability, drift and data continuity over extended periods.
Event Detection Test	Use controlled, safe reference signals to verify detection and classification.
Communication Test	Verify reliable timestamped transmission and local data backup.

12. Challenge → Mitigation

Challenge	Proposed Mitigation
Very small pressure signal	High-sensitivity diaphragm + optical readout + low-noise amplification

Challenge	Proposed Mitigation
Wind turbulence	Four-port inlet + spatial averaging + digital noise rejection
Temperature drift	Temperature sensing + calibration model + real-time compensation
Long-period baseline drift	Controlled pressure equalization + baseline tracking
Electrical interference	Low-noise PCB layout + shielding + filtered power
Mechanical vibration	Rigid enclosure + vibration isolation + signal validation
Large data volume	Edge processing + event-based transmission + local storage
Field contamination	Protected enclosure + serviceable pneumatic inlet design

13. Practical Prototype Roadmap

- Phase 1 – Concept: requirements, architecture and preliminary calculations.
- Phase 2 – Simulation: diaphragm, cavity and pneumatic-path modelling.
- Phase 3 – Bench Prototype: pressure chamber, diaphragm, optical readout and AFE.
- Phase 4 – Digital Prototype: ADC, STM32 firmware, filtering and data logging.
- Phase 5 – Noise Control: wind array, enclosure and vibration isolation.
- Phase 6 – Intelligence: event features, TinyML classification and multi-node correlation.
- Phase 7 – Field Prototype: weather-resistant packaging, remote communication and long-duration testing.
- Phase 8 – Validation: compare performance against defined sensitivity, bandwidth, stability and noise targets.

14. Expected Outcome

- A compact prototype capable of measuring low-frequency atmospheric pressure fluctuations.
- A documented sensing chain from mechanical pressure conversion to digital event output.
- Improved resistance to wind, temperature and electronic noise compared with a basic pressure-sensor approach.
- A scalable architecture that can be expanded into multiple synchronized sensor nodes.
- A practical foundation for atmospheric research, environmental monitoring and other legitimate infrasound sensing applications.

15. Final Proposed Solution - One-Line Concept

“A smart, low-noise microbarometer that combines MEMS-style diaphragm sensing, optical displacement measurement, wind-noise reduction, temperature compensation and edge intelligence to convert extremely small atmospheric pressure variations into reliable digital events.”

16. Recommended Prototype Block Diagram

ATMOSPHERIC INFRASOUND
↓
4-PORT WIND-NOISE REDUCTION ARRAY
↓
PRECISION DIAPHRAGM / MICROBAROMETER
↓
VCSEL + PHOTODIODE OPTICAL READOUT

↓
LOW-NOISE ANALOG FRONT END
↓
HIGH-RESOLUTION ADC
↓
STM32 MICROCONTROLLER
↓
DIGITAL FILTERING + TEMPERATURE COMPENSATION
↓
TINYML EVENT CLASSIFICATION
↓
LOCAL STORAGE + LoRa / GSM
↓
REMOTE MONITORING DASHBOARD

17. Documentation Notes

- All target performance values should be finalized after simulation and calibration testing.
- Optical readout, MEMS fabrication and pneumatic equalization are proposed engineering approaches and should be validated experimentally.
- Prototype component selection should be based on measured noise, sensitivity, stability, availability and environmental requirements.
- The final enclosure should be designed only after the pneumatic and mechanical response has been characterized.

Prepared as a professional proposed-solution document for prototype planning and technical presentation.

InfraSense — Comparison, Architecture & Disaster-Management Relevance

Technical Comparison & Disaster-Management Relevance Report

High-Sensitivity Microbarometer Infrasonic Sensor — InfraSense

Problem Statement ID: SIH26144

Organization: National Technical Research Organisation (NTRO)

Team Name: [Your Team Name]

Contents

1. Proposed Solution vs. Existing Systems — Point-Wise Comparison
2. System Block Diagram
3. Working Flowchart — Step-by-Step Solution Plan
4. Disaster-Management Relevance
5. Summary

1. Proposed Solution vs. Existing Systems

1.1 Existing Systems — Quick Overview

A. Imported Professional (CTBTO-grade) Sensors

- Use electromagnetic (LVDT), capacitive, or piezoelectric transduction (e.g., MB2000, MB2005, MB3, Hyperion, Chaparral).
- Highly sensitive and field-proven over decades of deployment.
- Cost several lakh rupees per unit; long import lead times; high annual maintenance cost.

B. DIY Strain-Gauge Microbarometers (Hobbyist)

- Metal diaphragm with bonded foil strain gauges in a Wheatstone-bridge configuration.
- Calibrated capillary leak for reference-chamber equalization.
- Requires precise manual fabrication; no built-in wind-noise suppression or documented calibration.

C. Basic MEMS-Only Sensors

- Commodity MEMS barometric chips used without a reference chamber or differential technique.
- Inexpensive but limited to strong, nearby events; vulnerable to wind noise and drift outdoors.

1.2 Point-Wise Comparison Table

Parameter	Imported Professional	DIY Strain-Gauge	Basic MEMS-Only	Proposed InfraSense
Sensing method	Electromagnetic / capacitive / piezoelectric	Strain gauge on metal diaphragm	Bare MEMS die	Differential MEMS + reference chamber
Cost per unit	Several lakh ₹	Low-moderate	Very low	~₹6,500–8,000 (prototype)
Wind-noise handling	Dedicated hose arrays	None	None	Porous-hose spatial filter
Pressure equalization	Precision engineered	Manual capillary leak	None	Calibrated capillary-leak path
Temperature compensation	Factory-calibrated	Partial	Minimal	Sensing + firmware correction
Calibration method	Certified lab	Manual, ad hoc	None	Documented multi-point + reference
DAQ / software	Proprietary	Manual Arduino logging	Basic logging	Open-market digitizer + Python/ObsPy
Import dependency	High	Low	Low	None — indigenous

Parameter	Imported Professional	DIY Strain-Gauge	Basic MEMS-Only	Proposed InfraSense
Deployment density feasible	Very low	Low	High but unreliable	High — low cost + practical

1.3 Why the Proposed Solution Is Better — Key Points

- **Cost advantage:** Over 99% cheaper than imported professional sensors, enabling dense, wide-area deployment that cost currently prevents.
- **Completeness:** The only option in this comparison combining differential sensing, wind-noise filtering, temperature compensation, and documented calibration together.
- **Indigenous:** Fully sourceable and buildable in India — zero dependency on imported sensing technology.
- **Practical to build:** Unlike DIY strain-gauge designs, uses a ready-made differential MEMS element instead of precision manual strain-gauge bonding.
- **Reproducible:** Built on open-market components and open-source software, unlike closed proprietary station systems.

Honest limitation: raw sensitivity of the chosen MEMS element is not expected to match certified professional capacitive/optical instruments — a deliberate cost/practicality trade-off validated through the calibration protocol in Section 3.

2. System Block Diagram

The block diagram below shows the complete InfraSense signal chain, from atmospheric infrasound input to dashboard alerts.

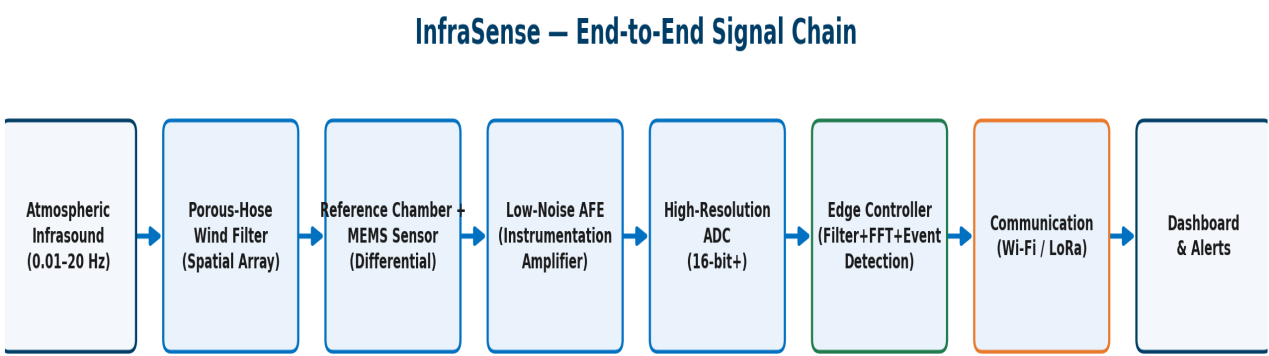


Figure 1: InfraSense end-to-end system block diagram.

2.1 Block-Wise Function Summary

Block	Function
Porous-Hose Wind Filter	Spatially averages turbulent wind fluctuations before they reach the sensor
Reference Chamber + MEMS Sensor	Differential pressure sensing against a calibrated reference volume
Low-Noise AFE	Amplifies the weak differential signal with minimal added noise
High-Resolution ADC	Digitizes the conditioned analog signal (16-bit class)
Edge Controller	Digital filtering, FFT analysis, event detection, temperature compensation
Communication	Transmits data/alerts via Wi-Fi / LoRa to a central system
Dashboard & Alerts	Real-time waveform display, event logs, and threshold-based notifications

3. Working Flowchart — Step-by-Step Solution Plan

The project is executed as five sequential, practically achievable phases, shown in the flowchart below, each producing a demonstrable, testable output.

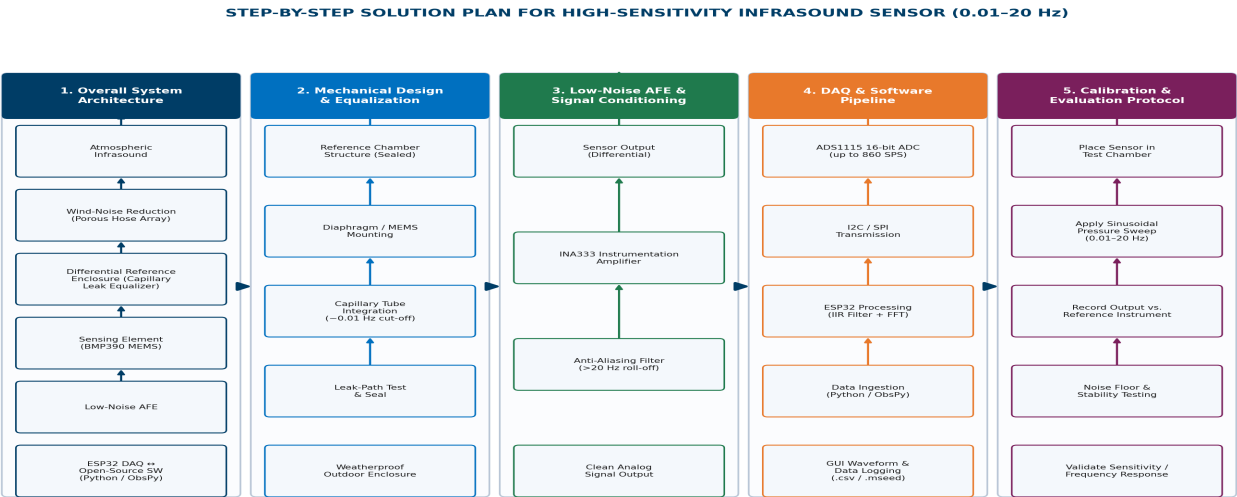


Figure 2: Five-phase step-by-step working flowchart for the InfraSense prototype.

3.1 Phase-Wise Key Points

Phase 1 — Overall System Architecture

- Atmospheric infrasound enters through the porous-hose wind filter.
- Differential reference enclosure equalizes slow drift via a capillary leak.
- BMP390 MEMS sensor measures differential pressure; ESP32 handles DAQ + open-source software.

Phase 2 — Mechanical Design & Equalization

- Sealed reference chamber built and MEMS sensor mounted with vibration isolation.
- Capillary tube tuned to ~ 0.01 Hz cut-off; leak-tested and sealed.
- Assembly housed in a weatherproof, insulated outdoor enclosure.

Phase 3 — Low-Noise AFE & Signal Conditioning

- INA333-class instrumentation amplifier boosts the weak differential signal.
- Anti-aliasing filter (>20 Hz roll-off) limits out-of-band noise before digitization.

Phase 4 — DAQ & Software Pipeline

- 16-bit ADC samples the signal; ESP32 applies IIR filtering and FFT analysis.
- Python/ObsPy pipeline provides real-time waveform GUI and data logging (.csv / .mseed).

Phase 5 — Calibration & Evaluation Protocol

- Sensor placed in a test chamber; a known sinusoidal pressure sweep (0.01–20 Hz) is applied.
- Output recorded against reference; noise floor, sensitivity, and stability validated.

4. Disaster-Management Relevance

Infrasound signatures precede or accompany a wide range of hazardous events, often arriving well before other confirmation channels. The diagram below shows how InfraSense fits into the broader disaster early-warning chain.

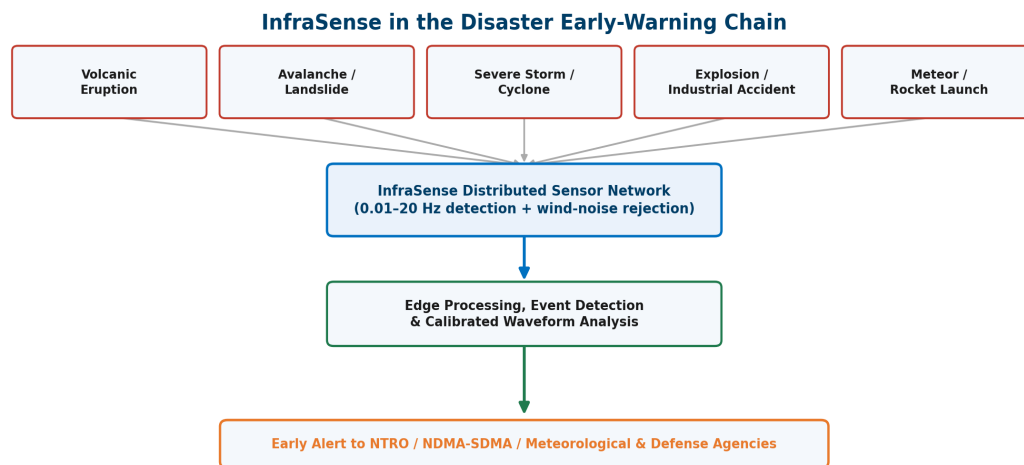


Figure 3: InfraSense position within the disaster early-warning chain.

4.1 Hazard Events Detectable via Infrasound

- Volcanic eruptions — eruption columns and lahars generate strong, sustained low-frequency signatures.
- Avalanches and landslides — large mass movements produce detectable infrasonic energy.
- Severe storms and cyclones — microbarom activity from ocean-storm interaction is a well-documented infrasound source.
- Industrial and accidental explosions — including mining blasts and pipeline/structural failures.
- Meteor entries and rocket/missile launches — relevant to both civil safety and national security monitoring.

4.2 Why Infrasound Enables Early Warning

- Infrasonic waves travel very long distances (hundreds to thousands of kilometres) with low atmospheric attenuation.
- Detection can occur before a hazard's direct physical effects (ash fall, debris, storm surge) reach a populated area.
- A dense sensor network enables triangulation of the source location and estimation of event severity from signal amplitude and duration.
- Historical precedent: the 1883 Krakatoa eruption's pressure wave was recorded on barometers in London, thousands of miles away, and the 1908 Tunguska explosion was similarly detected at a distance — demonstrating the fundamental reach of this sensing modality.

4.3 Relevance to Indian Disaster-Management Stakeholders

Stakeholder	Relevance
NDMA / SDMA	Early alerts for volcanic tremors, avalanches, and severe storms to trigger regional disaster-response protocols
NTRO & Defense Agencies	Detection of unauthorised explosions and rocket/missile launches; border-area surveillance
Meteorological Departments	Supplementary data source for severe-weather and cyclone-related atmospheric monitoring
Geophysical Research Institutions	Correlating infrasound with seismic data for volcanic and landslide hazard research
Industrial Safety Regulators	Pipeline and structural leak/explosion detection in industrial zones

4.4 Why a Low-Cost, Dense Network Matters for Disaster Management

- A single expensive imported station can only cover one location; India's disaster-prone geography (Himalayan belt, coastal cyclone zones, volcanic-adjacent regions) needs many stations, not one.
- At an estimated cost of ₹3,500–8,000 per unit, dozens of InfraSense nodes can be deployed for the cost of a single imported sensor — directly enabling the dense coverage that early warning requires.
- Networked nodes (Section 2) allow triangulation and cross-validation of events, reducing false alarms compared to a single isolated sensor.
- Solar/battery-ready power design supports deployment in remote, off-grid, disaster-prone terrain where continuous grid power is unavailable.

Scope note: InfraSense is presented here as a monitoring and early-alert research platform. Any operational disaster-warning deployment would require domain-specific threshold studies, integration with existing NDMA/SDMA protocols, and field validation against reference instruments before being relied upon for public safety decisions.

5. Summary

- InfraSense offers a low-cost, indigenous alternative to imported infrasound sensors, at a small fraction of the cost while covering wind-noise suppression, temperature compensation, and documented calibration — features often missing in DIY or basic-MEMS alternatives.
- The system architecture follows a clear, modular block diagram from atmospheric input to dashboard alerts, and the working flowchart breaks the build into five practically achievable phases.
- Its low cost and networkable design make it directly relevant to India's disaster-management ecosystem, enabling denser infrasound monitoring coverage than currently feasible with imported instruments alone.

InfraSense — Comparison, Architecture & Disaster-Management Relevance

Technical Comparison & Disaster-Management Relevance Report

High-Sensitivity Microbarometer Infrasonic Sensor — InfraSense

Problem Statement ID: SIH26144

Organization: National Technical Research Organisation (NTRO)

Team Name: [Your Team Name]

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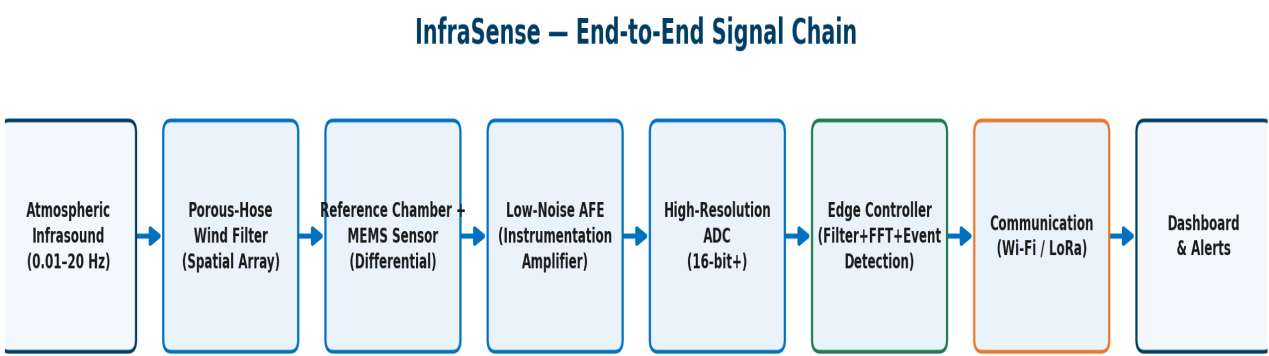


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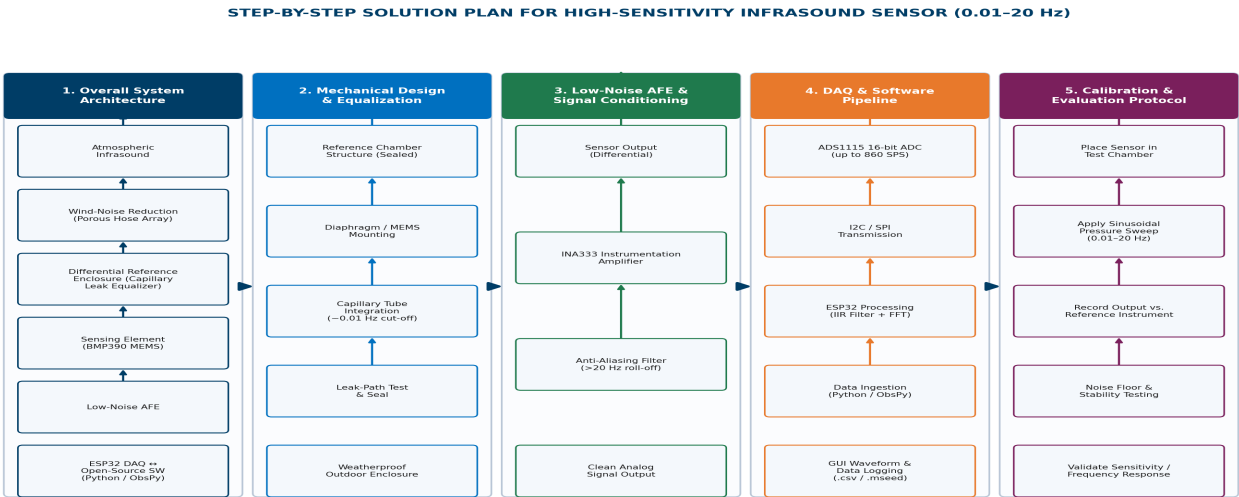


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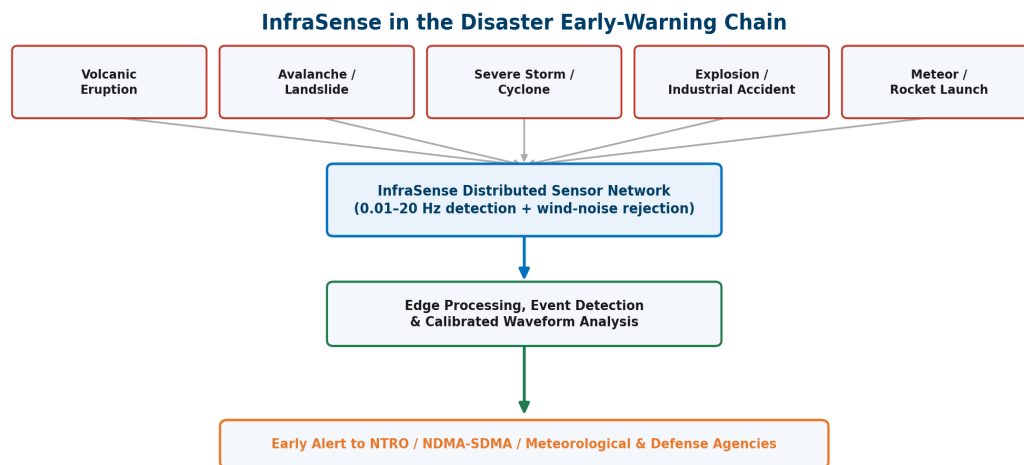


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InfraSense: Proposed Solution vs. Existing Systems

A Comparative Technical Analysis

Comparative Analysis Report

*High-Sensitivity Microbarometer Infrasonics Sensor — Proposed
Solution vs. Existing Systems*

Problem Statement ID: SIH26144

Organization: National Technical Research Organisation (NTRO)

Team Name: [Your Team Name]

1. Introduction

This report presents a focused comparison between the proposed InfraSense high-sensitivity microbarometer infrasound sensor and the infrasound-sensing solutions that currently exist — spanning professional/imported instruments, documented hobbyist designs, and basic commodity MEMS sensors. The objective is to clearly position what the proposed solution does differently, and why.

2. Existing Systems — Overview

2.1 Professional / Imported Electromagnetic Microbarometers

The CTBTO's International Monitoring System (IMS) uses professional-grade stations (e.g., MB2000, MB2005, MB3) based on an aneroid capsule read out via an electromagnetic (LVDT) transducer. Commercial vendors such as Hyperion, Chaparral, and CET offer similar instruments using piezoelectric, differential-capacitive, and LVDT sensing respectively. These are highly sensitive and field-proven, but expensive and import-dependent.

2.2 DIY / Hobbyist Strain-Gauge Microbarometers

Documented amateur designs use a metal diaphragm instrumented with bonded foil strain gauges in a Wheatstone-bridge configuration, with a calibrated capillary leak venting the reference cavity. This is more sensitive than a bare MEMS sensor and cheaper than professional instruments, but requires precise manual fabrication and has no built-in wind-noise suppression, temperature compensation, or documented calibration.

2.3 Basic MEMS-Only Sensors

Commodity MEMS barometric sensors (e.g., BMP388/BMP390-class) are inexpensive and widely available, but used alone — without a reference chamber, differential technique, or wind filtering — they are typically limited to strong, nearby events and are vulnerable to wind-induced noise and baseline drift outdoors.

3. Side-by-Side Comparison

Parameter	Imported Professional (CTBTO-grade)	DIY Strain-Gauge (Hobbyist)	Basic MEMS-Only Sensor	Proposed InfraSense
Sensing method	Electromagnetic (LVDT) / capacitive / piezoelectric	Strain gauge on metal diaphragm	Bare MEMS capacitive die	Differential MEMS + reference chamber
Approx. cost / unit	Several lakh ₹ (imported)	Low-moderate (DIY parts)	Very low (single sensor)	~₹6,500–8,000 (prototype); ₹3,500–5,000 at scale
Wind-noise handling	Dedicated hose/pipe arrays at station level	None built-in	None	Porous-hose spatial-averaging filter
Long-period equalization	Precision engineered	Capillary leak (manual tuning)	Not addressed	Calibrated capillary-leak reference path
Temperature compensation	Factory-calibrated	Partial (extra gauges)	Minimal	Dedicated sensing + firmware correction
Calibration methodology	Certified lab facilities	Manual, ad hoc	None	Documented multi-point + reference comparison
Data acquisition / software	Proprietary station systems	Arduino-class, manual	Basic logging only	Open-market digitizer + Python/ObsPy waveform GUI
Fabrication complexity	Very high (factory-grade)	High (precision manual bonding)	Very low	Moderate — achievable with standard workshop/lab tools
Import dependency	High	Low	Low	None — fully indigenous
Deployment density feasible	Very low (cost-limited)	Low (build complexity)	High but unreliable	High — low cost + practical build

4. Analysis by Dimension

4.1 Sensitivity & Accuracy

Imported professional sensors remain the most sensitive option, benefiting from decades of refinement and factory calibration. DIY strain-gauge designs achieve moderate sensitivity sufficient for regional-scale events. Basic MEMS-only sensors are the weakest, lacking any reference-chamber or differential technique. InfraSense targets the middle-to-upper range of this spectrum by combining a differential MEMS element with a purpose-built reference chamber — a combination that neither the DIY nor basic-MEMS approach implements together.

4.2 Cost & Accessibility

This is where InfraSense differentiates most sharply. Imported instruments cost several lakh rupees per unit, making dense deployment financially impractical. InfraSense is estimated at ₹6,500–8,000 per prototype unit, falling to ₹3,500–5,000 at production scale — a reduction exceeding 99% relative to imported systems, while still incorporating engineering safeguards (wind filtering, calibration, temperature compensation) that basic MEMS and most DIY designs omit.

4.3 Wind-Noise Rejection

Wind turbulence is one of the dominant sources of false or degraded readings in any outdoor infrasound sensor. Professional stations address this with dedicated hose/pipe arrays. Neither typical DIY builds nor basic MEMS sensors include this feature. InfraSense adopts the same principle as professional stations — a distributed porous-hose wind filter — scaled to a single-unit, low-cost implementation.

4.4 Calibration & Long-Term Stability

Professional systems are calibrated at certified facilities; DIY builds rely on manual, ad hoc calibration; basic MEMS sensors are typically used uncalibrated. InfraSense introduces a documented, repeatable multi-point calibration and periodic zero-check methodology — directly satisfying the NTRO requirement for a defined calibration methodology, and going beyond what either the DIY or basic-MEMS category typically provides.

4.5 Data Acquisition & Software

Professional stations rely on proprietary acquisition systems. DIY builds typically use manual, one-off Arduino-based logging. InfraSense uses an open-market digitizer combined with Python/ObsPy-based real-time waveform display and analysis — satisfying the NTRO requirement for open-market data-acquisition and analysis software while remaining fully reproducible and extensible.

5. Positioning Summary

InfraSense is positioned as the practical middle path between two extremes: the very expensive, import-dependent professional station, and the inexpensive but unreliable basic-MEMS or manually-built DIY sensor. It borrows the differential-reference and capillary-leak principle proven in both professional and DIY designs, adds systematic wind-noise suppression and documented calibration — largely absent from DIY and basic-MEMS approaches — and remains buildable at a small fraction of imported system cost.

5.1 Key Differentiators

- **Cost:** Over 99% cheaper than imported professional-grade sensors.
- **Completeness:** Only the proposed solution combines differential sensing, wind-noise filtering, temperature compensation, and documented calibration in a single low-cost design.
- **Indigenuity:** Fully sourceable and buildable within India, with no dependency on imported sensing technology.
- **Reproducibility:** Built on open-market components and open-source software (Python/ObsPy), unlike proprietary professional systems.
- **Practicality:** Unlike DIY strain-gauge designs, does not require precision manual strain-gauge bonding — uses a ready-made differential MEMS sensing element instead.

5.2 Honest Limitations

- Raw sensitivity of the chosen MEMS sensing element is not expected to match professional capacitive/optical instruments — this is a deliberate cost/practicality trade-off for the prototype stage.
- Long-term field reliability (multi-year drift, extreme-weather survivability) of professional stations is backed by decades of deployment history that a new design cannot yet claim.
- All InfraSense performance figures in this report are design targets to be confirmed through the calibration and validation protocol, not yet field-proven results.

6. Conclusion

Compared against imported professional systems, DIY strain-gauge builds, and basic MEMS-only sensors, the proposed InfraSense solution offers the most balanced combination of low cost, engineering completeness, and practical buildability. It does not claim to exceed the raw sensitivity of certified professional instruments, but it is the only option in this comparison that combines wind-noise suppression, temperature compensation, and documented calibration methodology at a cost low enough to enable the dense, indigenous deployment that India's infrasound-monitoring needs currently lack.

HIGH-SENSITIVITY MICROBAROMETER INFRASOUND SENSOR

Proposed Solution & Prototype Development Document

A compact, low-noise and intelligent atmospheric sensing platform for detecting very-low-frequency pressure fluctuations from approximately 0.01 Hz to 20 Hz.

Project Theme	Precision Infrasound Monitoring
Core Innovation	MEMS + optical readout + low-noise electronics + digital intelligence
Target Frequency	~0.01 Hz to 20 Hz
Deployment	Outdoor and distributed monitoring environments

Professional project-document format • Simple English • Prototype-oriented • Human-readable

1. Executive Summary

The proposed system is a high-sensitivity microbarometer designed to detect low-frequency atmospheric pressure variations that are difficult to capture with conventional pressure sensors. The design combines a sensitive mechanical transducer, differential pressure measurement, optical displacement readout, low-noise analog electronics, temperature compensation, pressure equalization and wind-noise reduction.

- Designed for infrasonic pressure fluctuations in the approximate 0.01–20 Hz range.
- Uses a precision diaphragm-based sensing mechanism to convert pressure changes into mechanical displacement.
- Uses an optical readout concept with VCSEL and photodiode to improve displacement sensitivity.
- Uses a custom low-noise analog front end and digital processing chain for clean signal acquisition.
- Uses temperature compensation to reduce sensor drift caused by environmental temperature changes.
- Uses a multi-port wind-noise reduction structure to reduce turbulence-related pressure fluctuations.
- Uses an STM32-class microcontroller for acquisition, filtering, event detection and communication.
- Supports LoRa/GSM-based remote monitoring for distributed deployments.

2. Problem Definition

Infrasound consists of very-low-frequency atmospheric pressure waves. The challenge is not only detecting a pressure variation, but extracting a very small signal from temperature drift, wind turbulence, mechanical vibration, electronic noise and long-term pressure changes.

Key Design Challenges

- Very small pressure variations require high sensitivity.
- The target band extends to very long periods, so baseline drift must be controlled.
- Outdoor wind turbulence can be much larger than the desired infrasonic signal.
- Temperature changes can alter mechanical and electronic characteristics.
- Low-frequency measurements require stable pressure equalization without suppressing the signal.
- The prototype must combine sensitive mechanics with practical electronics and field protection.

3. Proposed Solution - System Architecture

The proposed architecture follows a complete sensing chain from atmospheric pressure disturbance to a validated digital event record.

Stage	Purpose
Atmospheric Pressure Wave	External infrasonic disturbance
Wind-Noise Reduction Array	Four-port spatial averaging / cancellation concept
Microbarometer Diaphragm	Pressure-to-displacement conversion
Differential / Optical Readout	Sensitive measurement of diaphragm motion
Low-Noise Analog Front End	Amplification and signal conditioning
Temperature Compensation	Drift correction using temperature sensing
ADC + STM32	High-resolution digitization and embedded processing
Digital Filtering + TinyML	Noise rejection and event classification
Communication	LoRa / GSM remote transmission
Monitoring Dashboard	Time-series signal and event visualization

4. Core Technical Design

4.1 High-Sensitivity Pressure Sensing Unit

- Use a precision pressure-sensing chamber with a very small effective volume.
- Use a mechanically optimized diaphragm to obtain measurable displacement from small pressure changes.
- Control diaphragm stiffness, thickness, geometry and stress during design simulation.
- Use stable materials and mechanical mounting to reduce long-term drift.

4.2 Mechanical Diaphragm Transducer

- A thin diaphragm acts as the primary mechanical transducer.
- Atmospheric pressure variation produces controlled diaphragm deflection.
- MEMS fabrication can be explored using an SOI wafer and DRIE-based micromachining.
- COMSOL simulation can be used to study deformation, resonance, stress and sensitivity before fabrication.

4.3 Differential Pressure Measurement

- A differential configuration helps reject common environmental pressure changes and mechanical disturbances.
- The reference side can be coupled to a controlled reference volume through a long-period equalization path.
- The pneumatic path must be designed carefully so the required low-frequency response is preserved.

4.4 Optical Readout

- Use a VCSEL light source and photodiode receiver as a contactless displacement measurement concept.
- A Fabry-Pérot cavity can convert diaphragm displacement into an optical intensity or phase-related signal.
- Optical sensing reduces direct electrical loading of the mechanical element.
- The optical path should be mechanically isolated from vibration and protected from contamination.

5. Low-Noise Electronics

Analog Front End

- Use a low-noise preamplifier close to the sensing element.
- Use controlled gain stages to avoid saturation during strong atmospheric disturbances.
- Use anti-alias filtering before ADC conversion.
- Separate analog and digital power/ground paths where practical.
- Use shielding and careful PCB layout to reduce electromagnetic interference.

Digital Acquisition

- Use a high-resolution ADC suitable for low-frequency precision measurement.
- Use an STM32-class microcontroller for real-time acquisition and processing.
- Store raw and processed measurements for later validation.
- Add timestamping so events can be compared across multiple sensor nodes.

6. Environmental Compensation & Noise Control

Temperature Compensation

- Measure temperature near the mechanical and electronic sensing region.
- Characterize sensor output at different temperatures during calibration.

- Apply a temperature-dependent correction model in firmware.
- Continuously monitor temperature so compensation remains active during field operation.

Long-Period Pressure Equalization

- Provide a controlled pneumatic path for slow static pressure equalization.
- Use a high-flow-resistance path to avoid rapidly short-circuiting the target infrasonic signal.
- Optimize tubing length, restrictor geometry and chamber volume during prototype testing.
- Validate the frequency response experimentally rather than relying only on theoretical calculations.

Wind-Noise Reduction

- Use a four-port spatial inlet structure around the pressure sensor.
- Route each port through matched pneumatic paths before combining the signals.
- Use spatial averaging to reduce localized turbulent fluctuations.
- Place the sensing enclosure away from sharp edges and strong direct airflow.
- Use field calibration to compare wind-noise performance with and without the inlet array.

7. Intelligent Signal Processing

- Apply high-pass, low-pass and band-limited digital filters according to the target frequency band.
- Use adaptive noise estimation to identify changing background conditions.
- Use beamforming / spatial processing when multiple sensor channels are available.
- Use TinyML for lightweight event classification at the edge.
- Classify events using signal features such as amplitude, duration, spectral content and multi-channel correlation.
- Transmit only important event summaries when communication bandwidth is limited, while retaining raw data locally.

8. Proposed Prototype – Physical Construction

Mechanical Stack

- Weather-resistant outer enclosure.
- Four-port wind-noise reduction inlet.
- Internal acoustic / pneumatic chamber.
- Precision diaphragm assembly.
- Optical emitter-receiver module.
- Reference volume and pressure equalization path.
- Vibration-isolated internal mounting.

Electronics Stack

- Low-noise analog front-end PCB.
- Temperature sensor.
- High-resolution ADC.
- STM32 microcontroller board or custom control PCB.
- Local storage for data logging.
- LoRa / GSM communication module.
- Regulated power supply with battery/solar-ready option.

9. Innovative Features

Innovation	Implementation Idea
Hybrid Optical-MEMS Readout	Combines precision mechanical sensing with contactless optical displacement measurement.
Four-Port Wind-Noise Array	Uses spatially distributed pneumatic inputs to reduce turbulence-related noise.
Adaptive Digital Filtering	Changes filtering behaviour according to measured background noise.
Edge TinyML Classification	Performs lightweight event recognition locally before transmission.
Temperature-Aware Calibration	Uses continuous temperature data to correct sensor drift.
Distributed Sensor Networking	Multiple nodes can share timestamped events for spatial confirmation.
Energy-Aware Communication	Transmits event summaries while keeping detailed data locally available.
Simulation-to-Prototype Workflow	Uses multiphysics simulation before fabrication to reduce design iterations.

10. Design & Development Workflow

01. Define target sensitivity, bandwidth and environmental requirements.
02. Model the diaphragm, chamber and pressure path in COMSOL or equivalent simulation software.
03. Select materials and mechanical dimensions based on sensitivity and stability.
04. Develop the optical readout and validate displacement response.
05. Design the low-noise analog front end and ADC interface.
06. Build the first laboratory prototype.
07. Perform static, dynamic, temperature and vibration characterization.
08. Develop wind-noise reduction hardware and compare measured noise floors.
09. Implement digital filtering, calibration and TinyML event classification.
10. Integrate communication and remote monitoring.
11. Perform controlled field trials and compare results with reference instruments where available.
12. Freeze the validated design and prepare the final enclosure and documentation.

11. Testing & Validation Plan

Test	Validation Method
Sensitivity Test	Apply known small pressure variations and measure output response.
Frequency Response	Evaluate amplitude and phase response across the target band.
Noise Floor Test	Measure baseline noise in a controlled low-disturbance environment.
Temperature Test	Characterize output drift over the intended operating temperature range.
Wind Test	Compare signal noise under different airflow conditions.
Long-Duration Test	Monitor baseline stability, drift and data continuity over extended periods.
Event Detection Test	Use controlled, safe reference signals to verify detection and classification.
Communication Test	Verify reliable timestamped transmission and local data backup.

12. Challenge → Mitigation

Challenge	Proposed Mitigation
Very small pressure signal	High-sensitivity diaphragm + optical readout + low-noise amplification

Challenge	Proposed Mitigation
Wind turbulence	Four-port inlet + spatial averaging + digital noise rejection
Temperature drift	Temperature sensing + calibration model + real-time compensation
Long-period baseline drift	Controlled pressure equalization + baseline tracking
Electrical interference	Low-noise PCB layout + shielding + filtered power
Mechanical vibration	Rigid enclosure + vibration isolation + signal validation
Large data volume	Edge processing + event-based transmission + local storage
Field contamination	Protected enclosure + serviceable pneumatic inlet design

13. Practical Prototype Roadmap

- Phase 1 – Concept: requirements, architecture and preliminary calculations.
- Phase 2 – Simulation: diaphragm, cavity and pneumatic-path modelling.
- Phase 3 – Bench Prototype: pressure chamber, diaphragm, optical readout and AFE.
- Phase 4 – Digital Prototype: ADC, STM32 firmware, filtering and data logging.
- Phase 5 – Noise Control: wind array, enclosure and vibration isolation.
- Phase 6 – Intelligence: event features, TinyML classification and multi-node correlation.
- Phase 7 – Field Prototype: weather-resistant packaging, remote communication and long-duration testing.
- Phase 8 – Validation: compare performance against defined sensitivity, bandwidth, stability and noise targets.

14. Expected Outcome

- A compact prototype capable of measuring low-frequency atmospheric pressure fluctuations.
- A documented sensing chain from mechanical pressure conversion to digital event output.
- Improved resistance to wind, temperature and electronic noise compared with a basic pressure-sensor approach.
- A scalable architecture that can be expanded into multiple synchronized sensor nodes.
- A practical foundation for atmospheric research, environmental monitoring and other legitimate infrasound sensing applications.

15. Final Proposed Solution - One-Line Concept

“A smart, low-noise microbarometer that combines MEMS-style diaphragm sensing, optical displacement measurement, wind-noise reduction, temperature compensation and edge intelligence to convert extremely small atmospheric pressure variations into reliable digital events.”

16. Recommended Prototype Block Diagram

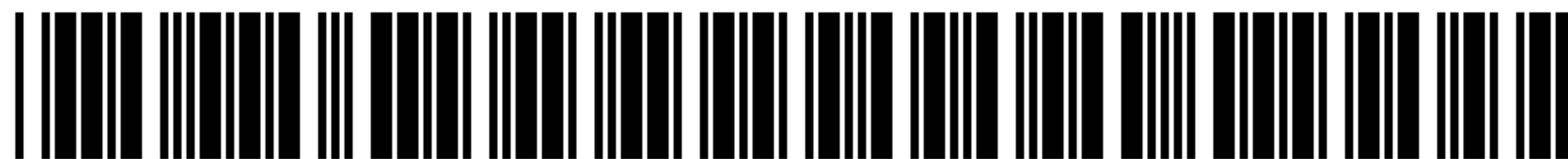
ATMOSPHERIC INFRASOUND
↓
4-PORT WIND-NOISE REDUCTION ARRAY
↓
PRECISION DIAPHRAGM / MICROBAROMETER
↓
VCSEL + PHOTODIODE OPTICAL READOUT

↓
LOW-NOISE ANALOG FRONT END
↓
HIGH-RESOLUTION ADC
↓
STM32 MICROCONTROLLER
↓
DIGITAL FILTERING + TEMPERATURE COMPENSATION
↓
TINYML EVENT CLASSIFICATION
↓
LOCAL STORAGE + LoRa / GSM
↓
REMOTE MONITORING DASHBOARD

17. Documentation Notes

- All target performance values should be finalized after simulation and calibration testing.
- Optical readout, MEMS fabrication and pneumatic equalization are proposed engineering approaches and should be validated experimentally.
- Prototype component selection should be based on measured noise, sensitivity, stability, availability and environmental requirements.
- The final enclosure should be designed only after the pneumatic and mechanical response has been characterized.

Prepared as a professional proposed-solution document for prototype planning and technical presentation.



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(12) **United States Patent**
Bayram et al.

(10) **Patent No.:** **US 11,889,264 B2**
(45) **Date of Patent:** **Jan. 30, 2024**

(54) **FIBER OPTIC MEMS MICROPHONE**

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Ozmen, Ankara (TR); Ekin Muharrem
Karaca, Ankara (TR); Doğa Buse
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patent is extended or adjusted under 35
U.S.C. 154(b) by 199 days.

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(86) PCT No.: **PCT/TR2019/050262**

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PCT Pub. Date: **Oct. 22, 2020**

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H04R 23/00 (2006.01)

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CPC **H04R 23/008** (2013.01); **H04R 2201/003**
(2013.01)

(58) **Field of Classification Search**

CPC H04R 23/008; H04R 2201/003
See application file for complete search history.

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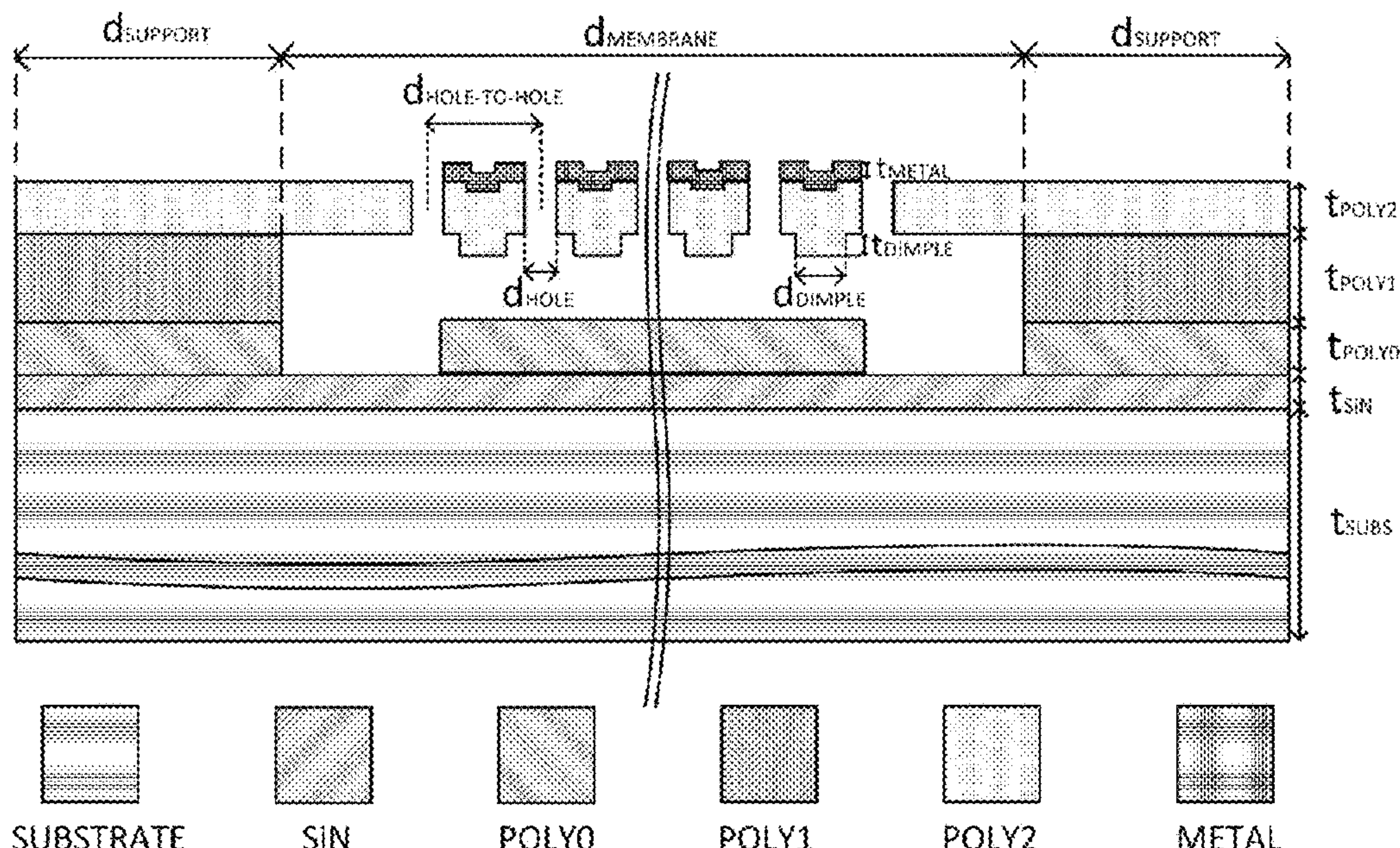
Primary Examiner — Suhan Ni

(74) *Attorney, Agent, or Firm* — Bayramoglu Law Offices
LLC

(57) **ABSTRACT**

A fiber optic MEMS microphone featuring an electrically
deflectable MEMS membrane via a conversion of an optical
energy propagating in an optical fiber cable to an electrical
energy with a photodiode chip. The fiber optic MEMS
microphone includes a MEMS device, the photodiode chip,
a voltage, a power adjustable laser beam and a light.

13 Claims, 13 Drawing Sheets



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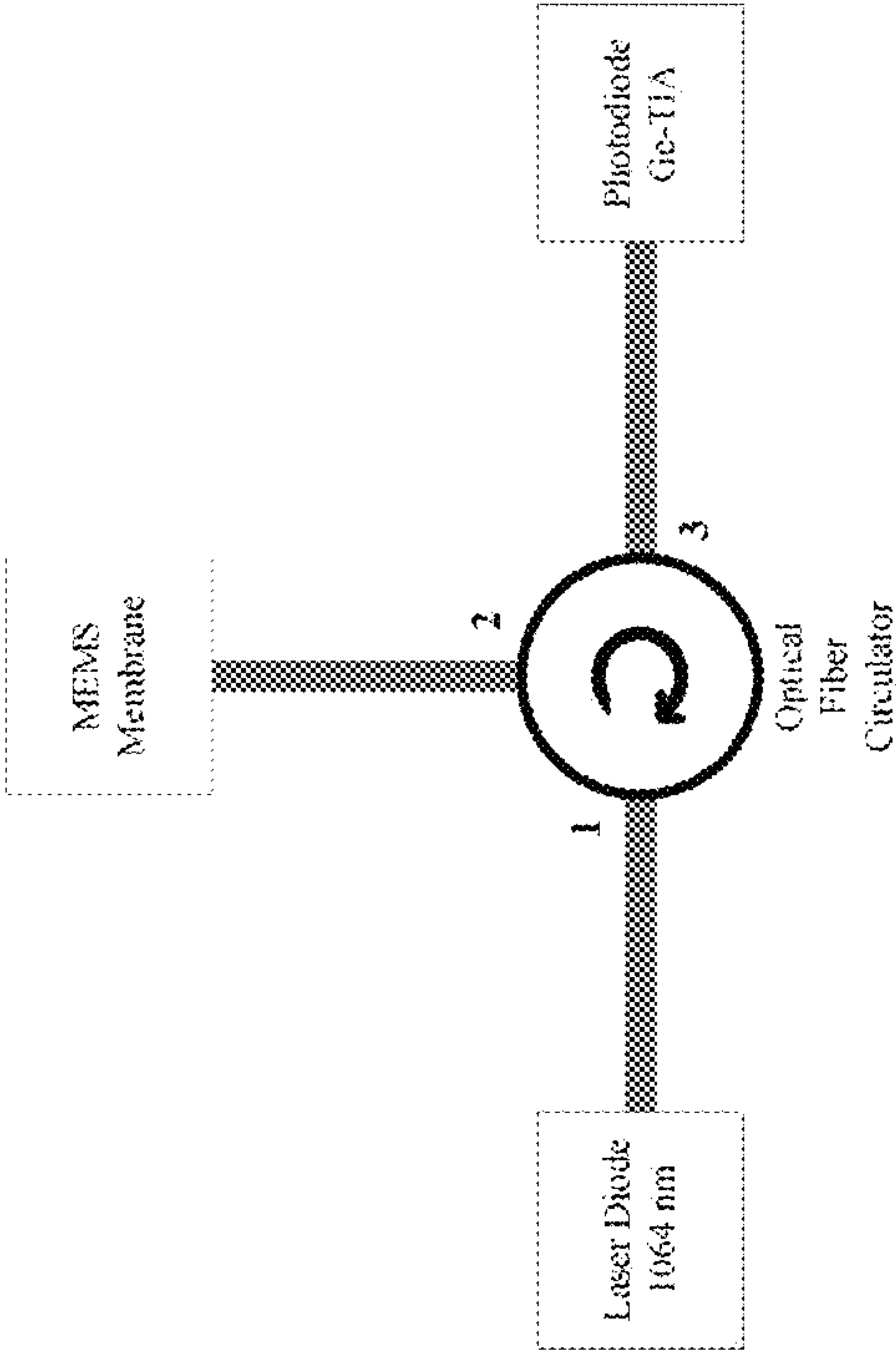


FIG. 1A

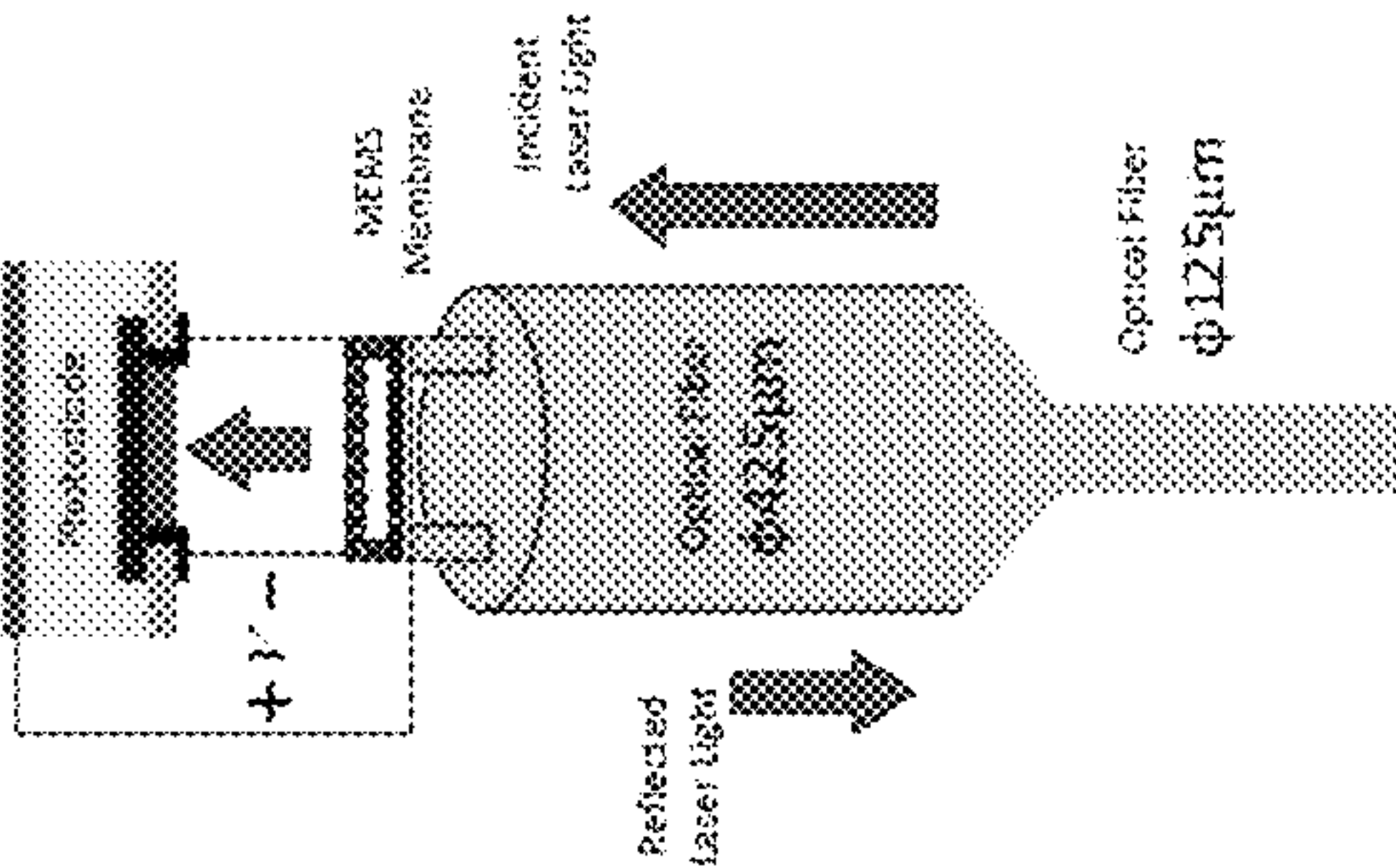


FIG. 1B

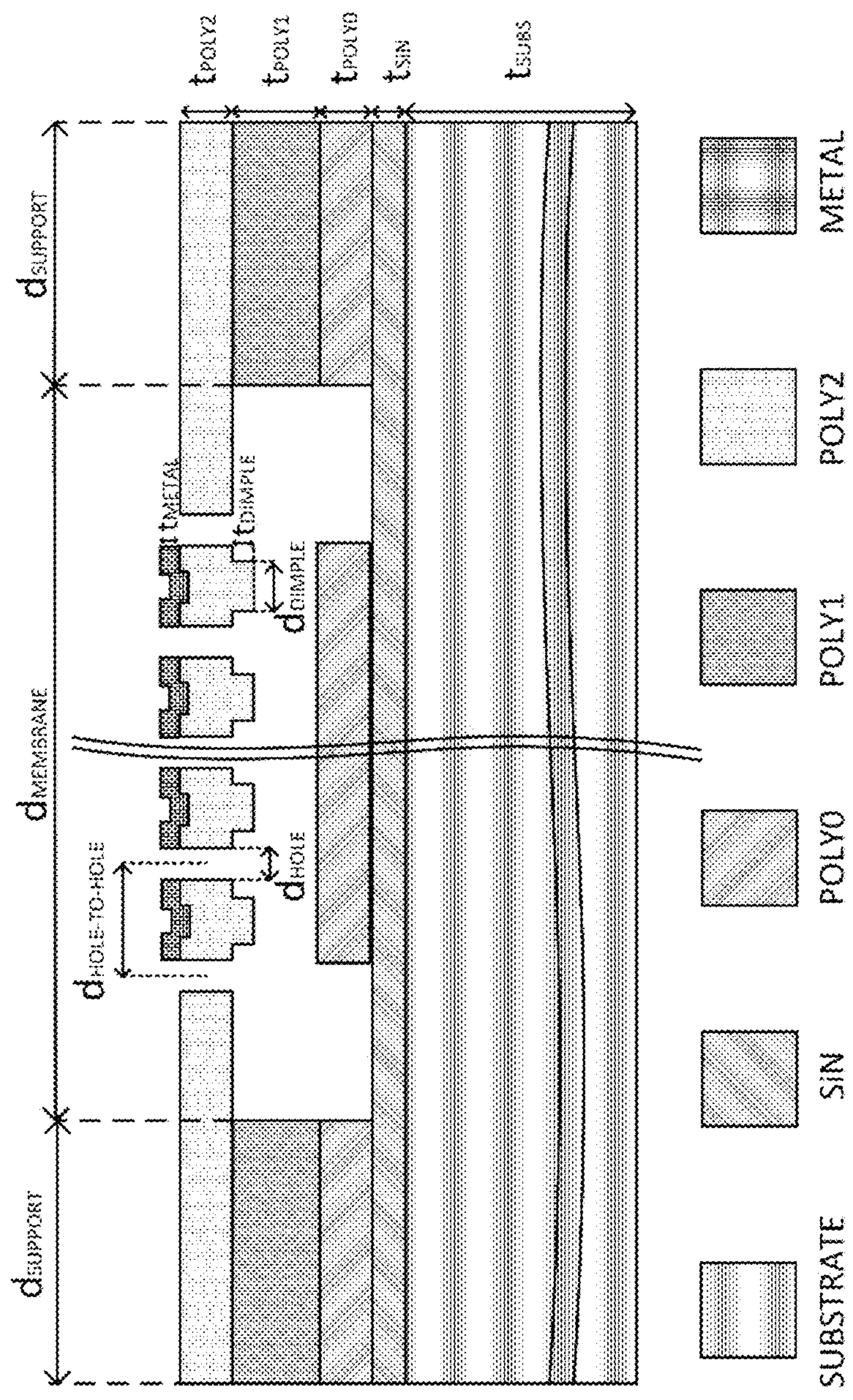


FIG. 2

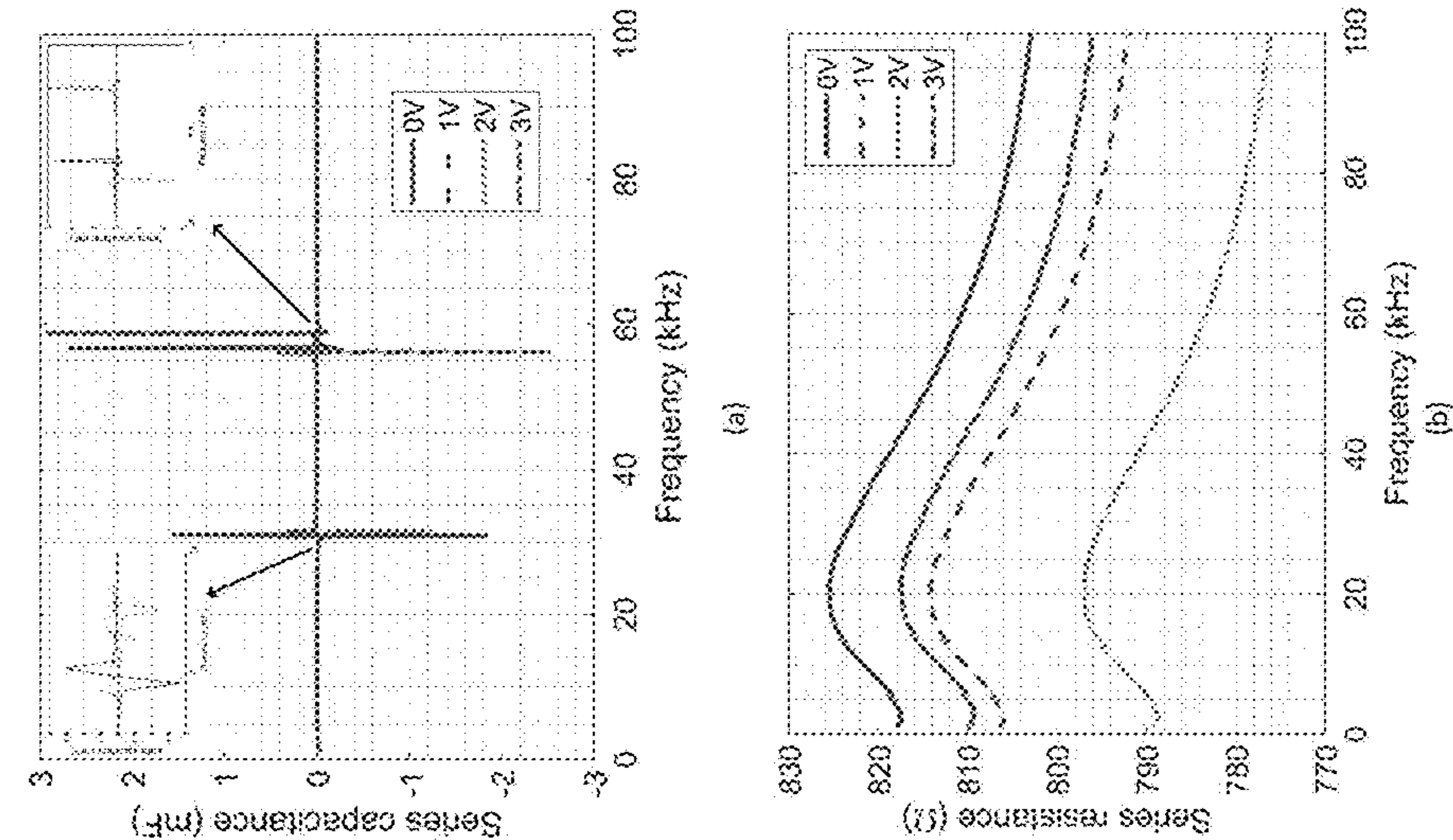


Figure 3B

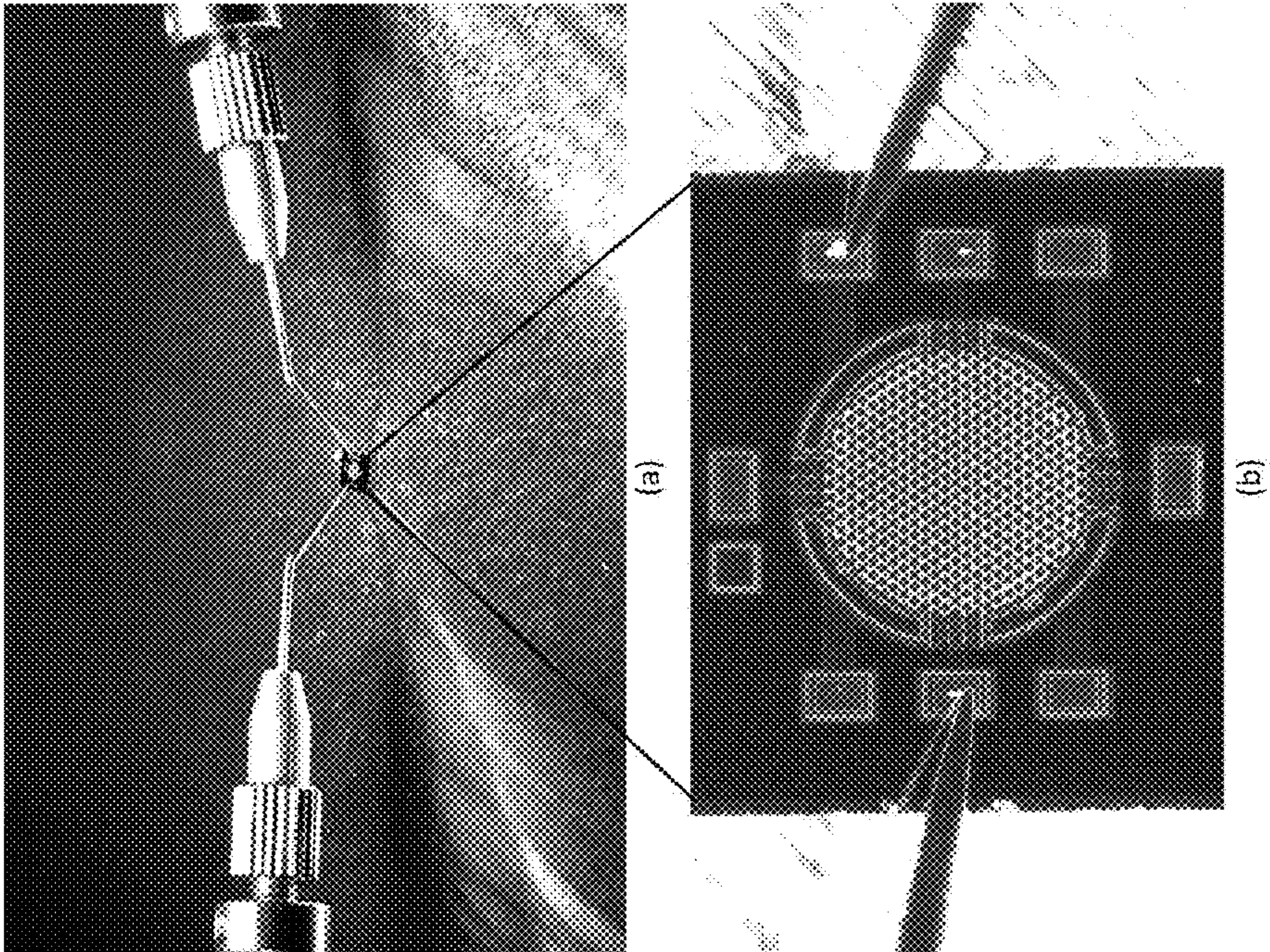


Figure 3A

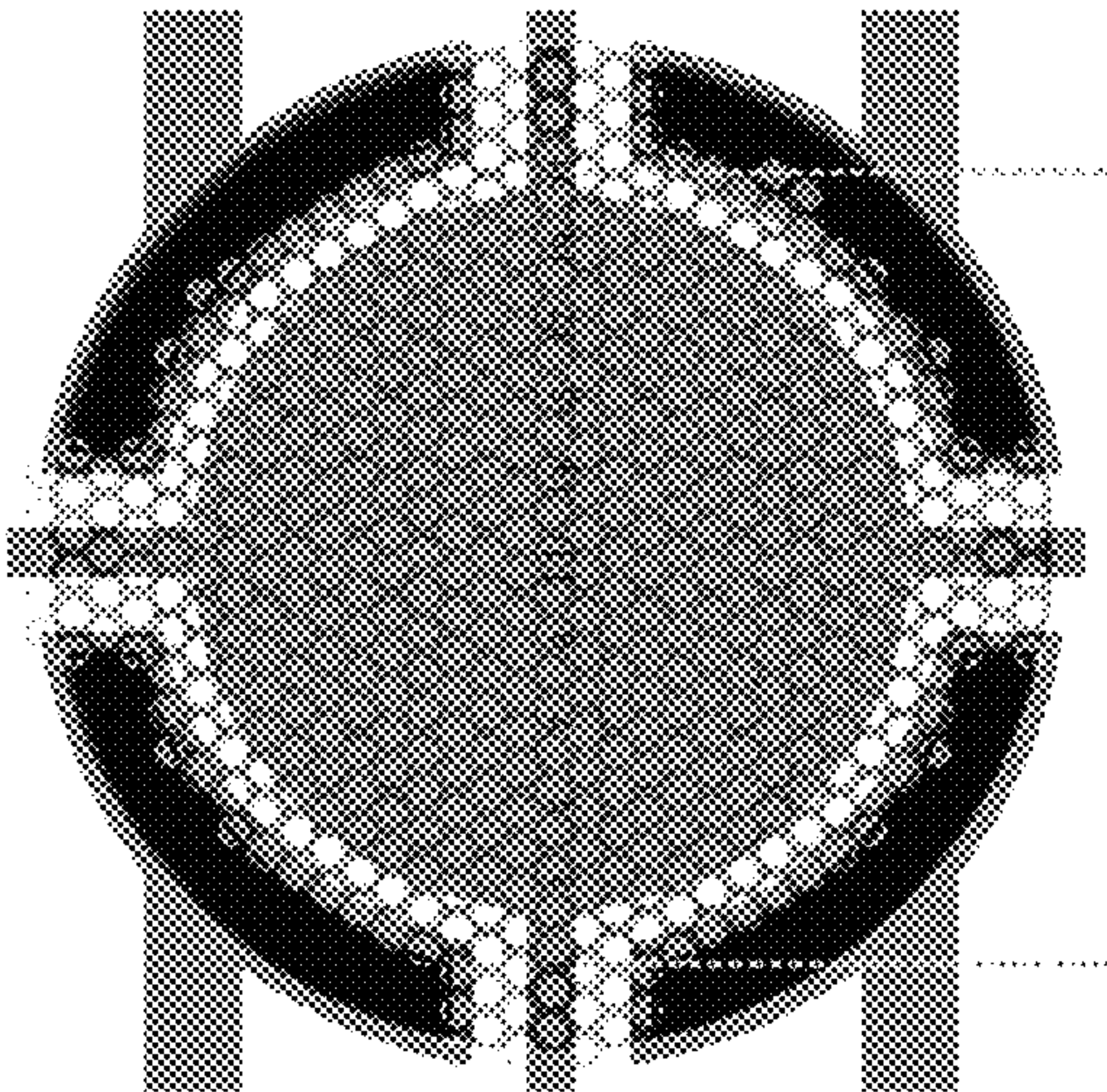
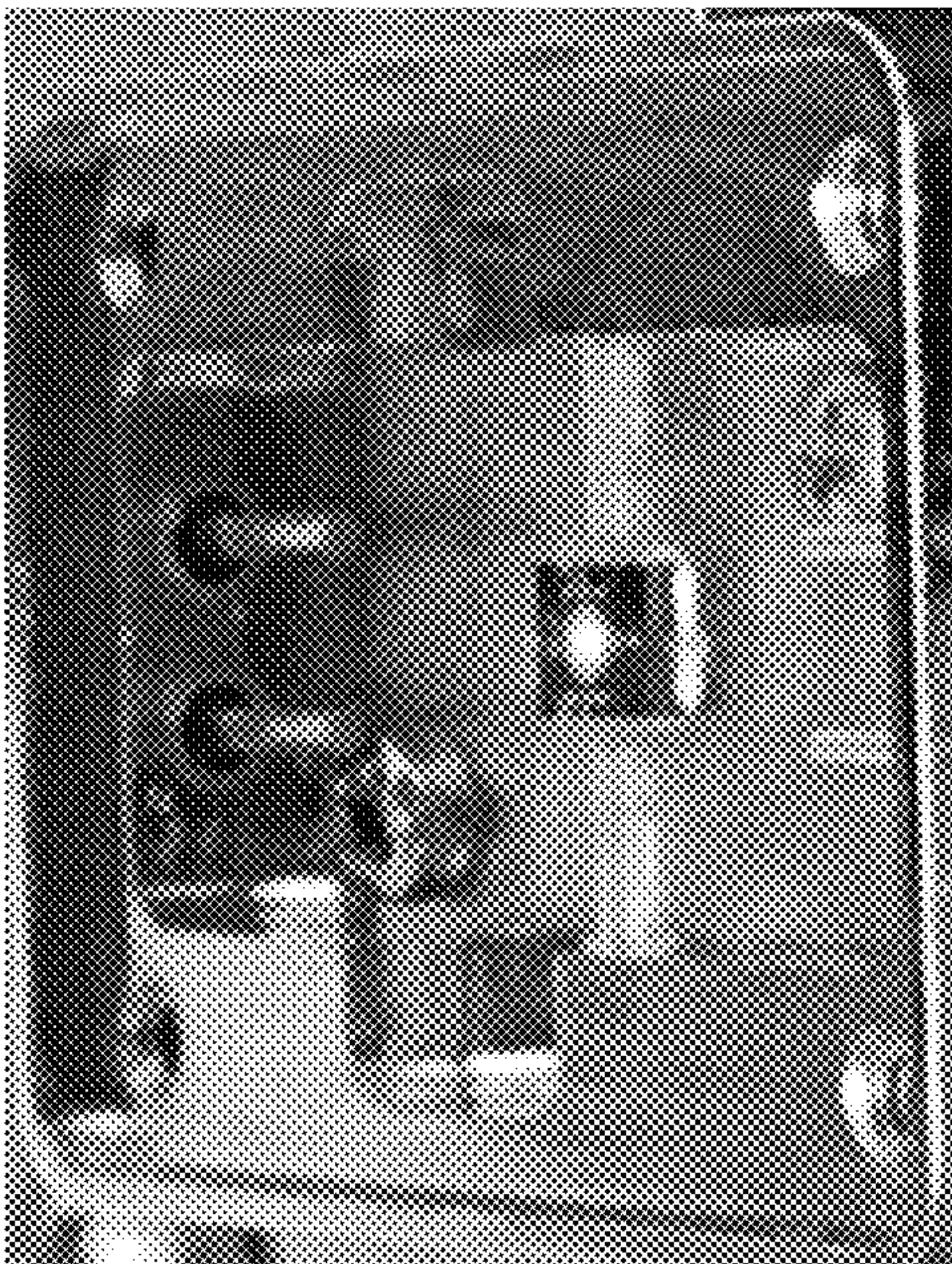


FIG. 4A



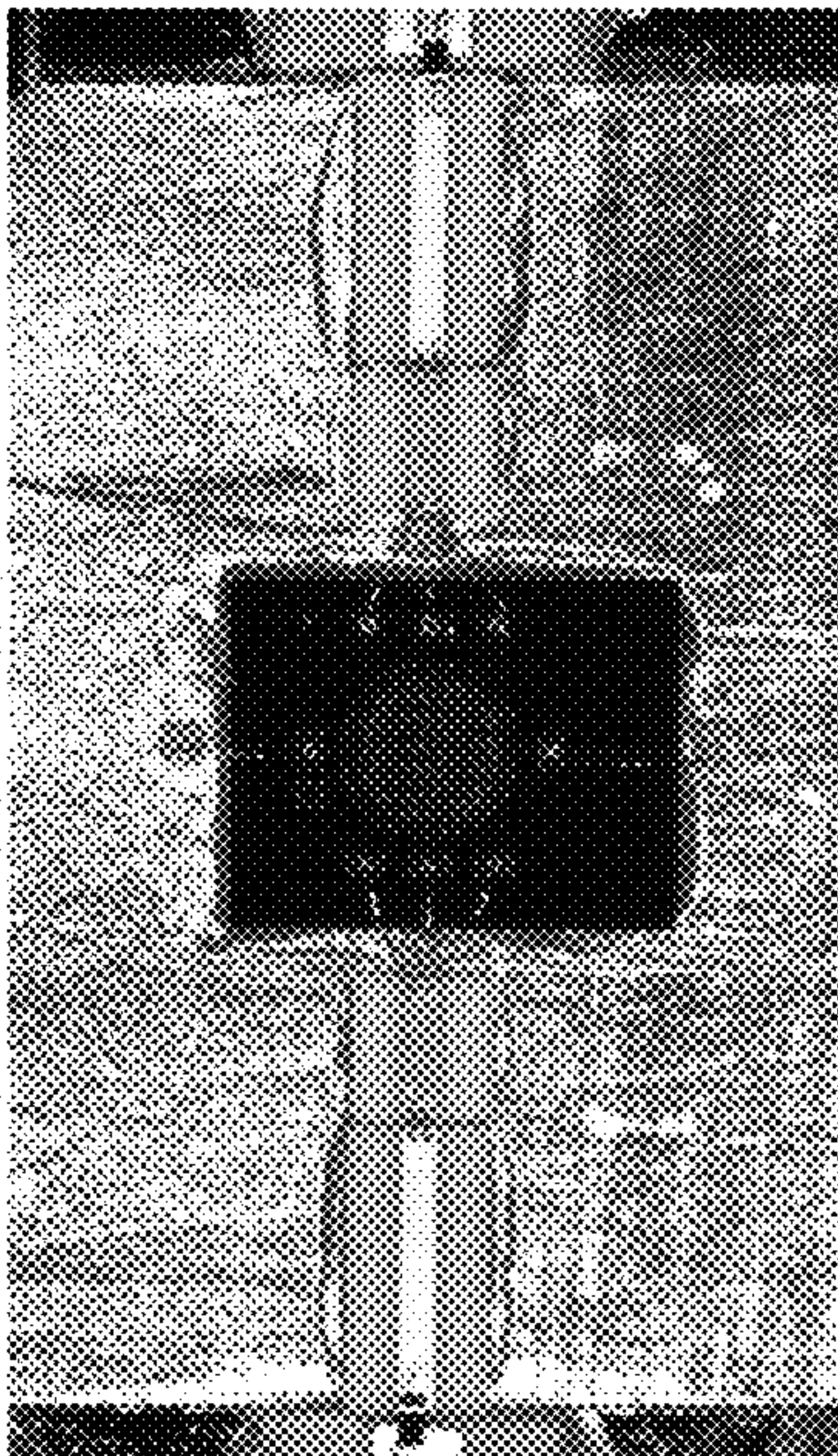
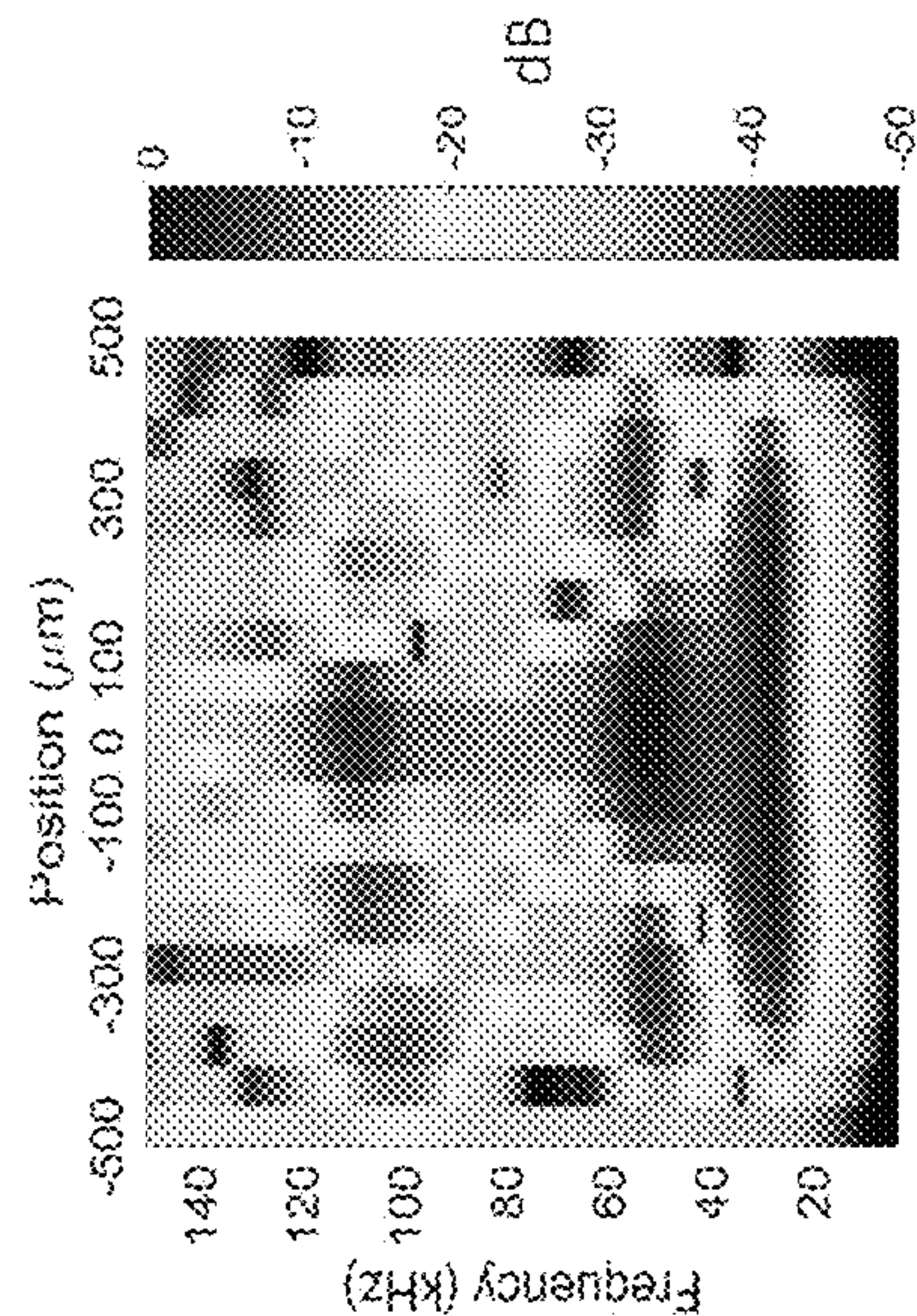


FIG. 4B

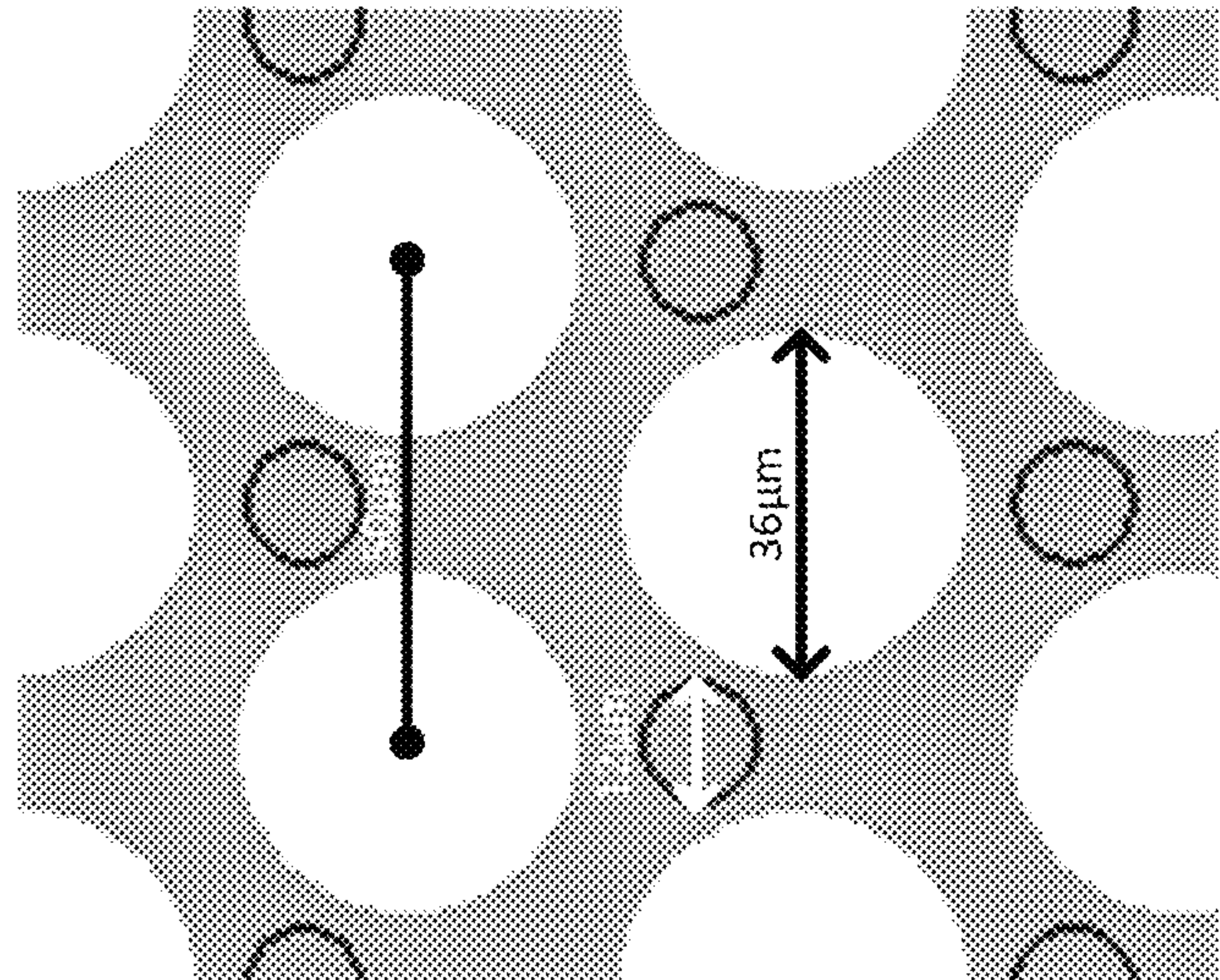


FIG. 5B

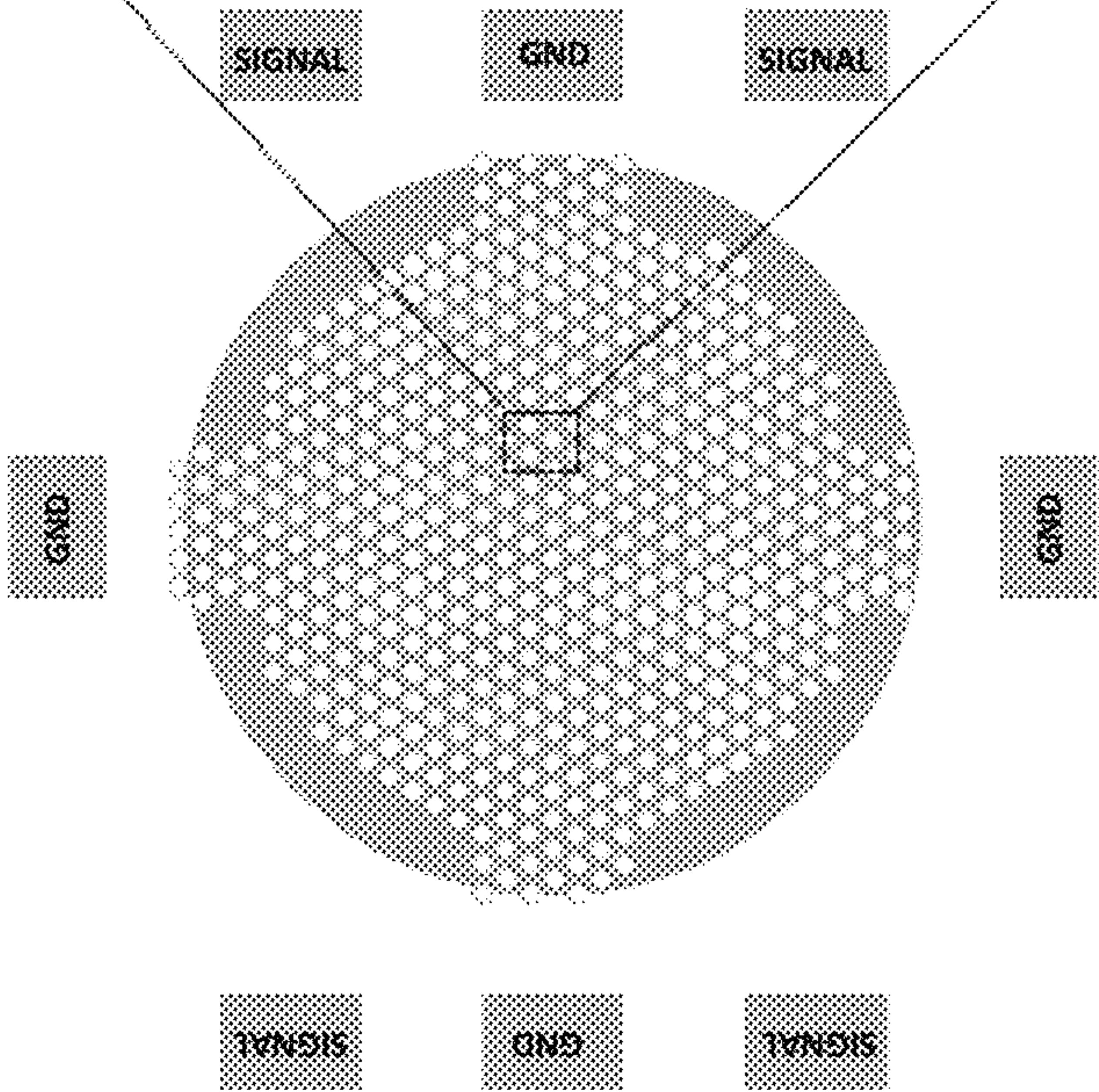


FIG. 5A

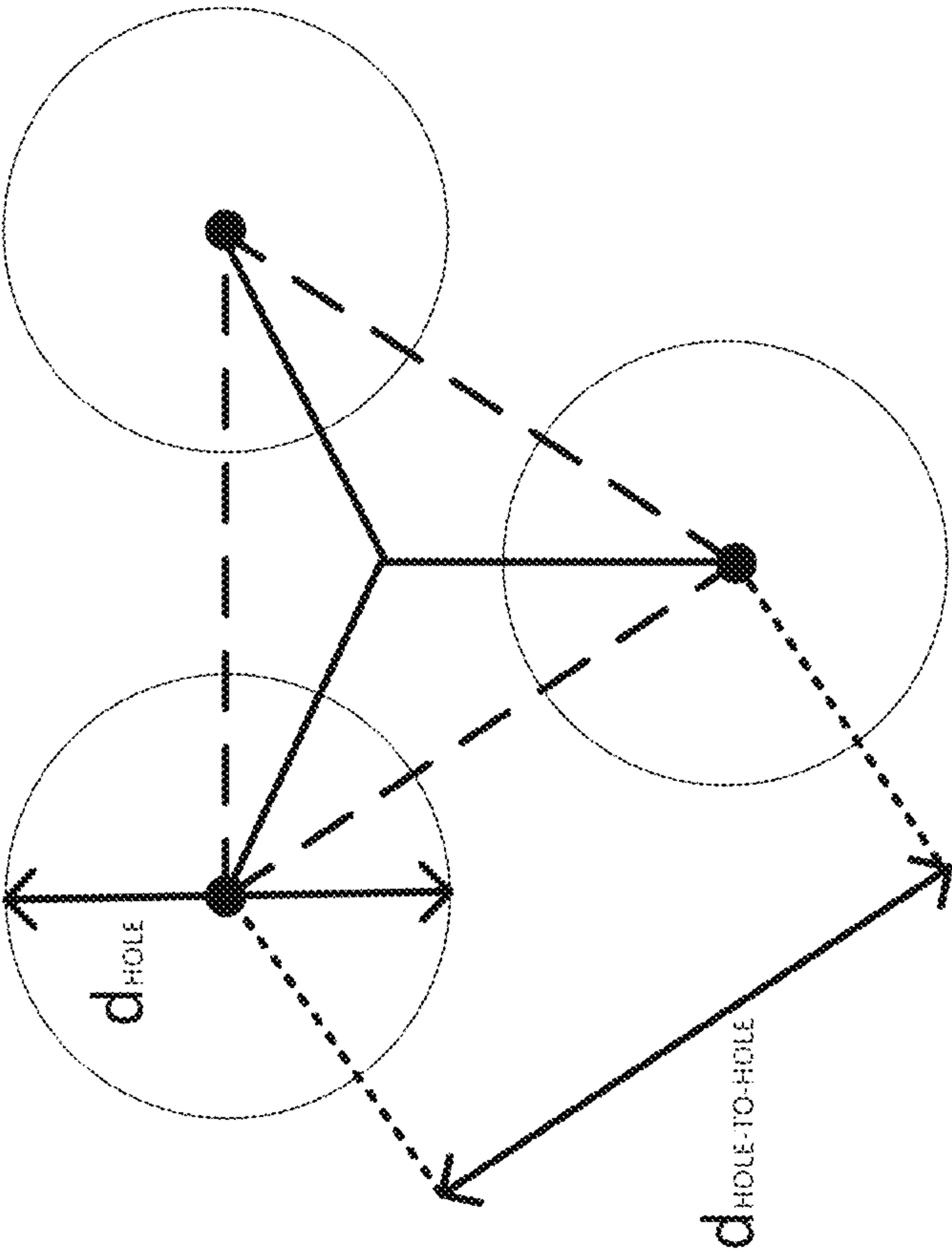


FIG. 6

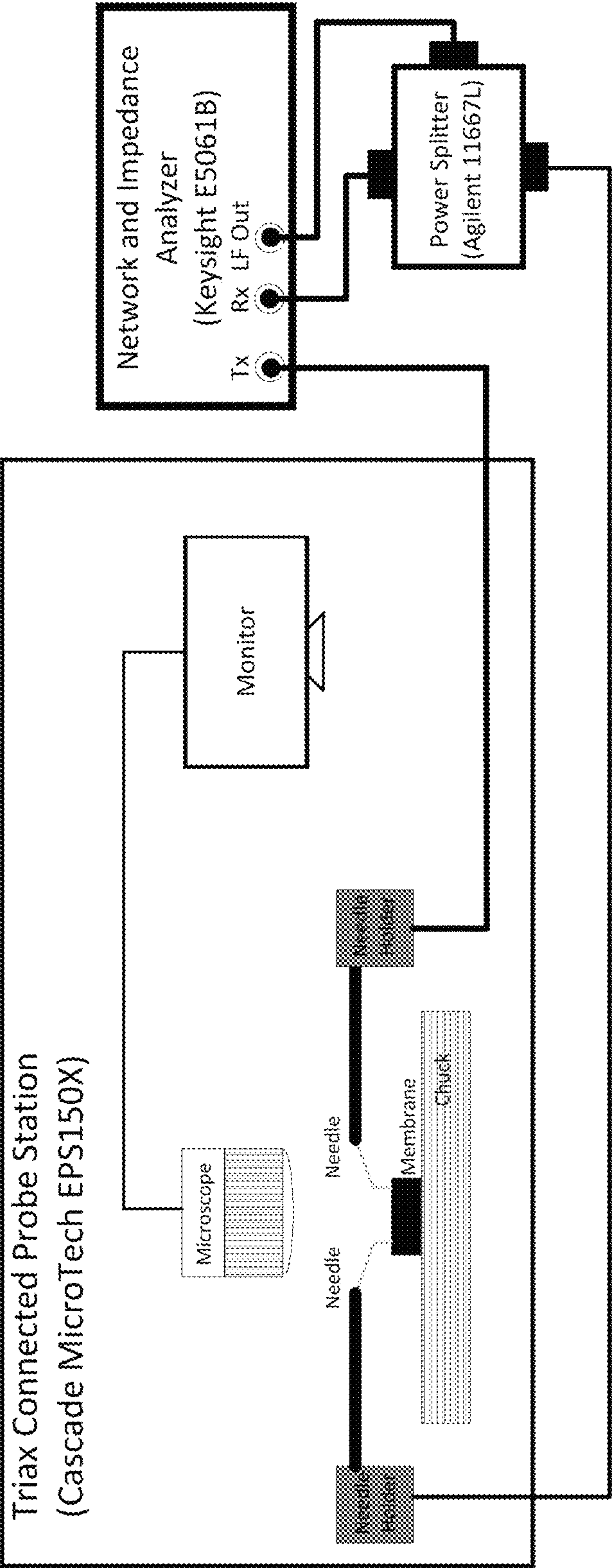


FIG. 7

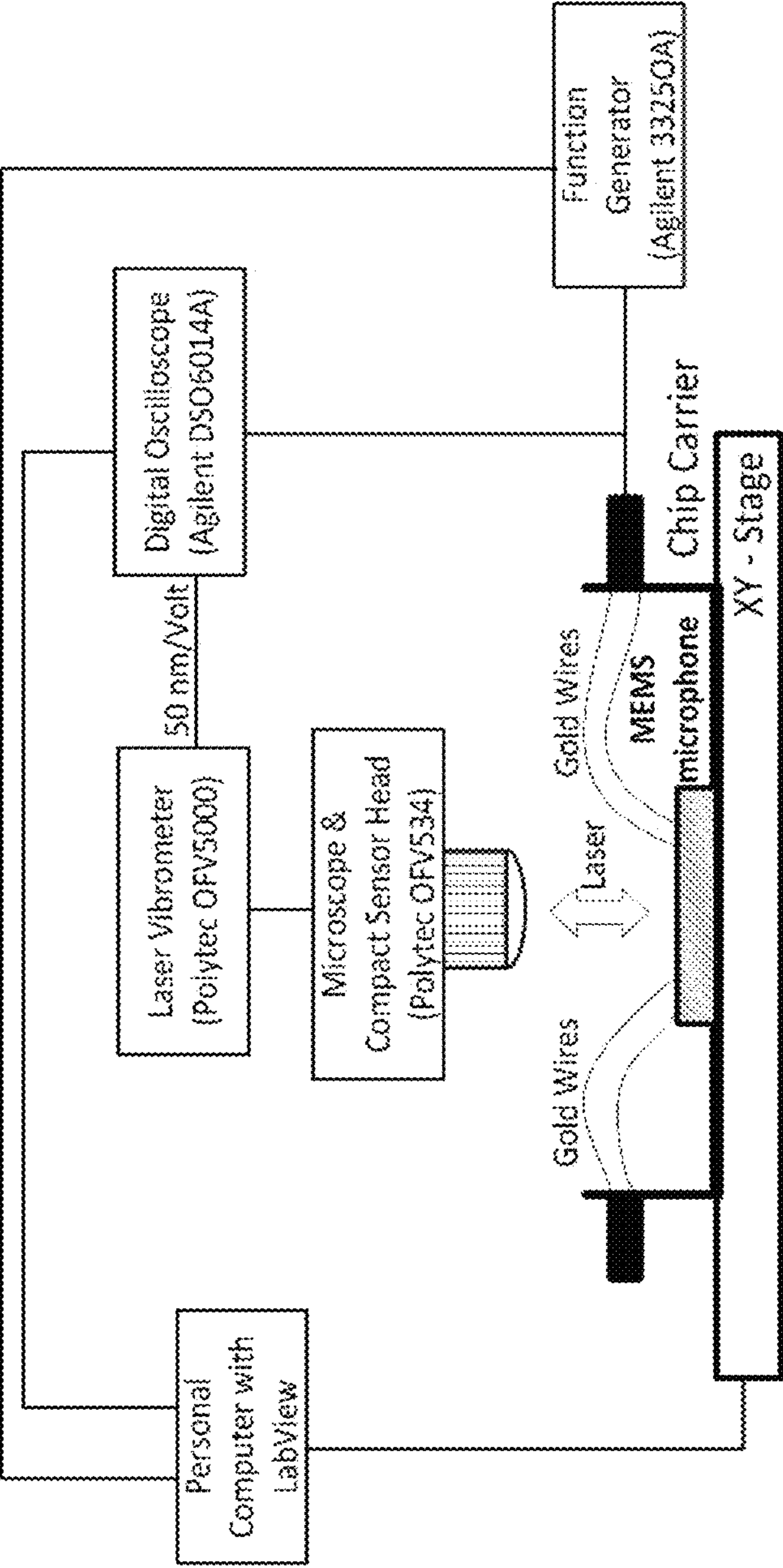


FIG. 8

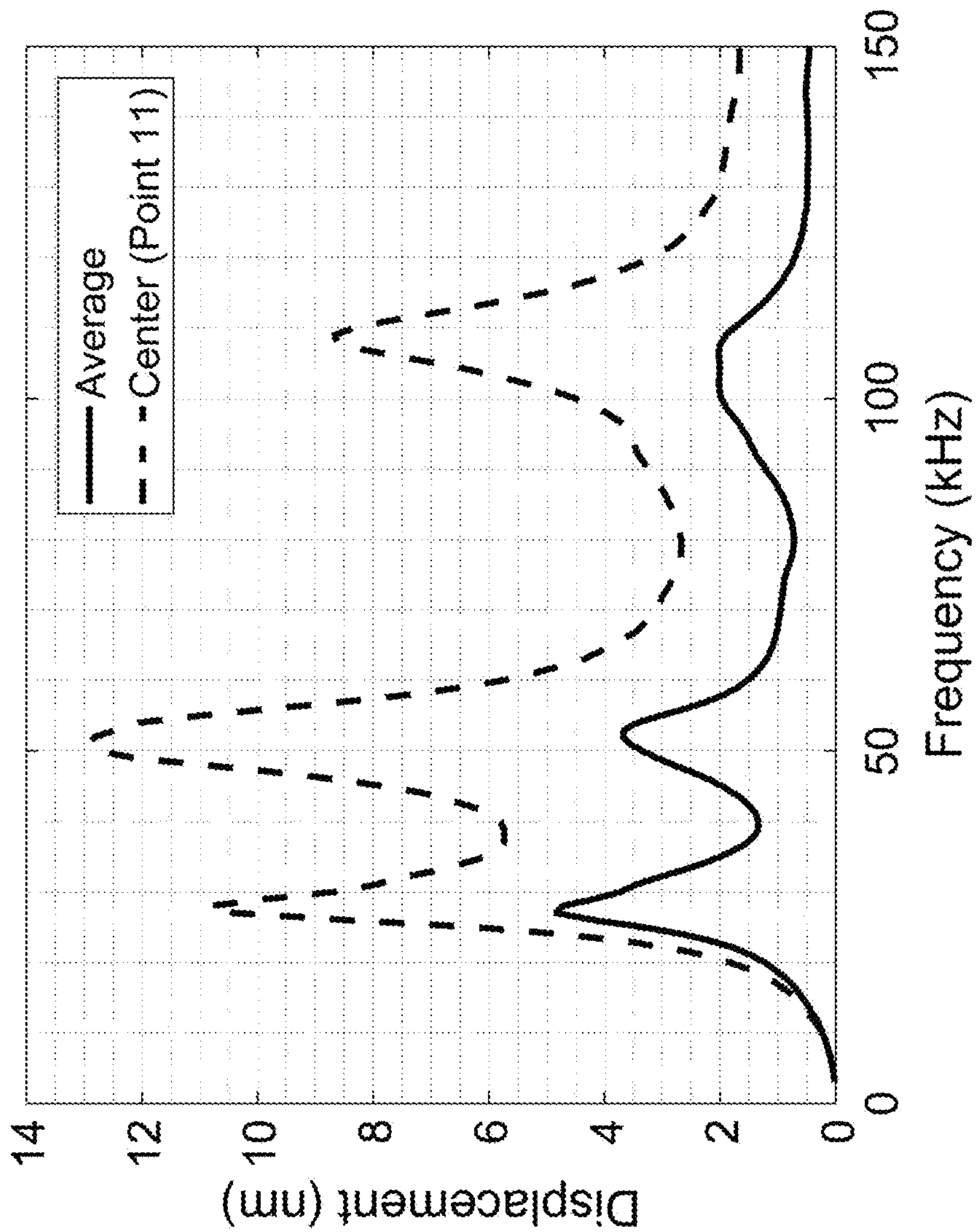


FIG. 9

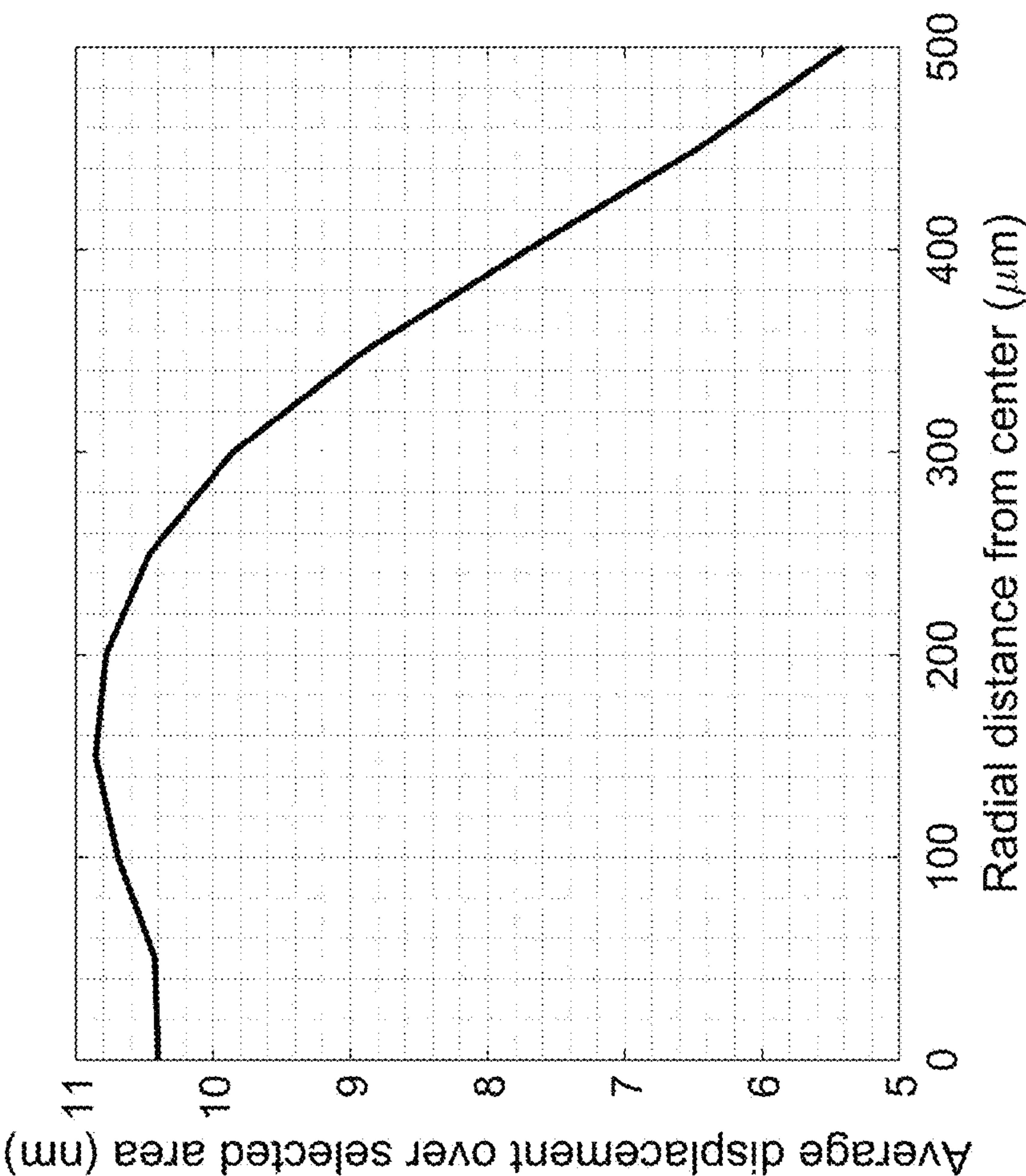


FIG. 10

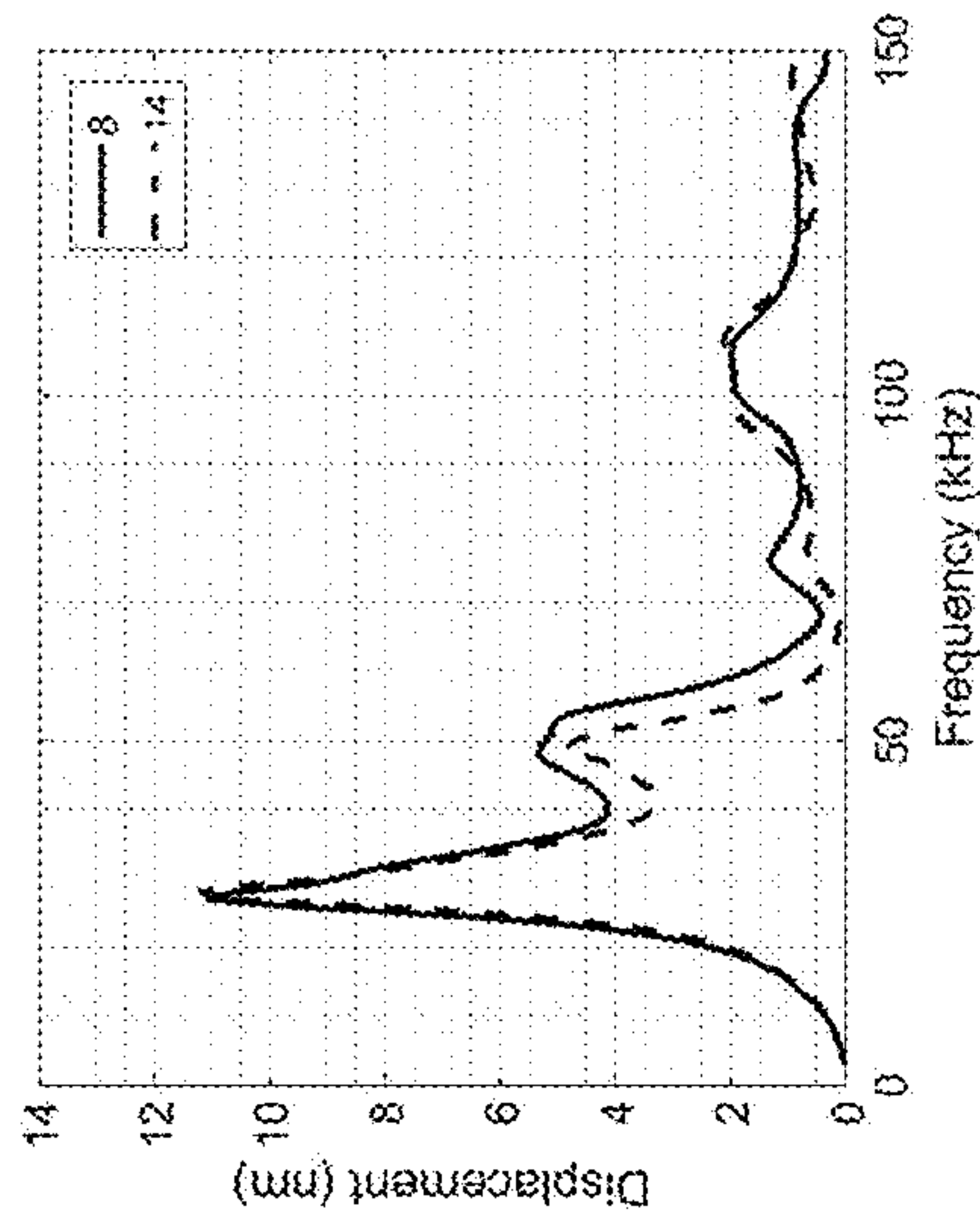


FIG. 11B

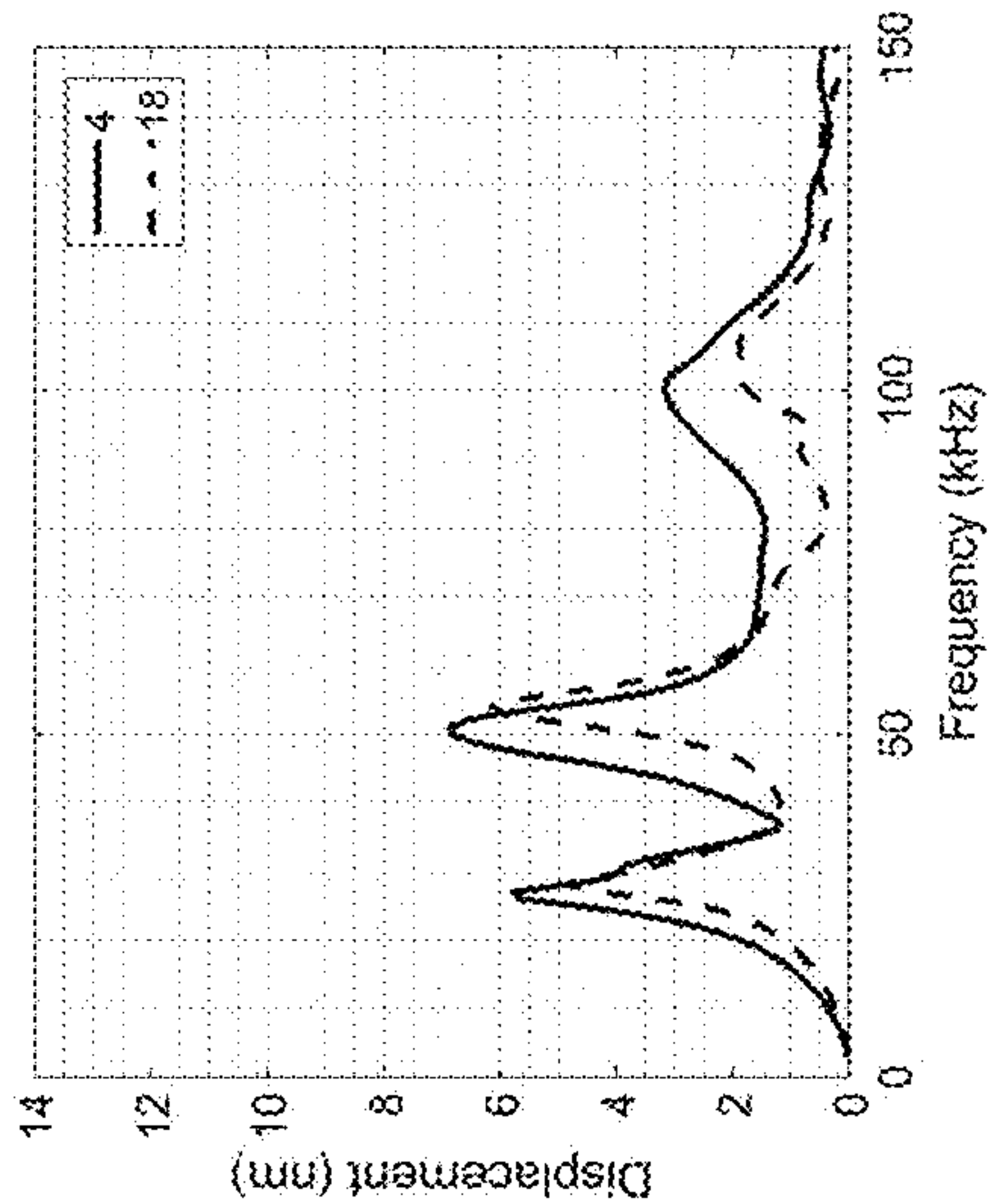


FIG. 11D

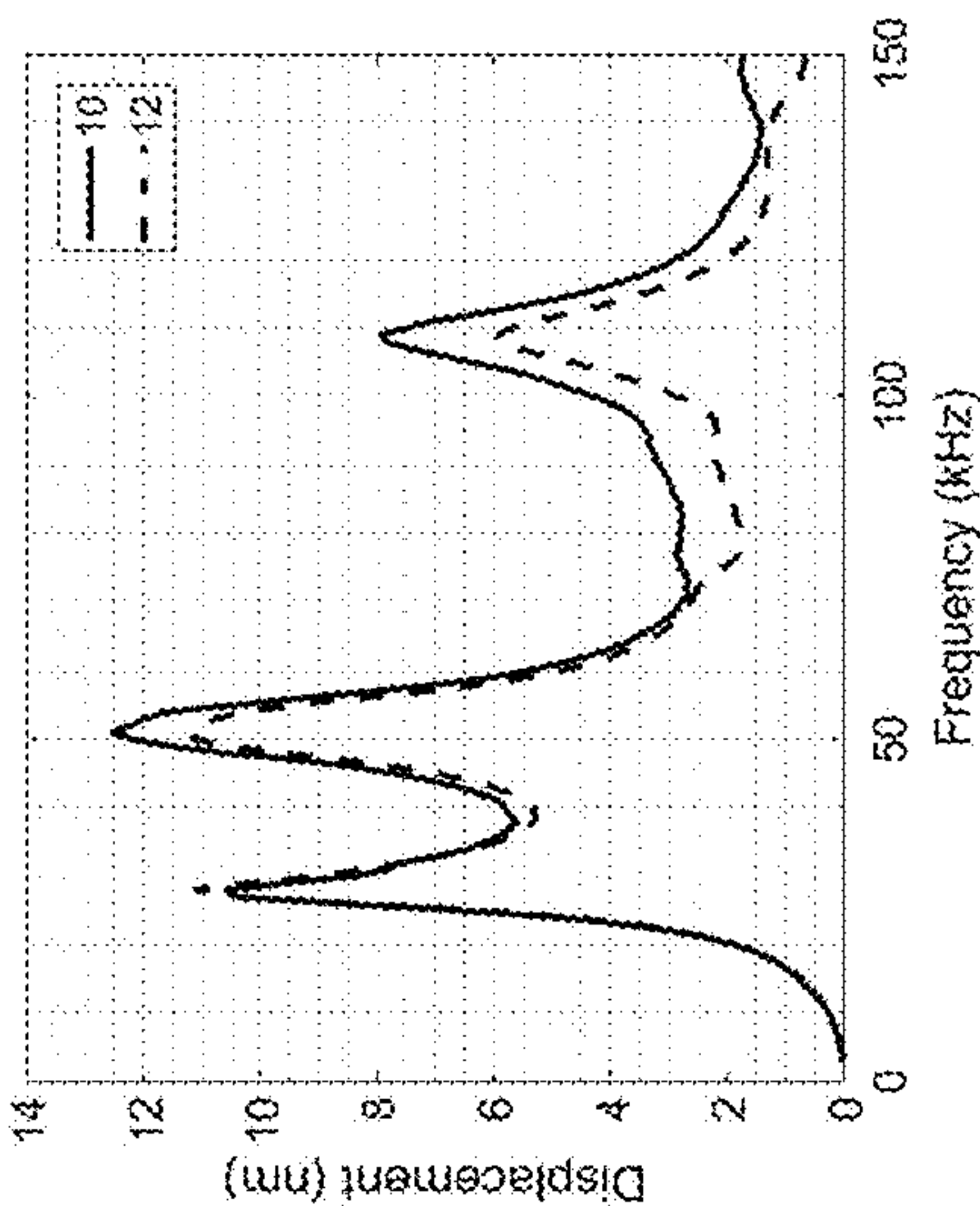


FIG. 11A

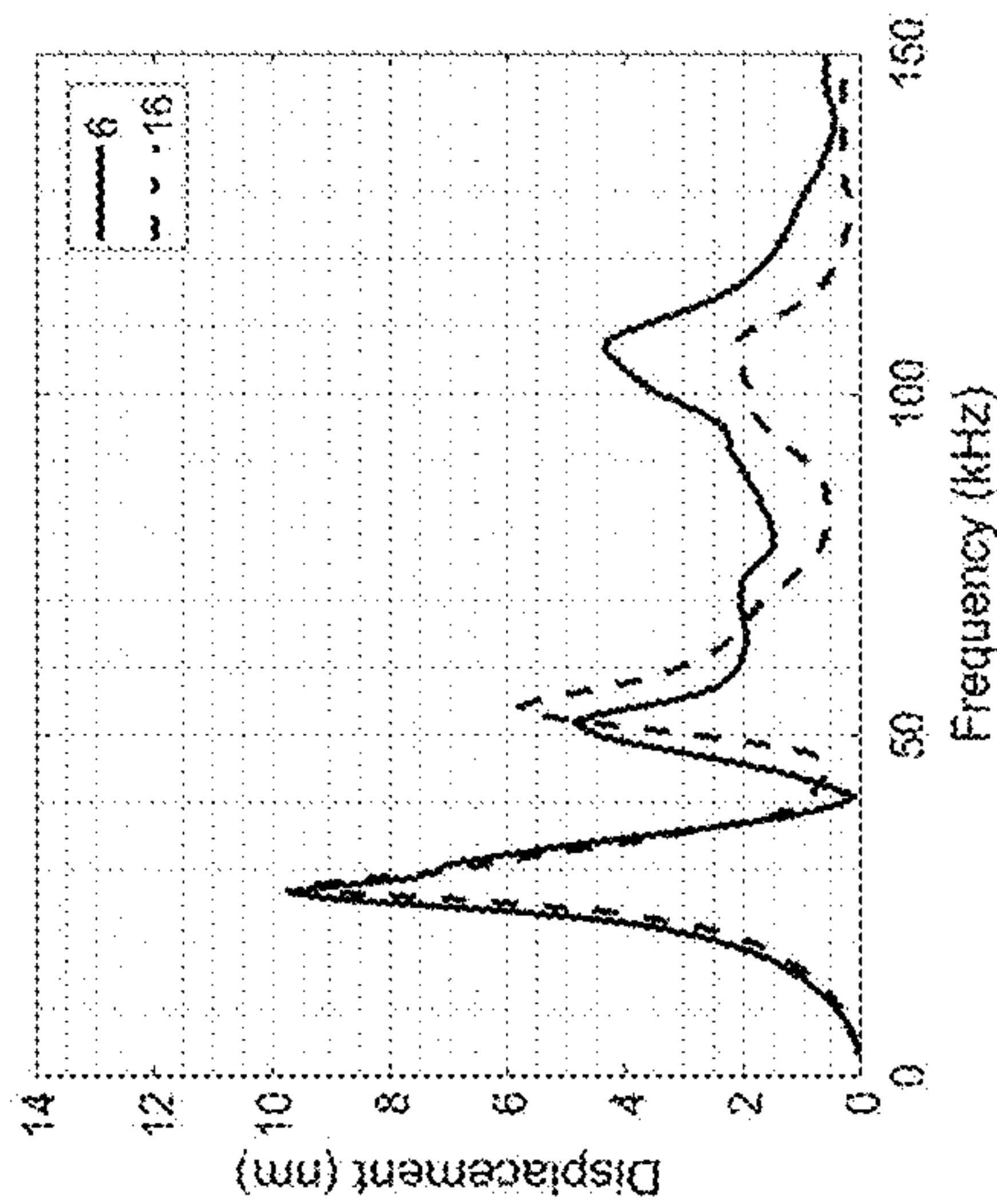


FIG. 11C

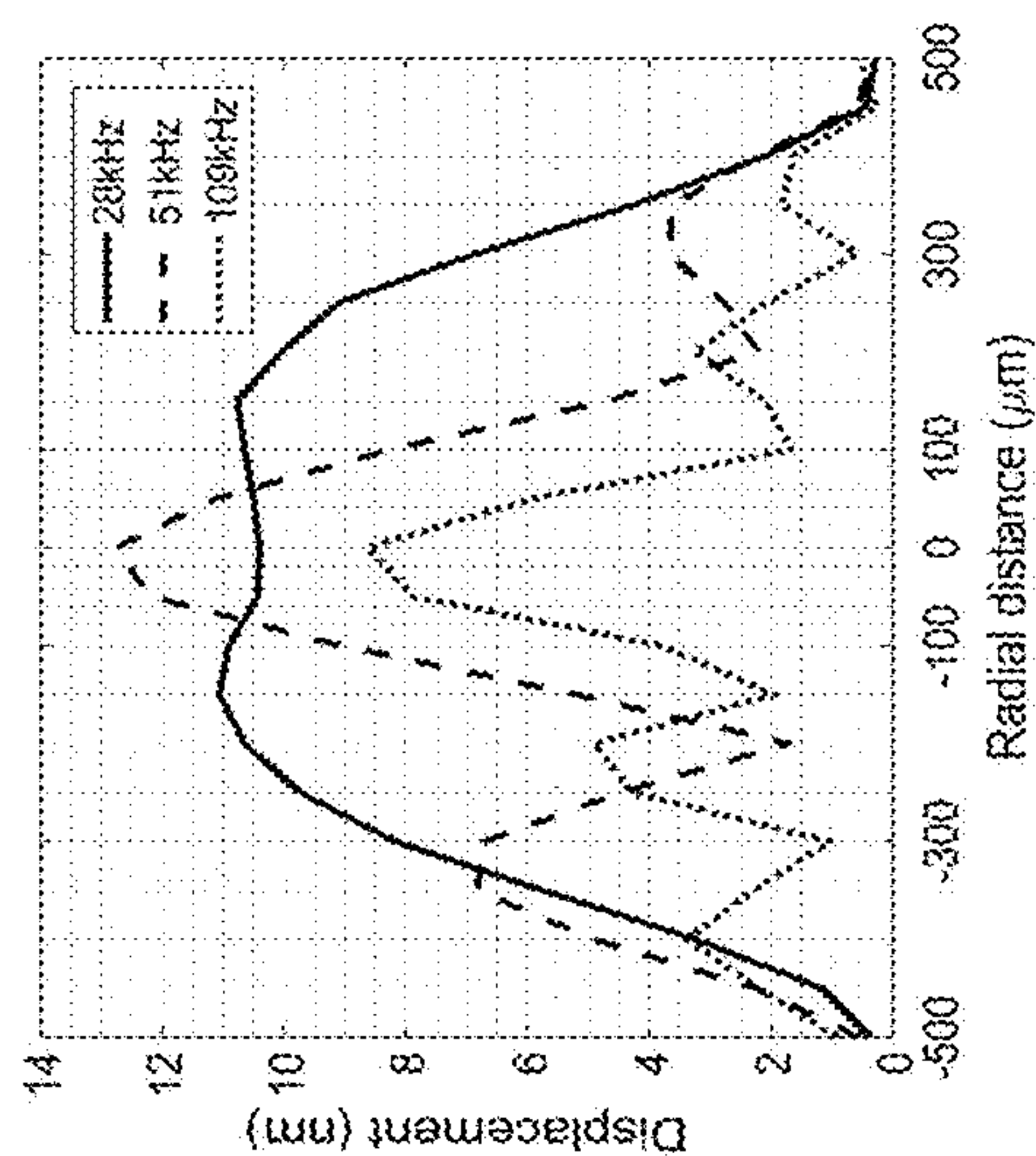


FIG. 12A

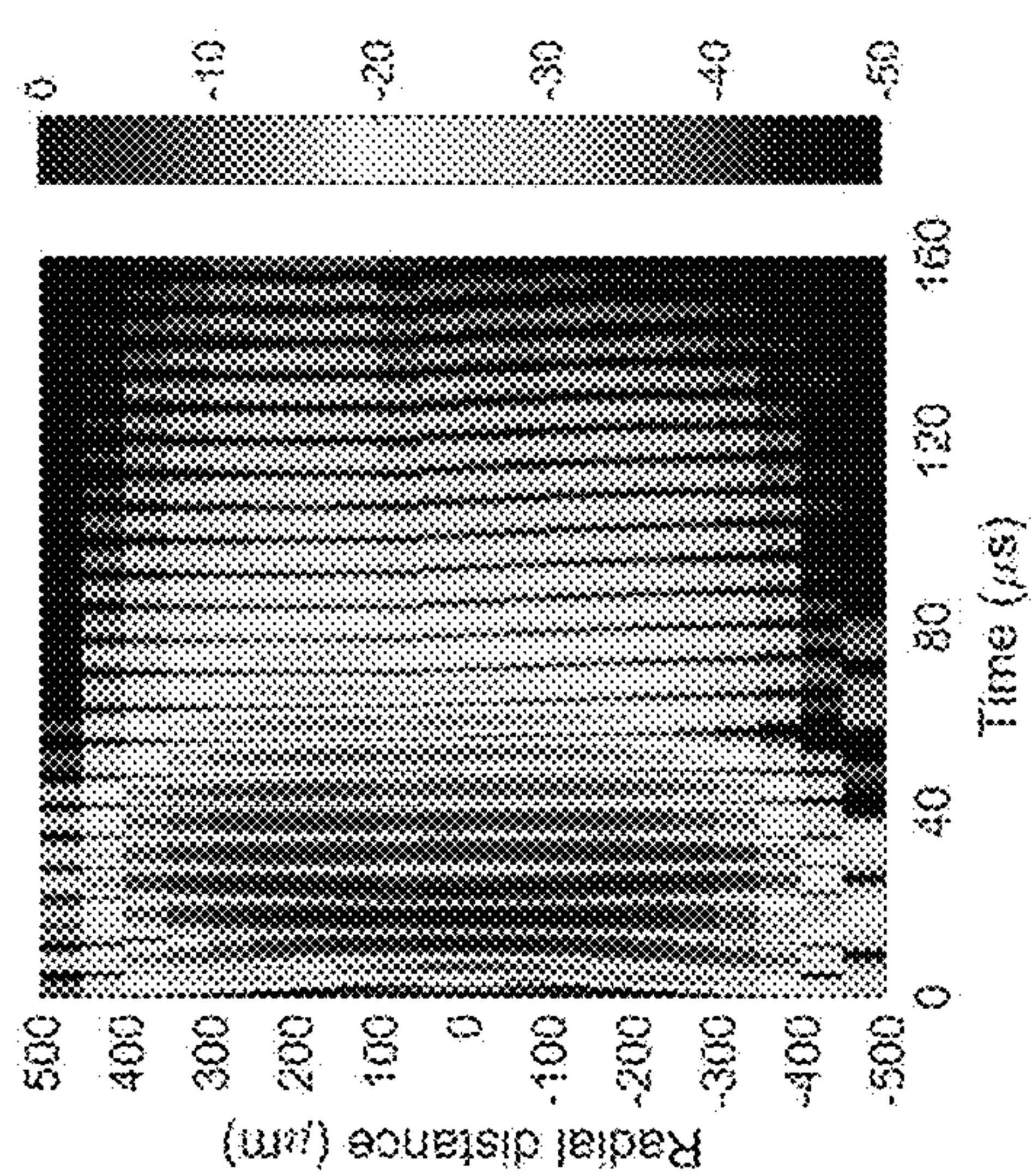


FIG. 12B

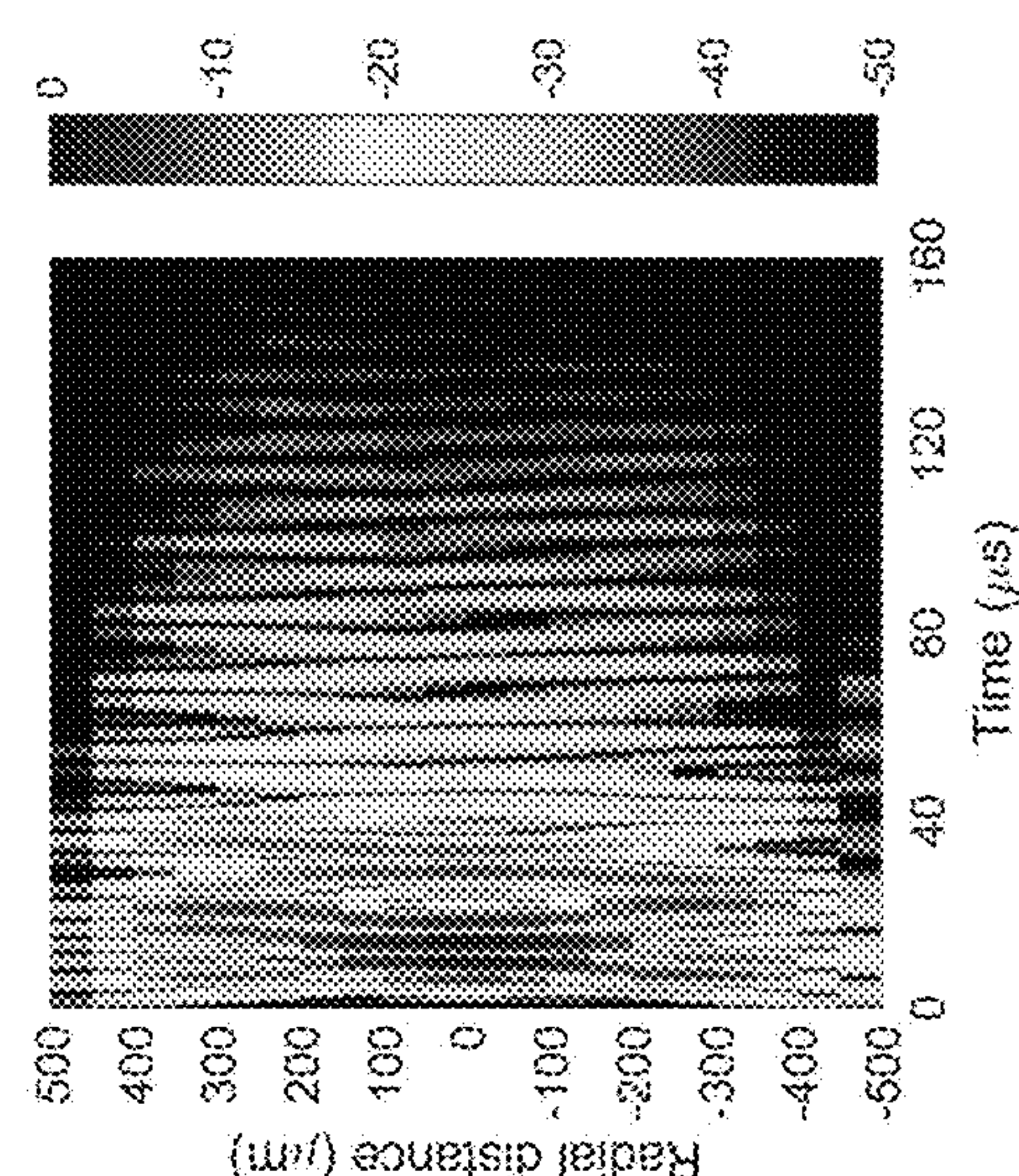


FIG. 12C

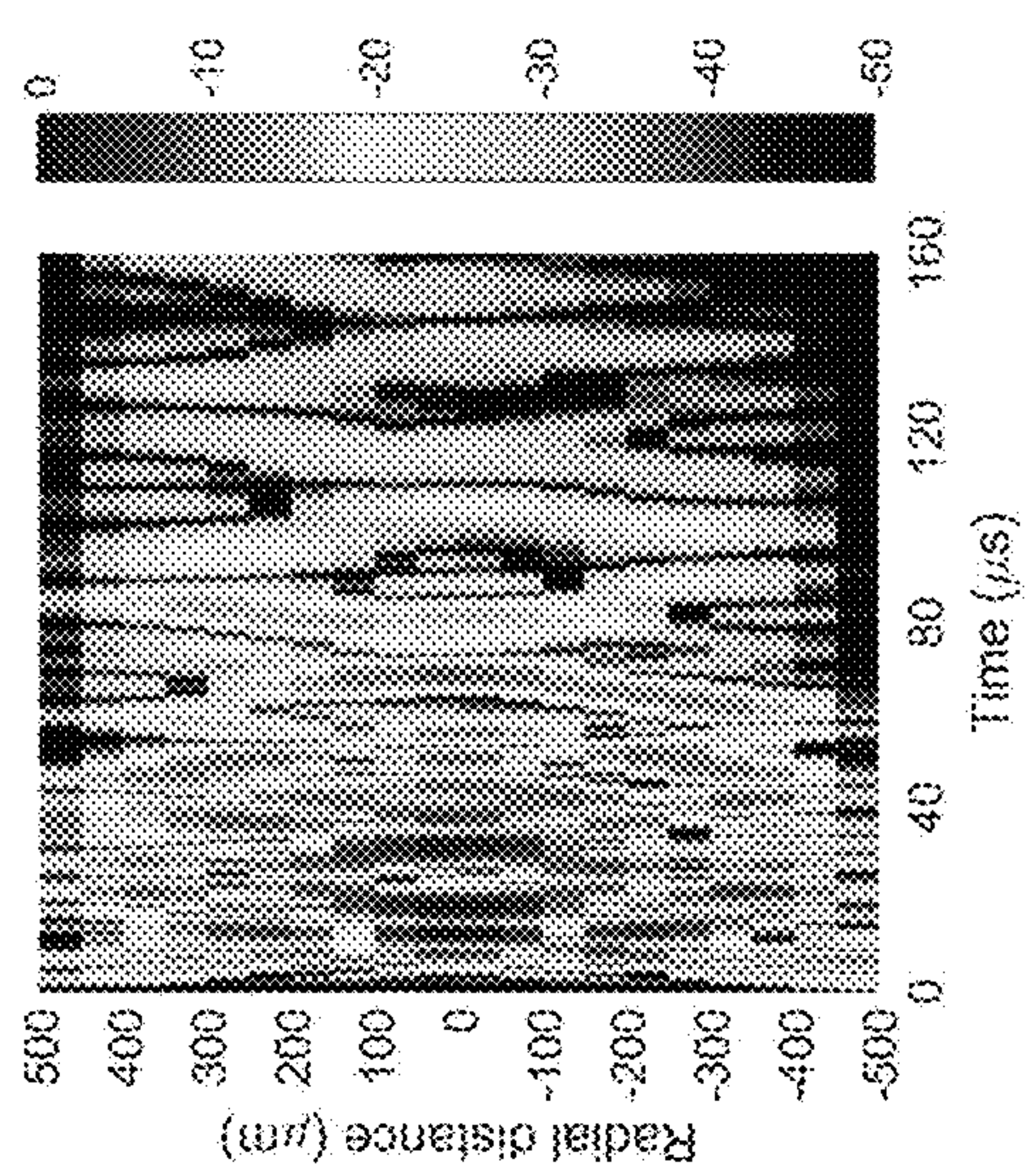


FIG. 12D

FIBER OPTIC MEMS MICROPHONE**CROSS REFERENCE TO THE RELATED APPLICATIONS**

This application is the national stage entry of International Application No. PCT/TR2019/050262, filed on Apr. 18, 2019, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The invention relates to fiber optic MEMS microphone that features electrically deflectable MEMS membrane via conversion of optical energy propagating in the optical fiber to electrical energy with photodiode chip.

BACKGROUND

Microphones, which are instruments that convert sound waves into electrical signals, are being developed with MEMS technology. The MEMS microphones have become ideal for use in portable technologies with their resistance to acceleration, low energy consumption, long-term stable performance, high sensitivity and small size. MEMS fiber optic microphones are used in distributed sensor networks and hazardous industrial environments because they are resistant to electromagnetic effects. With very low loss fiber optic cables, they can be carried away for miles away. When MEMS-based fiber optic microphone studies and patents in academic and industrial literature are examined, it is seen that MEMS devices are operated passively-without active voltage applied- and their sensitivities are limited and constant in operation. In commercially available microelectromechanical systems (MEMS) microphones, detection of the audible frequency range (20 Hz-20 kHz) is utilized through some electrical measurements such as current/voltage or capacitive measurements. These measurements are vulnerable to the electrical noise of the system and the environment. In addition, these microphones do not give accurate results under harsh environmental conditions such as high temperature and high pressure. Fiber optical sensors are used in a wide range of sensing applications for temperature, vibration, pressure, index of refraction sensing. Employing the microelectromechanical systems (MEMS) technology based on fiber optic sensing with extrinsic Fabry-Perot interferometry provides high sensitivity detection of high frequency ultrasound. Today, in some applications, fiber optic microphones are preferred over condenser microphones. The conventional condenser microphone is a high impedance device due to the necessity of a high input impedance preamplifier for further signal processing¹. For the condenser microphone, this additional high impedance preamplifier is an additional burden in the design which reduces the performance of the microphone even for some sophisticated electronic designs. Additionally, due to the cable capacitances, the distance between the microphone and the receiver electronics must be very small which restricts the application of the microphone in a confined space. The fiber optic MEMS microphone has very high immunity to electromagnetic interference since all of the electronic components that is used in the design are kept out of the sensor probe¹. The main advantage of MEMS technology is that it offers the flexibility to achieve the desired response range, bandwidth, and sensitivity by basically adjusting the size of the membrane^{1,2}. Also two different methods are used with fiber optic microphones; phase and

intensity modulation. Phase modulation technique gives better performance considering the measurement range and durability^{2,3}. For these reasons, the necessity of finding a new way of designing a MEMS microphone for audible frequency range detection has raised. It was shown that utilization of light as the way to detect sound waves gives much better performance than the microphones which senses the sound waves electrically⁴.

Optical sensing technique for small displacement measurement is vital for industrial non-destructive techniques, vibration and, design and testing of microstructures for their unique advantages, such as immunity to electromagnetic interferences, stability, repeatability, durability against harsh environments, high sensitivity, high resolution, and fast response³. The fiber optic microphone can enable innovative applications in a variety of applications due to the wide dynamic range, high sensitivity and flat frequency response over large bandwidth¹.

In the past two decades, extrinsic Fabry-Perot interferometer based pressure sensors have undergone a significant growth and substantial research has been carried out on it. Diaphragm-based extrinsic Fabry-Perot interferometer sensors have been successfully used for low pressure and acoustic wave detections.

To obtain high fidelity Fabry-Perot interferometer based pressure sensor, the most crucial part is the MEMS membrane. The light propagating through the fiber hits the membrane at the fiber end and then reflects back. At that point, the Fabry-Perot cavity is obtained between the fiber end and the membrane. Small displacements at the membrane are detected by analyzing the interference fringes from the cavity. The feasibility of phase modulation technique is experimentally proven by measurements⁴.

By adjusting size of the membrane, the electrical impedance of the membrane in air can be adjusted¹. Since both incident and reflected beams in fiber optic MEMS microphone share the same optical fiber, environmental effects at the fiber, such as changes in temperature, pressure and vibration do not significantly affect the interference signal obtained by Fabry-Perot interferometer¹. Fiber optic microphone has the potential capability of on-line and remote sensing. This type of sensor can also be extended to measure vibration and acceleration. Due to the simplicity of the structure and the ease of operation, this design can be used in variety of applications⁴. Pressure sensors can also be utilized via fiber optic idea. It is shown that linear pressure sensitivity between 0 kPa (94 dB) and 600 kPa (210 dB) is achieved in the temperature range of 20° C.-300° C. with these pressure sensors⁶.

Recently, different membrane materials, such as Parylene-C and graphene oxide, are proposed for a microphone application. Parylene-C gives strong response in the order of 2000 nm/Pa at 20 Hz. Due to the bio-compatibility of Parylene-C, these acoustic sensors are said to be very useful in biomedical applications⁷. With the developments in the fabrication of graphene oxide, reliable production of it with any desired thickness becomes possible. Graphene oxide based membrane is shown to be give flat response in 100 Hz-20 kHz range⁸. It is also shown that corrugated silver membranes can be used for optical fiber microphones. Introduced corrugations on the membrane was shown to improve the performance of the microphone and the new design gives response in the order of 50 nm/Pa in the range of 63 Hz-1 kHz⁹. In addition to the efforts of finding new membrane materials, there is ongoing research on the improvement of the shape and size of MEMS membranes. One suggested method is to microfabricate an annular

corrugated MEMS membrane. In this method, the proposed design can sense the minimum detectable pressure level of $3 \mu\text{Pa}/\text{Hz}^{1/2}$ at 1 kHz^{10} .

SUMMARY

A new approach for MEMS technology, the fiber optic MEMS microphone, is presented in this document. Our invention is a laser-powered active MEMS fiber-optic acoustic sensor microphone. Microphones have an important place in our lives. Traditional microphones are now replaced with MEMS microphones. Small size MEMS microphones are available in capacitive and piezoelectric models.

MEMS microphones are influenced by RF and microwave signals and high magnetic fields such as MRI because they are based on capacitive or piezoelectric electrical measurement techniques. Fiber-optic MEMS microphones are obtained by combining MEMS passive diaphragm and fiber optics. In this invention, an electrically adjustable MEMS membrane instead of a MEMS passive diaphragm is presented. Also, electrical power is generated from the incident laser light right on the spot of the MEMS membrane through a photodiode chip. In doing so, impractical electrical conduction along the fiber-optic cable is eliminated and the cost for large capacity use has been significantly reduced. At this point, the idea of transforming the energy of the light carried within the optical cable to the electrical voltage on the membrane by means of a photodetector is used. Sensing element is the most critical part of fiber optic MEMS microphone. This sensing element, membrane, is designed such that the microphone operates in the desired range with desired sensitivity. In this invention, design and characterization of a custom-designed MEMS membrane to be used in an optical fiber optic microphone is performed which will be responsive over the audible frequency range. This design features electrically deflectable membrane having symmetrically distributed air holes across it. Microfabrication is done using a commercially available multi-user multi-project service (POLYMUMPS, MEMSCAP Inc., France). Full electrical and optical characterizations of the membrane are done using an impedance analyzer and laser vibrometer, respectively. The design includes reflective gold-coated surface which makes it suitable for optical interferometry based microphones. The transient and steady state analysis of the membrane is utilized, and the overall response of the membrane as well as the spatial response of it is obtained. The fundamental resonance of the membrane is 28 kHz which is slightly above the frequency range of interest, 20 kHz . The peak displacement of 10 nm is obtained from vibrometer measurements under 100 mV peak-to-peak voltage and 1 V DC bias conditions. After converting the applied AC voltage to the pressure, the sensitivity of the membrane is calculated to be around $40 \text{ nm}/\text{Pa}$ at 28 kHz .

The symmetry of the design is also verified through spatial analysis. This invention offers a new aspect for the design of MEMS membrane for optical microphones. This invention employs MEMS technology to realize a novel membrane rather than finding new membrane materials for fiber optic microphone applications. This MEMS membrane is designed to respond to the audible frequency range. This new design is compatible with optical sensing for detection of membrane displacement. Utilization of a mature microfabrication processes makes the design reliable and reproducible. Extensive analyses of steady state and transient responses of the membrane to different excitations are performed and the performance of the design is verified.

One aspect of the invention, wherein MEMS device is respectively comprising of METAL, POLY2, POLY1, POLY0, SiN, and Si substrate from top to the bottom.

Another aspect of the invention, wherein the diameter of air holes and dimples are set to $36 \mu\text{m}$ and $12 \mu\text{m}$.

Another aspect of the invention, wherein a photodiode chip is Ge-TIA or InGaAs P-I-N photodiode.

Another aspect of the invention, wherein value for the laser operation wavelength is 1064 nm .

Another aspect of the invention, wherein the displacement of the large perforated membrane corresponding to the sound waves in the active MEMS-based fiber optic acoustic microphone is determined by phase modulation of the light.

Another aspect of the invention, wherein the laser beam from the fiber optic cable is used both as a remote power transfer and as an acoustic signal sensor via the MEMS device.

Another aspect of the invention, wherein the control range of the MEMS membrane against acoustic stimulation and the sensitivity of the measuring system are adjusted by controlling the bias of the MEMS device in the microphone.

Another aspect of the invention, wherein to obtain sufficiently large oxide etch under the membrane, any point on the membrane must be reachable by an air hole with a maximum distance of $15 \mu\text{m}$ in-between.

Another aspect of the invention, wherein membrane on a chip carrier with gold electrical paths and gold wirebond between them.

Another aspect of the invention, wherein dimensions of the membrane design are,

Substrate thickness (t_{subs}) is $>650 \mu\text{m}$

Membrane diameter (d_{membrane}) is $1000 \mu\text{m}$

hole-to-hole diameter ($d_{\text{HOLE-TO-HOLE}}$) is $50 \mu\text{m}$

dimple diameter (d_{dimple}) is $12 \mu\text{m}$

hole diameter (d_{hole}) is $36 \mu\text{m}$

metal thickness (t_{metal}) is $0.51 \mu\text{m}$

POLY2 thickness (t_{poly2}) is $1.5 \mu\text{m}$

Dimple thickness (t_{dimple}) is $0.75 \mu\text{m}$

POLY1 thickness (t_{poly1}) is $2.0 \mu\text{m}$

POLY0 thickness (t_{poly0}) is $0.51 \mu\text{m}$

SiN thickness (t_{SiN}) is $0.61 \mu\text{m}$.

Another aspect of the invention, wherein the MEMS device is remotely driven and controlled by laser power.

Another aspect of the invention, wherein the power of the laser beam is converted into a voltage source that will control the MEMS device via the photodiode integrated with the MEMS device.

BRIEF DESCRIPTION OF THE DRAWINGS

The figures used to better explain fiber optic MEMS microphone developed with this invention and their descriptions are as follows:

FIG. 1A A laser-powered active MEMS-based fiber-optic acoustic sensor-microphone showing Fabry Perot interferometer measurement system

FIG. 1B A laser-powered active MEMS-based fiber-optic acoustic sensor-microphone showing photodiode supplied active MEMS fiber optic microphone structure

FIG. 2 Cross-sectional view of the MEMS membrane design

FIG. 3A Electrical impedance measurements of MEMS structure showing the membrane on the chuck of the probe station as shown

FIG. 3B Electrical impedance measurements of MEMS structure showing the microscope view taken from the probe station as shown

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FIG. 4A Displacement measurements of MEMS structure using laser vibrometer

FIG. 4B Displacement measurements of MEMS structure using laser vibrometer

FIG. 5A The top view of the membrane showing a polysilicon layer of the membrane with air holes

FIG. 5B The top view of the membrane showing a zoomed view of the membrane which shows the air holes and dimples

FIG. 6 Hole arrangement for optimum membrane design

FIG. 7 Electrical characterization setup

FIG. 8 Optical characterization setup

FIG. 9 Deflection of the membrane as a function of frequency

FIG. 10 Average displacement over selected area for different radial distances from the membrane

FIG. 11A The analysis of the symmetry of the membrane. Response of two points that are separated by 50 μm from the center of the membrane

FIG. 11B The analysis of the symmetry of the membrane. Response of two points that are separated by 150 μm from the center of the membrane

FIG. 11C The analysis of the symmetry of the membrane. Response of two points that are separated by 250 μm from the center of the membrane

FIG. 11D The analysis of the symmetry of the membrane. Response of two points that are separated by 350 nm from the center of the membrane

FIG. 12A Spatial response of the membrane to the CW inputs at 28 kHz, 51 kHz and 109 kHz (steady-state)

FIG. 12B Spatial response of the membrane to one cycle of sine pulses at 28 kHz (transient)

FIG. 12C Spatial response of the membrane to one cycle of sine pulses at 51 kHz (transient)

FIG. 12D Spatial response of the membrane to one cycle of sine pulses at 109 kHz (transient)

DETAILED DESCRIPTION OF THE EMBODIMENTS

The present invention has been described in detail in the following. This invention offers a new aspect for the design of MEMS membrane for optical microphones.

In this section, a novelty is going to be demonstrated.

Our invention is a laser-powered active MEMS-based fiber-optic acoustic sensor-microphone (FIGS. 1A and 1B). The MEMS device is integrated with an optical fiber cable end structure and a photodiode chip. A portion of the laser beam from the fiber optic cable passes back through the MEMS substrate and membrane to form a Fabry-Perot interference and go to the fiber optic interferential motion sensor. The remaining part of the incoming laser beam illuminates the photodiode chip, creating the electrical power to activate the MEMS device. By changing the laser beam power, the MEMS device is driven at different voltage values. In this way, it is possible to obtain adjustable MEMS fiber optic microphone devices with high transfer factor and dynamic response intervals. The 1064 nm value for the laser operation wavelength is chosen because this wavelength can pass the MEMS device with a low loss of silicon substrate structure and illuminate the InGaAs P-I-N photodiode that will drive the MEMS device.

MEMS membrane, Laser diode, photodiode Ge-TIA and optical fiber circulator are shown in FIG. 1A. Laser diode which is shown in FIG. 1A as a 1064 nm value. MEMS membrane, photodiode, reflected laser light, incident laser

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light and Optical fiber are shown in 1b. Optical fiber which is shown in FIG. 1B as a $\Phi 125 \mu\text{m}$ and $\Phi 425 \mu\text{m}$.

MEMS fiber optic microphones have a vibrating diaphragm that corresponds to the acoustic sound wave. This diaphragm is not in any way an electrically controllable structure, i.e. it is passive. The diaphragm resonance frequencies are constant. With the fiber optic cable, the intensity or phase modulation of the incoming light is determined by the displacement of the diaphragm corresponding to the sound wave. In general, different materials of the diaphragm structure or diaphragm surface corrugation shapes are tried to increase sensitivity.

In our invention, the displacement of the large perforated membrane (FIG. 2) corresponding to the sound waves in the active MEMS-based fiber optic acoustic microphone is determined by phase modulation of the light. This allows the membrane to adjust its sensitivity to sound waves.

1) The MEMS device is remotely driven and controlled by laser power. The laser beam from the fiber optic cable is used both as a remote power transfer and as an acoustic signal sensor via the MEMS device. The power of the laser beam is converted into a voltage source that will control the MEMS device via the photodiode integrated with the MEMS device. In the literature, a MEMS study activated by laser power was not found.

2) In the active MEMS-based fiber-optic acoustic sensor, the control range of the MEMS membrane against acoustic stimulation and the sensitivity of the measuring system are adjusted by controlling the bias of the MEMS device in the microphone. This results in a wider dynamic response range than conventional MEMS fiber optic microphones. In the literature research, no MEMS fiber optic microphone working with active power supply has been found.

3) The MEMS device is integrated with an optical fiber cable end structure and a photodiode chip. The laser beam from the fiber optic cable will be used both as a remote power transfer and as an acoustic signal sensor via the MEMS device. A study using laser beam for remote power transfer in a sensor system has not been found in the literature.

Light is used not only for displacement, but also for energy transmission. The presence of an active membrane in which we can adjust the resonance frequency during use with this energy transmission is particularly important for microphones operating in a very wide frequency range (20 Hz to 20,000 Hz).

Membrane of a fiber-optic MEMS microphone is the sensing element of it. The design of membrane concerns several issues such as the reproducibility, durability, stable performance of the membrane. These issues are addressed to the microfabrication process. Based on commercially available multi-user multi-processes (MUMPS) offered by foundries, POLYMUMPS process (MEMSCAP Inc., France) is selected due to its suitability for microfabrication of airborne membranes supported by the non-limiting process design rules for our intended microphone application. Furthermore, reproducibility and consistency of this mature process is considered to be advantageous for the realization of high fidelity membrane, which is of utmost importance for the optical phase detection of light.

This process is based on polysilicon layers. The ability to design membranes and the ability to etch sacrificial oxide layers under the polysilicon layers makes this process valuable for our design. Obtain perfect etching of sacrificial oxide layers requires placement of holes in the polysilicon layers. The distance between any etching holes cannot be larger than 30 μm . In our design, we also take into account

the process variations in the design of our membrane. On top of the membrane, we put metal layer for the reflection of light from the membrane. CO₂ dry etch in addition to the standard HF wet etch for oxide removal is used. CO₂ dry is used to prevent the stiction of the adhesion between the membrane and the substrate for the large aspect ratio used in the membrane (1:500). Very low compressive stress (<7 MPa) of POLY2 membrane material with a thickness of 2 μm and very low tensile stress (<24 MPa) of METAL coverage of 0.51 μm on the membrane results in an almost ideal compensation of stress resulting an equivalent stress less than 1 MPa¹¹. This makes our large aspect-ratio membrane having negligible curvature due to residual stress. The diameter of the membrane is adjusted to obtain high response in the audible frequency range. The formula given by Equation 1 is used¹². This formula gives the first resonance frequency of a circular membrane by taking the membrane material properties and the membrane dimensions into account. For this design, the material properties and required dimensions are taken from the MEMSCAP.¹²

$$f \cong \frac{20.42}{\pi a^2} \sqrt{\frac{D}{h\rho}}, D = \frac{Eh^3}{12(1-\mu^2)} \quad (1)$$

where a is the diameter, h is the thickness, p is the mass density, which is taken as 2330 kg/m³, E is the Young's Modulus, which is taken as 169 GPa, and μ is the Poisson's ratio, which is taken as 0.22, of the membrane.

Since the detection of audible signals in 20 Hz-20 kHz frequency band is desired, the resonance frequency that is slightly above the upper limit of this band is required.

The resultant design of the membrane consists of a polysilicon membrane with air holes and it is called as a meshed-structured membrane. The top view of the membrane is shown in FIGS. 5A and 5B.

The diameter of air holes and dimples are set to 36 μm and 12 μm, respectively. The distance between the centers of two air holes is determined according to the design rules. To obtain sufficiently large oxide etch under the membrane, any point on the membrane must be reachable by an air hole with a maximum distance of 15 μm in-between by design rules. The geometric problem that is solved is represented in FIG. 6.

The design is fully symmetric, and the holes and their separations are identical in both directions on the membrane. There are two variables in this design, the diameter of the holes and the distance between the centers of the holes. Since the placement of holes is fully symmetric, an equilateral triangle is obtained with the corners taken as any three air hole centers. After calculations, it is found that $2d_{HOLE-TO-HOLE} - \sqrt{3}d_{HOLE} \leq 30\sqrt{3}\mu\text{m}$. Another objective is to maximize the fill factor. Fill factor (FF) of the design is given as

$$FF = 1 - \frac{\pi}{2\sqrt{3}} \left(\frac{d_{HOLE}}{d_{HOLE-TO-HOLE}} \right)^2.$$

Since the amount of the light reflected from the membrane should be sufficiently large for the detection of the displacement of the membrane, at least 50% fill factor is aimed. Therefore, the constraint becomes

$$\frac{d_{HOLE}}{d_{HOLE-TO-HOLE}} < 0.75.$$

To optimize both constraints, $d_{HOLE}=36 \mu\text{m}$ and $d_{HOLE-TO-HOLE}=50 \mu\text{m}$ are chosen with safety margins. In this case, the fill factor becomes 53% and hole-to-hole distance becomes 50 μm. The maximum distance between any point on the membrane and the air hole is 14 μm.

The cross-sectional view of the resultant is shown in FIG. 2 and the values of the parameters in this figure are given in Table I.

TABLE I

Values of the representative dimensions of the design.	
Dimension parameter	Value
Membrane diameter ($d_{MEMBRANE}$), μm	1000
Support length ($d_{SUPPORT}$), μm	150
Hole-to-hole diameter ($d_{HOLE-TO-HOLE}$), μm	50
Dimple diameter (d_{DIMPLE}), μm	12
Hole diameter (d_{HOLE}), μm	36
Metal thickness (t_{METAL}), μm	0.51
POLY2 thickness (t_{POLY2}), μm	1.5
Dimple thickness (t_{DIMPLE}), μm	0.75
POLY1 thickness (t_{POLY1}), μm	2.0
POLY0 thickness (t_{POLY0}), μm	0.51
SiN thickness (t_{SiN}), μm	0.61
Substrate thickness (t_{SUBS}), μm	>650

Electrical and Optical Measurement Setup

To characterize and verify the performance of the micro-fabricated MEMS membrane, electrical impedance and optical laser vibrometer measurements are employed. Cumulative response of the whole membrane is achieved via electrical whereas spatial response of spots as small as 2 μm can be collected via optical measurements. To characterize the membrane electrically, network and impedance analyzer (5061B, Keysight Technologies, California, USA) and triax connected probe station (EPS150X, Cascade MicroTech, Oregon, USA) are used as shown in FIG. 7. Series Gain-Phase mode using LF-Out port is used.

In this setup, membrane is placed on the chuck of the probe station. Electrical connections from ground and signal pads of the MEMS chip are made using tungsten needles (PTT-120-/4-25, Cascade MicroTech, Oregon, USA) followed by triax-to-BNC adapter to network and impedance analyzer. A power splitter (Agilent 11667L) is used to connect the device to the network analyzer properly to implement the suggested configuration of low frequency measurement of high impedance device.

The series capacitance (C_s) and series resistance (R_s) measurements are taken to detect any resonance frequency in the frequency range of 1 kHz-100 kHz. During measurements Intermediate Frequency Band Width (IFBW) is set to 500 Hz with an averaging of 32. 10 dBm ($1V_{peak-to-peak}$) sinusoidal signal is used and 1601 data points are taken in the given frequency range. DC voltage is changed from 0 V up to 3 V.

Optical characterization is done with the setup as shown in FIG. 8. Laser Vibrometer (OFV5000, Polytec, Germany) is used together with the digital oscilloscope (DSO6014A, Agilent, California, USA), the function generator (33250A, Agilent, California, USA) and a personal computer with LabView (National Instruments, Texas, USA) on it to control the devices in the setup.

Optical characterization is based on the detection of the displacement of the MEMS membrane as a result of the electrical excitation. A laser light at 633 nm wavelength is sent to the membrane and the reflected light is used to understand the deflection of the membrane via the interferometer that is utilized between the membrane and the laser light. In fact, this way of displacement measurement is very similar to the way that is used in fiber optic microphones.

To generate an acoustical pressure on the membrane, the electrical excitation is done by a function generator which generates 0.1 V peak-to-peak sinusoidal signal with a DC bias voltage of 1 V. The excitation frequency is swept from 3 kHz to 150 kHz. Besides the general response of the membrane, this optical characterization setup enables the spatial inspection of the membrane. In other words, by directing the laser light on different points on the membrane, spatial response of the membrane to any excitation can also be obtained.

Results

The electrical characterization setup is prepared by putting the membrane on the chuck of the probe station as shown in FIG. 3A. By using needles, electrical connections are utilized. The microscope view taken from the probe station is as shown in FIG. 3B.

To avoid the collapse of the membrane on the ground plate, maximum of 3V DC bias is applied between the plates of the microphone. The measured C_s and R_s values are shown in FIGS. 3A and 3B. Two resonance frequencies are observed at 30 kHz and 56 kHz, respectively.

In optical measurement setup, the membrane is placed in a chip carrier as shown in FIG. 4B and the electrical pads are connected to electrically conductive gold lines on the chip carrier by gold wire bonds as shown in FIG. 4B. Chip carrier has two SMA (SubMiniature version A) connectors which are used to excite the membrane electrically through its pads. The signal generator is connected to the MEMS device and it is also connected to the oscilloscope as a triggering signal for the output of the vibrometer.

In the optical measurement, all data were taken from 21 points on the membrane to understand its overall response as well as its spatial response. Those 21 points are 50 μm apart from each other with the 11th point at the center of the membrane as shown in FIG. 4A. Input frequency is swept from 3 kHz to 150 kHz with a continuous wave (CW) sinusoidal excitation. The response of the membrane in both spatial and frequency domains are shown in FIG. 4B. It is observed that there are three resonance frequencies at 28 kHz, 51 kHz and 109 kHz. Region around the center of the membrane gives the highest response to any input.

Total response of the membrane considering the amount of deflection is also obtained by processing the spatial data. The overall response and the response of the central point are shown in FIG. 9. At 28 kHz, the average displacement of the membrane is 5 nm in average and 10 nm in the center. The fractional 3-dB bandwidth of this resonance is 32%.

To determine where to focus the laser light to obtain the maximum deflection per area the responses of data points at 28 kHz are analyzed. Average displacement over selected area is obtained for different radial distances from the center of the membrane and the resultant characteristics is shown in FIG. 10. The optimum area to focus the light has a radius of 150 μm . The average displacement is almost constant for circular areas with radius ranging from 50 μm to 250 μm .

To analyze the symmetry in the response of the membrane, the data from the points that are 50 μm , 150 μm , 250 μm and 350 μm distant from the center of the membrane are used. This characteristics is shown in FIGS. 11A-D.

For the points that are 50 μm distant from the center of the membrane, it is realized that their peak responses are at the same frequencies. This situation occurs also for 100 μm distant points. For 250 μm distant points, the location of the second resonance frequency changes slightly by 3.8%. Further small deviations are observed for 350 μm distant points. The first, second and third resonance frequencies change by 3.6%, 5.9% and 4%, respectively.

The spatial steady state responses of the membrane to the input signals at 28 kHz, 51 kHz and 109 kHz are shown in FIG. 12A. To obtain the temporal transient response of the membrane, single cycle sinusoidal signals at the resonance frequencies, 28 kHz, 51 kHz and 109 kHz are used. The responses are as shown in FIGS. 12B-D. It is observed that the transient response of the membrane to the input at 28 kHz has longest ringing response. All of the transient responses decay with time. Electrical characterization is applicable to the membrane since it can be modelled as a series capacitor and a series resistor. A parallel plate capacitor can be visualized by considering the membrane and the substrate as parallel plates. In the design of the microphone, a 450 μm -radius, 0.5 μm -thick polysilicon layer is added on the substrate which is just below the membrane by 2 μm . This part is called the ground plate and it is connected to the ground pads by polysilicon lines. The movable membrane is also connected to the signal pads via polysilicon lines.

From electrical measurements as shown in FIGS. 3A and 3B, first resonance frequency at 30 kHz is close to the expected one which was calculated from the Equation (1) as 26 kHz. From series capacitance data, it is observed that this frequency changes from 29.4 kHz to 30.6 kHz when the applied DC voltage is changed from 0V to 3V. The resonance frequency is observed around 56 kHz and it changes by 3 kHz with the applied DC voltage. Therefore, the resonance frequency of the membrane is fine tunable with the applied DC voltage. From 0V to 2V, the first resonance frequency increases by 750 Hz which may be due to the stress stiffening and from 0V to 3V, it decreases by 250 Hz which may be due to the effect of spring softening dominating stress stiffening¹⁴.

The measured capacitance values in FIG. 3A are not actual capacitance values of the MEMS membrane, which is theoretically calculated to be 1.4 pF, due to the effects of cabling and lack of calibration at the end of probes. They only show the existence of resonance frequencies, and high capacitance values indicate that the resonances around 30 kHz and 56 kHz are strong. The measured resistance values are relatively high due to the resistive lightly-doped polysilicon lines compared to the highly conductive metal.

Modal shapes of the resonances are shown in FIG. 4B. It can be observed that the first resonance frequency is the fundamental resonance frequency since it has no nulls in its mode shape. The second resonance frequency at 51 kHz and the third resonance frequency at 109 kHz has one and two nulls along the radial direction as expected, respectively. This data supports that these are the first three resonances of the membrane.

In addition to the operation of the device around 28 kHz, from FIG. 4B, this membrane can be used to operate at 58 kHz and 109 kHz. By focusing the light on circular areas with 150 μm and 100 μm radius, this membrane can operate at 58 kHz and 109 kHz, respectively.

If the parallel plate capacitor assumption is used, the applied voltage can be used to model the force acting on the plates of the capacitor, namely the membrane. This information is valuable since the membrane is required to

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respond sound pressure. From the average displacement data in FIG. 9, this calculation is done by

$$P = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \epsilon \left(\frac{V}{d_{eff}} \right)^2$$

formula which represents the amount of pressure that acts on the plates of a parallel plate capacitor. In this formula, ϵ is actually ϵ_0 , which is the permittivity of free space, and gap between the plates and 1.25 μm for the regions with dimples. First, the gap between the plates should be calculated. For a parallel plate capacitor, deflection of the plate as a function of applied voltage is given as

$$V_n^2 = \frac{27}{4} x_n (1 - x_n)^2,$$

where

$$V_n = \frac{V}{V_{collapse}}, V_{collapse}$$

being the collapse voltage of the membrane,

$$x_n = \frac{x}{d_{eff}}, d_{eff}$$

being the effective gap between the plates of the capacitors¹³. In this design, $V_{collapse}$ is observed to be larger than 3V. d_{eff} can be calculated from different regions of the membrane, namely the polysilicon region with dimples, region-1 and the polysilicon regions without dimples, region-2. The air gap of region-1 is $g_1=1.25 \mu\text{m}$ and the air gap of region-2 is $g_2=2 \mu\text{m}$. Effective gap can be calculated by

$$\frac{1}{9} \left(\frac{1}{g_1} \right)^2 + \left(\frac{1}{g_2} \right)^2 = \frac{10}{9} \left(\frac{1}{d_{eff}} \right)^2$$

resulting in d_{eff} being 1.86 μm . If we insert this value into the equation, we can find that for 1V DC bias, as in the optical measurement setup, $x_n=0.01$ which results in $x=18.6 \text{ nm}$. Since this value is negligible compared to $d_{eff}=1.86 \mu\text{m}$, it can be ignored. The applied voltage is the sum of DC and ac voltages as $V=V_{DC}+V_{ac} \cos(\omega t)$. By taking square of it, we obtain three terms: the purely DC term, which is already responsible for the change of the effective gap between the plates, the ac term with frequency two times the excitation frequency, which is neglected and the ac term with frequency same as the excitation frequency. This final term should be used in the pressure formula. By taking $d_{eff}=1.86 \mu\text{m}$, it is found that the average displacement per pressure of the membrane at 28 kHz is approximately 40 nm/Pa. Highly reflective surface and high pressure sensitivity of the membrane make it suitable for microphone applications. The average displacement of the membrane is predicted to be increase with the larger DC bias voltages. Therefore, operating this device around 3V DC bias gives the highest response and this makes the detection of the displacement

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easier. The area that the laser light to be spotted is important since width of the light beam increases with distance. The amount of power that is coupled to the fiber after reflection starts decreasing after the critical width of the beam which is equal to the diameter of the fiber. By taking the distance light travels constant, the only parameter to change is the width of the beam when it hits the membrane. This width should be large enough to obtain the sufficient amount of deflection information from the membrane. From FIG. 10, the optimum width is calculated as 300 μm (150 μm -radius circle). Therefore, focusing light to the area with 150 μm radius would be the best solution considering both the sensitivity and the amount of light that couples back to the fiber. The membrane shows a highly symmetric response within the area with radius 250 μm as shown in FIGS. 11A-D. The symmetry can also be verified from the transient response of the membrane to the one cycle sinusoidal excitations as shown in FIGS. 12B-D. From symmetry property, the central operation of the membrane is verified; the membrane has circular symmetry with respect to the center of it.

The characteristics of a custom designed 1300 $\mu\text{m} \times 1300 \mu\text{m}$ featuring 1.5 μm -thick, 1000 μm -diameter MEMS membrane with 36 μm -diameter air holes which is designed as the sensing element for a fiber optic microphone is microfabricated by POLYMUMPS process and its characteristics are investigated. Air holes are used in the design to obtain sacrificial etch of oxide layers under the membrane. These holes are designed such that the fill factor of the design is 53%. The surface of the membrane is gold-coated which utilizes a reflective surface for the light. The resonance frequencies of the design are obtained by electrical and optical measurement setups. The fundamental resonance frequency of the design changed by 3% with the change of the applied dc voltage from 0V to 3V. Higher order modes are also observed and the center of the membrane gives strong response also to the higher modes. The response of the design is spatially symmetric and the mode shapes suggests that the resonance at 28 kHz is the fundamental mode of the membrane. By obtaining the resonance frequency slightly above the audible frequency range, strong response and almost no phase reversal have achieved. High sensitivity of the design, 40 nm/Pa, makes it suitable for fiber optic microphones. High circular symmetry of the membrane is also important due to the circular spots of laser lights. This design also offers the flexibility of selecting the area to focus light on the membrane since it gives almost the same response within the 250 μm -radius area. By the characteristics of the membrane, this design is verified to be suitable for fiber optic microphones.

From the above detailed description, a fiber optic MEMS microphone, comprising;

A MEMS device integrated at the end of an optical fiber cable,

A photodiode chip placed on top of the MEMS device, Voltage generated by the photodiode chip which is electrically wired to the SIGNAL and GND pads of the MEMS device,

The MEMS device which is capable of reflecting part of the incident laser light from the membrane back to the optical fiber whereas allowing the remaining part of the incident laser light transmit through the membrane onto the photodiode chip,

The MEMS device which features a membrane coated with optically reflective material that reflects the incident laser light,

The MEMS device which features a membrane having air holes that transmit the incident laser light,
 Power adjustable laser beam by laser diode which is biasing the MEMS device at different voltage values,
 Light which is used for detection of the membrane displacement via phase modulation or intensity modulation of the incident laser light,
 Light which is also used in energy transmission through the optical fiber for the generation of voltage across the photodiode chip.

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What is claimed is:

1. A fiber optic MEMS microphone, comprising:
 a MEMS device integrated at an end of an optical fiber cable,
 a photodiode chip placed on a top of the MEMS device, a voltage generated by the photodiode chip, wherein the photodiode chip is electrically wired to SIGNAL and GND pads of the MEMS device,
 wherein the MEMS device reflects a part of an incident laser light from a membrane back to the optical fiber cable, and allows a remaining part of the incident laser light to transmit through the membrane onto the photodiode chip,
 the membrane is coated with an optically reflective material for reflecting the incident laser light,
 the membrane has air holes for transmitting the incident laser light,
 a power adjustable laser beam by a laser diode, wherein the laser diode is biasing the MEMS device at different voltage values,
 wherein a light is used for a detection of a membrane displacement via a phase modulation or an intensity modulation of the incident laser light, and
 the light is also used in an energy transmission through the optical fiber cable for a generation of the voltage across the photodiode chip.
2. The fiber optic MEMS microphone according to claim 1, wherein the MEMS device is respectively comprising of a METAL, a POLY2, a POLY1, a POLY0, a SiN, and a Si substrate from the top to a bottom of the MEMS device.
3. The fiber optic MEMS microphone according to claim 1, wherein a diameter of each of the air holes and a diameter of a dimple are set to 36 μm and 12 μm .
4. The fiber optic MEMS microphone according to claim 1, wherein the photodiode chip is a Ge-TIA photodiode or an InGaAs P-I-N photodiode.
5. The fiber optic MEMS microphone according to claim 1, wherein a value for a laser operation wavelength is 1064 nm.
6. The fiber optic MEMS microphone according to claim 1, wherein a displacement of a perforated membrane corresponding to sound waves in an active MEMS-based fiber optic acoustic microphone is determined by the phase modulation of the incident laser light.
7. The fiber optic MEMS microphone according to claim 1, wherein the power adjustable laser beam from the fiber optic cable is used both as a remote power transfer and as an acoustic signal sensor via the MEMS device.
8. The fiber optic MEMS microphone according to claim 1, wherein a control range of the membrane against an acoustic stimulation and a sensitivity of a measuring system are adjusted by controlling a bias of the MEMS device in the fiber optic MEMS microphone.
9. The fiber optic MEMS microphone according to claim 1, wherein to obtain an oxide etch under the membrane, a point on the membrane must be reachable by an air hole with a maximum distance of 15 μm in-between.
10. The fiber optic MEMS microphone according to claim 1, wherein the membrane is placed on a chip carrier with gold electrical paths and gold wire bond between the membrane and the chip carrier.
11. The fiber optic MEMS microphone according to claim 1, wherein dimensions of a membrane design are as follows:
 a substrate thickness (t_{subs}) is $>650 \mu\text{m}$,

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a membrane diameter (d_{membrane}) is 1000 μm ,
 a hole-to-hole diameter ($d_{\text{HOLE-TO-HOLE}}$) is 50 μm ,
 a dimple diameter (d_{dimple}) is 12 μm ,
 a hole diameter (d_{hole}) is 36 μm ,
 a metal thickness (t_{metal}) is 0.51 μm , 5
 a POLY2 thickness (t_{poly2}) is 1.5 μm ,
 a dimple thickness (t_{dimple}) is 0.75 μm ,
 a POLY1 thickness (t_{poly1}) is 2.0 μm ,
 a POLY0 thickness (t_{poly0}) is 0.51 μm , and
 a SiN thickness (t_{SiN}) is 0.61 μm . 10

12. The fiber optic MEMS microphone according to claim
 1, wherein the MEMS device is remotely driven and con-
 trolled by a laser power.

13. The fiber optic MEMS microphone according to claim
 1, wherein a power of the power adjustable laser beam is 15
 converted into a voltage source controlling the MEMS
 device via the photodiode chip integrated with the MEMS
 device.

* * * * *

16

SMART INDIA HACKATHON 2026

InfraSense

High-Sensitivity Microbarometer Infrasonic Sensor

Project Documentation Report

*Design & Development of a High-Sensitivity Microbarometer
Infrasonic Sensor*

Problem Statement ID: SIH26144

Theme: Smart Automation | Category: Hardware

Organization: National Technical Research Organisation (NTRO)

Team ID: [Your Team ID] | Team Name: [Your Team Name]

Smart India Hackathon 2026

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1. Executive Summary

Infrasound — atmospheric pressure waves below 20 Hz, imperceptible to human hearing — carries early, long-range signatures of volcanic eruptions, avalanches, severe storms, explosions, meteors, and rocket launches. Detecting these signals reliably is central to disaster early-warning, geophysical research, and national security surveillance. Today, India depends almost entirely on imported microbarometer stations (₹15–25 lakhs per unit) to perform this function, which severely limits the density and reach of infrasound monitoring across the country and creates a long-term strategic dependency on foreign suppliers.

This report documents InfraSense, a proposed indigenous, high-sensitivity microbarometer infrasound sensor developed in response to National Technical Research Organisation (NTRO) problem statement SIH26144. The design directly addresses all nine mandatory hardware-architecture requirements specified in the problem statement — pressure sensing, mechanical transduction, differential measurement, low-noise electronics, temperature compensation, long-period pressure equalization, environmental enclosure, wind-noise reduction, and calibration methodology — while keeping the system realistic to build, test, and scale.

The prototype is built around a capacitive MEMS pressure sensor (BMP390) combined with a corrugated diaphragm structure, low-noise instrumentation amplifier, dual-vent pressure equalization, on-chip temperature compensation, and a hermetically sealed enclosure. Beyond meeting the baseline requirements, the design incorporates several innovations — networked multi-sensor beamforming for source triangulation, solar-powered autonomous operation, adaptive self-calibration, and LoRa-based remote health monitoring — that collectively enable a dense, low-maintenance, field-deployable sensor network suited to India's remote and border terrain.

At an estimated prototype cost of approximately ₹6,500 per unit — falling to an estimated ₹3,500–₹5,000 at a production scale of 100+ units — InfraSense offers a cost reduction of over 99% relative to imported alternatives, while remaining grounded in mature, proven MEMS and embedded-systems technology. This report presents the complete technical rationale, architecture, component specification, cost model, feasibility analysis, and implementation roadmap for the system.

2. Introduction & Background

2.1 What is Infrasound?

Infrasound refers to acoustic pressure waves with frequencies below approximately 20 Hz — the lower boundary of human hearing. Despite being inaudible, infrasonic waves carry substantial energy and can propagate over hundreds to thousands of kilometres through the atmosphere with relatively little attenuation, making them a powerful tool for remote sensing of energetic natural and man-made events.

Sources of infrasound include volcanic eruptions, large avalanches and landslides, severe thunderstorms and tornadoes, meteor entries, industrial and accidental explosions, mining blasts, rocket launches, and nuclear tests. Because these sources generate distinctive infrasonic signatures, a sufficiently sensitive and well-distributed sensor network can detect, localize, and classify such events well before other monitoring modalities (seismic, visual, or acoustic-audible) provide confirmation.

2.2 Global Context — The CTBTO Network

The Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) operates the world's most established infrasound monitoring network — the International Monitoring System (IMS) — comprising infrasound arrays distributed globally, each using multiple co-located microbarometers to enable beamforming, direction finding, and noise rejection. The standard instruments used in this network (e.g., the MB2000, MB2005, and MB3 series) rely predominantly on electromagnetic (LVDT-based) transduction of an aneroid capsule's displacement — a mature but comparatively expensive and import-dependent technology for a country such as India seeking to build indigenous, dense-coverage capability.

Emerging research — notably fiber-optic Fabry-Pérot interferometric sensing explored by institutions in France (CEA/iXblue) and China (Tianjin University, Wuhan National Laboratory) — has demonstrated that optical readout can, in principle, improve on the sensitivity and noise floor of conventional electromagnetic transduction. However, as of the current state of the art, this remains a research and prototype-stage technology; it has not been adopted as the standard, commercially deployed approach in operational infrasound networks. This context directly informs the design philosophy adopted in this report: build on proven, provable capacitive MEMS sensing today, while keeping a clear technology roadmap toward optical readout for future, higher-performance iterations.

2.3 The Indian Context and Motivation

India's current reliance on imported infrasound sensors constrains deployment in three ways: high upfront cost (₹15–25 lakhs per unit) restricts the number of stations that can be procured; long import lead times (6–12 months, subject to export restrictions) slow deployment and replacement; and high annual maintenance costs (₹2–3 lakhs per year) burden long-term operating budgets. Collectively, these factors mean that India's infrasound monitoring coverage — critical for disaster early-warning, especially in the Himalayan, coastal, and seismically active regions, as well as for strategic and security

surveillance along sensitive borders — remains sparse relative to what dense, low-cost, indigenous sensors could provide.

This project responds directly to this gap by proposing a sensor architecture that is technically sound, buildable with current component technology, and priced to enable the wide-area deployment that current imported alternatives make impractical.

3. Problem Statement

Problem Statement ID: SIH26144
Title: Design & Development of a High-Sensitivity Microbarometer Infrasonnd Sensor
Organization: National Technical Research Organisation (NTRO)
Category: Hardware | Theme: Smart Automation

3.1 Background (as stated in the official problem statement)

Infrasonnd sensors are precision instruments designed to detect and measure low-frequency atmospheric pressure waves — infrasonnd — that fall below the range of human hearing, typically under 20 Hz. These waves can travel long distances through the atmosphere and are produced by natural and human-made sources including distant industrial explosions, volcanic eruptions, severe weather systems, meteors, rocket launches, and other energetic phenomena. Detection and analysis of these signals are important for atmospheric monitoring, geophysical research, disaster warning systems, and security applications.

3.2 Required Hardware Architecture (a-i)

The problem statement requires that the sensor design address the complete hardware architecture, spanning nine specific elements:

Ref.	Requirement
(a)	Pressure sensing mechanism
(b)	Mechanical transducer design
(c)	Differential pressure measurement technique
(d)	Low-noise analog front-end electronics
(e)	Temperature compensation
(f)	Long-period pressure equalization system
(g)	Environmental enclosure
(h)	Wind-noise reduction interface
(i)	Calibration methodology

The frequency range of interest is approximately 0.01 Hz to 20 Hz. The design must achieve detection of very small pressure variations while maintaining long-term stability, low drift, and high signal fidelity, with data acquisition and real-time waveform display/analysis software included to demonstrate a complete, functional sensor system.

3.3 Expected Solution

The expected prototype must demonstrate: (a) detection of low-frequency pressure signals; (b) laboratory characterization of frequency response; (c) noise floor measurement; (d)

sensitivity estimation; and (e) stability testing. Evaluation is conducted against defined performance parameters, using digitizer and data-acquisition software (available in the open market) to demonstrate the complete functional sensor system.

4. Literature Review & Existing Technologies

4.1 Electromagnetic (LVDT) Microbarometers — The Current Standard

The majority of operational infrasound stations worldwide, including those in the CTBTO's IMS network, use an aneroid capsule whose pressure-induced displacement is converted to an electrical signal via an electromagnetic (LVDT-type) transducer. This approach is well understood, field-proven over decades, and supported by an established calibration and maintenance ecosystem — but it is also the primary driver of the high unit cost and import dependency that motivates this project.

4.2 Optical (Fabry-Pérot) Readout — Emerging Research

Several research groups have explored replacing electromagnetic transduction with optical interferometric readout to improve sensitivity and dynamic range:

- CEA / iXblue (France, 2017) presented a prototype optical microbarometer using an interferometric transducer in place of the conventional electromagnetic coil, explicitly describing the work as a step toward a future optical instrument rather than a deployable product.
- Tianjin University (China, IEEE Photonics Journal, 2021) developed a fiber-optic Fabry-Pérot infrasound sensor using a mechanical filter structure, achieving a minimum detectable frequency of 0.005 Hz — within the target band for this project.
- Wuhan National Laboratory (China, Photonics Research, 2024) reported a miniaturized fiber-optic Fabry-Pérot sensor targeting mHz-range infrasound detection, addressing miniaturization and sensitivity trade-offs.
- Scripps Institution of Oceanography (USA) tested an Optical Fiber Infrasound Sensor (OFIS) in a limited field deployment at the Piñon Flat Observatory — a CTBT IMS-designated site — as a research validation, not a standard replacement instrument.

These results confirm that optical readout is scientifically validated and an active, promising research direction, but not yet a mainstream, commercially available, or field-deployed standard. This positions optical MEMS sensing as an appropriate long-term roadmap item for this project rather than a component of the initial prototype.

4.3 Capacitive MEMS Pressure Sensing — Proven, Practical Foundation

Capacitive MEMS pressure sensors — including commercially available devices such as the Bosch BMP390 — are mature, mass-manufactured, low-cost components with well-characterized performance (RMS noise as low as 0.02 Pa, sampling rates up to 200 Hz). Related MEMS fabrication research, including diaphragm designs built on SOI wafers using DRIE (deep reactive-ion etching), has independently demonstrated that low-cost, standardized MEMS components can achieve sub-milliPascal-class sensitivity when combined with appropriate diaphragm geometry and signal conditioning. This body of work

forms the technical foundation for the capacitive-sensing approach adopted in this project.

5. Proposed Solution — Overview

InfraSense is a capacitive MEMS-based, high-sensitivity infrasound sensor designed to detect pressure fluctuations in the 0.01–20 Hz band, built entirely from proven, commercially available components and mature fabrication techniques. The design directly satisfies every hardware-architecture requirement in the NTRO problem statement while adding a set of networking, autonomy, and self-management innovations that make the system practical to deploy at scale across India's remote and border terrain.

5.1 Core Design Elements

- Optimized corrugated diaphragm — a thin, flexible membrane geometry that increases sensitivity to low-frequency pressure changes while retaining mechanical robustness.
- Capacitive readout (BMP390-based) — high sensitivity, low power consumption, and long-term stability using proven, mass-manufactured MEMS sensing technology.
- Dual-vent pressure equalization — balances slow-changing static atmospheric pressure (weather drift) while preserving the sensor's response to genuine infrasound events.
- On-chip temperature compensation — corrects thermal drift in real time for stable, accurate readings across field operating temperatures.
- Hermetically sealed environmental enclosure — protects the sensor and electronics from moisture, dust, and temperature extremes for reliable long-term field deployment.
- Complete data acquisition system — auto-calibration and real-time waveform display and analysis using an ESP32-based digitizer and open-market software.

5.2 Problem-Solution Mapping

Identified Problem	InfraSense Response
Low sensitivity for tiny pressure variations	Corrugated diaphragm + capacitive readout + low-noise amplification
High import cost restricts deployment	Indigenous design at an estimated ~₹6,500/unit prototype cost
Electronic noise reduces signal accuracy	Low-noise instrumentation amplifier + digital filtering
Poor long-term stability / calibration drift	On-chip temperature compensation + auto-calibration system
Temperature drift affects output	Real-time temperature sensing and correction
Limited real-time monitoring capability	LoRa-based remote health monitoring + real-time waveform display

5.3 Uniqueness Statement

Networked beamforming, solar-powered autonomy, self-calibration, and LoRa-based remote monitoring together enable a low-cost, deployable infrasound sensor network suited to remote and border regions — a combination not offered by existing imported single-unit microbarometer systems.

6. System Architecture & Technical Approach

6.1 Hardware Signal Chain

The sensor's signal chain follows a logical progression from mechanical sensing through to real-time display, illustrated in the block flow below:

- Step 1: Corrugated Diaphragm — pressure sensing and mechanical transduction
- Step 2: Differential Capacitive Cell (BMP390) — differential pressure measurement
- Step 3: Low-Noise Amplifier (INA333) — analog signal conditioning
- Step 4: ESP32 ADC + MCU — digitization, temperature compensation, and control
- Step 5: Data Acquisition & Real-Time Waveform Display — Python GUI with auto-calibration

6.2 Supporting Subsystems

Subsystem	NTRO Ref.	Description
Long-Period Equalization	(f)	Micro-leak vent balances slow atmospheric drift while preserving infrasound response
Environmental Enclosure	(g)	Hermetic, IP65-rated housing protecting against moisture, dust, temperature extremes
Wind-Noise Interface	(h)	Porous inlet plus multi-sensor beamforming for spatial noise rejection
Calibration System	(i)	Built-in reference-pressure source enabling automated 24-hour calibration cycles

6.3 Implementation Methodology (Process Flow)

1. Design & Simulation — diaphragm and cavity modelling using COMSOL Multiphysics / ANSYS
2. MEMS Fabrication — SOI-wafer bulk micromachining using DRIE and wafer bonding (for the custom diaphragm variant)
3. Assembly & Packaging — integration of sensing element, amplifier, MCU, and enclosure
4. Calibration — reference-pressure-source-based zero-point and sensitivity calibration
5. Lab Characterization — noise floor, sensitivity, and frequency-response measurement
6. Field-Ready Prototype — final integration and validation for deployment testing

6.4 Key Technologies Used

Layer	Technology / Tool
Design & Simulation	COMSOL Multiphysics, ANSYS
Sensing	BMP390 capacitive MEMS pressure sensor
Signal Conditioning	INA333 low-noise instrumentation amplifier (50 nV/√Hz)
Processing	ESP32 (ADC + MCU), 4th-order digital IIR filtering, TinyML (TensorFlow Lite for Microcontrollers)
Communication	LoRa (long-range, low-power) module for remote data transmission
Timing	GPS module for precise ($\pm 1 \mu\text{s}$) event timestamping
Power	Solar panel (5W) + Li-Po battery (2000 mAh), ultra-low-power sleep mode
Data Acquisition	Python-based GUI for real-time waveform display and analysis

7. Component Specification & Bill of Materials

The following table lists every major component used in the prototype, its function within the system, and its estimated cost at single-unit (prototype) pricing.

#	Component	Function	Est. Cost (₹)
1	BMP390 (MEMS capacitive pressure sensor)	Core infrasound pressure sensing	400-500
2	INA333 (low-noise instrumentation amplifier)	Signal amplification & conditioning	300-400
3	ESP32 (MCU + ADC + Wi-Fi/BT)	Data processing, TinyML, system control	500-600
4	LoRa module	Long-range remote data transmission	500-600
5	Solar panel (5W) + Li-Po battery (2000 mAh)	Off-grid power supply	1000-1200
6	GPS module	Precise timestamping & synchronization	300-400
7	Temperature sensor	Temperature compensation	100-150
8	IP65 enclosure (hermetically sealed)	Environmental protection	800-1000
9	Micro-leak vent assembly (porous tube/hose)	Long-period pressure equalization	150-200
10	Custom-fabricated PCB	Circuit board for all components	700-900
11	Wiring, connectors, misc. hardware	Assembly essentials	200-300
12	Assembly & labor	Manual prototype assembly	300-400
	TOTAL (Prototype, 1 unit)		~₹6,500

7.1 Component Rationale

- BMP390 was selected over custom optical MEMS sensing for the prototype stage because it is commercially available, well-characterized (0.02 Pa RMS noise, 200 Hz sampling), and directly integrable via I2C/SPI with the STM32/ESP32 architecture — enabling the full signal chain and algorithm set to be validated without requiring cleanroom fabrication access.
- INA333 provides the low-noise gain stage needed to bring the BMP390's weak differential signal to a usable digitisation level without introducing significant additional electronic noise.

- ESP32 was chosen for its balance of processing capability (sufficient for TinyML inference and digital filtering), low power consumption, and low cost, over higher-power alternatives.
- LoRa and GPS modules address the NTRO requirement for real-time monitoring capability and enable future multi-station correlation consistent with CTBTO-style array processing.

8. Innovation & Uniqueness

Beyond satisfying the baseline hardware requirements, InfraSense incorporates four principal innovations that differentiate it from a conventional single-unit microbarometer design.

Multi-Sensor Beamforming Array

- Multiple sensor nodes (5–10) connected via a mesh network enable triangulation to localize an infrasound source and estimate its bearing angle.
- Beamforming across nodes improves signal-to-noise ratio and enables discrimination between genuine infrasound events and local wind noise.
- Network redundancy ensures continued operation if an individual node fails.

Solar-Powered Field Autonomy

- A 5W solar panel with 2000 mAh Li-Po battery backup enables fully autonomous, off-grid operation.
- An ultra-low-power deep-sleep mode (microamp-level) extends operational endurance between charge cycles.
- This makes the sensor practical for remote, hard-to-access, or border deployment sites with no grid power infrastructure.

Adaptive Self-Calibration

- Automatic gain control adjusts sensitivity in real time, preventing saturation during strong events while preserving sensitivity during quiet periods.
- A built-in reference-pressure source enables on-demand calibration without external laboratory equipment, directly addressing the NTRO calibration-methodology requirement.

Remote Health Monitoring

- Periodic built-in self-tests monitor sensor health, battery status, and drift, enabling predictive maintenance rather than reactive repair.
- LoRa connectivity transmits diagnostic and sensor data from remote sites without dependency on cellular or Wi-Fi infrastructure.

8.1 What Makes This Solution Unique

- One of the few proposed solutions targeting a per-unit cost under ₹10,000.
- 100% indigenous design and manufacturing intent, consistent with the Make in India initiative.
- Combines edge AI (TinyML) inference with automatic self-calibration in a single low-cost device.
- Explicitly addresses every one of the nine NTRO hardware-architecture requirements (a)–(i).

9. Implementation Methodology

The project is structured into distinct, sequential phases, allowing progress to be validated at each stage before advancing to the next.

Phase 1 — Design & Simulation

Diaphragm and pressure-cavity geometry are modelled in COMSOL Multiphysics / ANSYS to predict sensitivity, resonant frequency, and mechanical stress distribution before any physical fabrication is undertaken.

Phase 2 — Component Sourcing & Circuit Design

The BMP390 sensor, INA333 amplifier, ESP32 controller, LoRa/GPS modules, and power subsystem are sourced and integrated into a custom PCB design.

Phase 3 — Assembly & Packaging

The sensor, electronics, and support subsystems (vent assembly, temperature sensor) are assembled and housed within the hermetically sealed IP65 enclosure.

Phase 4 — Firmware Development

ESP32 firmware is developed to implement digital filtering, temperature compensation, TinyML-based event classification, auto-calibration logic, and LoRa communication.

Phase 5 — Calibration

The assembled unit undergoes zero-point and sensitivity calibration against a known reference pressure source, with correction coefficients stored on-device.

Phase 6 — Laboratory Characterization

Noise floor, frequency response, sensitivity, and short-term stability are measured in controlled laboratory conditions and compared against target specifications.

Phase 7 — Field Testing

The prototype is deployed at a test site to validate real-world performance, including response to genuine environmental pressure events and wind-noise conditions.

Phase 8 — Iteration & Scale-Up Planning

Results from field testing inform design refinements and the transition plan toward production-scale manufacturing and, longer-term, custom optical MEMS sensing.

10. Feasibility & Viability Analysis

10.1 Feasibility Analysis

- **Technical:** SOI-DRIE MEMS fabrication and capacitive transduction are mature, industry-proven technologies with an established supply chain.
- **Cost:** A BMP390-based design costs an estimated ~₹6,500 per unit at prototype scale — over 99% cheaper than imported sensors (₹15–25 lakhs).
- **Operational:** Low component count and reliance on open-market data-acquisition software keep the prototype realistic within typical hackathon and early R&D; cost, time, and power budgets.

10.2 Potential Challenges & Mitigation Strategies

Category	Key Risk	Mitigation Strategy
Technical	Diaphragm fabrication yield issues; sensor drift under temperature cycling	Corrugated diaphragm design + on-chip temperature compensation
Operational	Wind noise masking weak infrasound signals; remote-site power constraints	Porous-inlet beamforming array; solar-assisted duty-cycled operation
Regulatory	Deployment near sensitive/border zones may require government clearance	Early coordination with NTRO / NDMA for pilot-site approvals

Prototype-stage demonstration uses the BMP390 (an off-the-shelf capacitive MEMS sensor) to validate the complete signal chain, algorithms, and system architecture. Numeric performance targets — sensitivity, noise floor, and drift — are design estimates based on component datasheets and are to be confirmed through laboratory characterization (see Section 13).

11. Cost Analysis — Prototype vs. Production

Component costs at single-unit prototype pricing are significantly higher than at production scale, where bulk sourcing, mass PCB fabrication, and injection-molded enclosures substantially reduce per-unit cost. The table below compares estimated costs at both scales.

Component	Prototype (₹)	Production, 100+ units (₹, est.)
BMP390 sensor	400-500	250-300
INA333 amplifier	300-400	150-200
ESP32 module	500-600	300-350
LoRa module	500-600	300-350
Solar panel + battery	1000-1200	600-800
GPS module	300-400	180-220
Temperature sensor	100-150	60-80
Enclosure	800-1000	300-400 (injection-molded)
Micro-leak vent assembly	150-200	80-100
PCB	700-900	250-350 (mass manufactured)
Wiring / connectors / misc.	200-300	100-150
Assembly / labor	300-400	150-200 (automated)
TOTAL per unit	~₹6,500	~₹3,500 - ₹5,000

11.1 Scale Comparison

Scale	Cost per Unit	Total for 100 Units
Prototype (1 unit)	~₹6,500	—
Production (100+ units)	~₹3,500 - ₹5,000	₹3,50,000 - ₹5,00,000
Imported sensors (reference)	₹15,00,000 - ₹25,00,000	₹15 - 25 Crores

All production-scale figures are estimated, based on bulk component pricing and standard economies of scale typical in electronics manufacturing. Actual production cost will depend on vendor negotiations, order volume, and component availability at the time of procurement.

12. Risk Analysis & Mitigation Strategies

A structured risk assessment was undertaken across technical, operational, and regulatory dimensions, consistent with the challenges identified in Section 10.2. Each risk is paired with a specific, actionable mitigation measure integrated into the design.

Risk	Likelihood	Impact	Mitigation
Diaphragm fabrication yield loss	Medium	Medium	Corrugated design distributes mechanical stress, improving yield
Sensor drift from temperature cycling	High	High	On-chip real-time temperature compensation
Wind noise masking weak signals	High	High	Porous inlet + multi-sensor beamforming
Remote-site power interruption	Medium	Medium	Solar-assisted duty-cycled operation with battery backup
Deployment clearance delays	Medium	Low	Early coordination with NTRO / NDMA for approvals
Component supply-chain disruption	Low	Medium	Design uses widely available, multi-sourced components

13. Testing, Calibration & Validation Plan

The expected-solution criteria in the NTRO problem statement require the prototype to demonstrate five specific test outcomes. The planned validation approach for each is described below.

Detection of low-frequency pressure signals

A controlled pressure-generation setup (e.g., a calibrated speaker/piston chamber) will generate known infrasonic signals across the 0.01–20 Hz band, confirming the sensor's ability to register these signals in real time.

Laboratory characterization of frequency response

The sensor's output amplitude and phase will be measured against a swept-frequency reference input to construct a frequency-response curve across the target band.

Noise floor measurement

With the sensor in a quiescent (no deliberate input) state, output noise will be recorded and analyzed spectrally to determine the effective noise floor in Pa/√Hz.

Sensitivity estimation

Known small-amplitude pressure steps will be applied to estimate the smallest reliably detectable pressure change (minimum detectable signal).

Stability testing

The sensor's zero-point output will be logged over an extended period (target: 30 days) under varying ambient temperature to quantify long-term drift.

13.1 Calibration Methodology

- Zero-point calibration is performed automatically every 24 hours using the built-in reference-pressure source.
- Temperature-dependent correction coefficients are derived through controlled thermal-chamber testing and applied via an on-device polynomial correction algorithm.
- Calibration traceability is maintained through logged calibration history, referencable via the on-device diagnostic system.

14. Impact & Benefits

14.1 Potential Impact on Target Audience

- National Technical Research Organisation (NTRO) and defense/security agencies — remote surveillance and infrasound-based event detection.
- Disaster management agencies (NDMA/SDMA) — early alerts for volcanic tremors, avalanches, and severe storms.
- Meteorological & seismic observatories — augmenting existing monitoring networks with dense, low-cost sensor coverage.
- Industrial safety applications — pipeline and structural leak/explosion detection.

14.2 Benefits of the Solution

Social

- Early warning of hazardous events enables faster disaster response.
- Protects human lives near volcanic, seismic, and storm-prone regions.
- Strengthens national security through improved surveillance capability.

Economic

- Indigenous design reduces dependency on costly imported sensors.
- Low unit cost (~₹6,500 prototype; ~₹3,500-5,000 at production scale) enables wide-scale deployment.
- Supports the Make in India initiative and local manufacturing growth.

Environmental

- Low-power operation minimizes energy consumption.
- Compact design requires minimal infrastructure footprint.
- Solar-powered autonomy reduces reliance on grid electricity.

Strategic

- Builds indigenous infrasound-sensing capability, reducing reliance on imported CTBTO-grade systems.
- Enables dense border and sensitive-zone monitoring through networked, low-cost sensor arrays.
- Strengthens NTRO's early-warning and surveillance capacity with self-reliant, field-ready technology.

15. Regulatory & Deployment Considerations

Given the strategic and security-sensitive nature of infrasound monitoring — particularly for NTRO's surveillance mandate — deployment planning must account for the following considerations from the earliest project stages.

- **Site clearance:** Deployment near international borders or other sensitive zones will require coordination with NTRO and relevant local authorities before installation.
- **Data security:** Data transmitted via LoRa and stored on-device or centrally must follow applicable government data-security protocols, particularly for defense-relevant deployments.
- **Standards alignment:** Where feasible, timestamping (GPS-synchronized) and calibration practices are designed to remain broadly compatible with CTBTO-style multi-station correlation conventions, easing future interoperability.
- **Environmental permitting:** Solar and enclosure installations in ecologically sensitive or high-altitude areas should be reviewed against local environmental guidelines.

16. Future Roadmap

Phase 1 (Current) — Proof-of-Concept Prototype

BMP390-based capacitive sensing prototype validating the complete signal chain, firmware, calibration methodology, and networking architecture at an estimated cost of ~₹6,500 per unit.

Phase 2 — Field Pilot Deployment

Deployment of a small multi-node network (5–10 units) at a representative test site to validate beamforming, wind-noise rejection, and long-term stability under real environmental conditions.

Phase 3 — Production Scale-Up

Transition to bulk component sourcing, mass PCB fabrication, and injection-molded enclosures, targeting a per-unit cost of ₹3,500–5,000 at 100+ unit scale.

Phase 4 — Custom MEMS Diaphragm Integration

Development of a custom-designed, corrugated capacitive MEMS diaphragm (via SOI-DRIE fabrication) optimized specifically for the 0.01–20 Hz infrasound band, improving on the generic BMP390's sensitivity.

Phase 5 — Optical Readout Research Track

Longer-term research collaboration to evaluate fiber-optic or on-chip Fabry-Pérot optical readout, tracking the global research trajectory (Section 4.2) toward sub-milliPascal sensitivity, once the technology matures toward field-deployable readiness.

Phase 6 — National Network Integration

Integration of deployed sensor clusters into a unified, NTRO-accessible monitoring dashboard, enabling nationwide infrasound event correlation and early-warning dissemination.

17. Conclusion

InfraSense presents a technically grounded, cost-realistic, and fully indigenous response to NTRO problem statement SIH26144. By building on proven capacitive MEMS sensing rather than unproven, high-cost optical readout, the design remains buildable within realistic prototype timelines and budgets, while still addressing every mandatory hardware-architecture requirement specified in the problem statement. The addition of networked beamforming, solar autonomy, adaptive self-calibration, and remote health monitoring extends the solution beyond a single sensing unit into a genuinely deployable, low-maintenance sensor network architecture.

At an estimated cost of approximately ₹6,500 per prototype unit — falling to ₹3,500-5,000 at production scale — the solution offers a cost reduction exceeding 99% relative to imported alternatives, directly addressing the core problem of limited deployment density that currently constrains India's infrasound monitoring capability. This positions InfraSense as a credible, scalable, Make-in-India pathway toward strengthening the country's disaster early-warning and national security surveillance infrastructure.

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8. CEA / iXblue, "New Optical Microbarometer," presented at CTBTO Science and Technology Conference, 2017.
9. National Technical Research Organisation (NTRO) — Official Smart India Hackathon 2026 Problem Statement SIH26144: Design & Development of a High-Sensitivity Microbarometer Infrasound Sensor.

Annexure A: NTRO Hardware Requirement Cross-Reference

The table below cross-references each of the nine mandatory hardware-architecture elements from the official problem statement against the corresponding InfraSense design element and the section of this report where it is discussed in detail.

Ref.	Requirement	InfraSense Design Element	Section
(a)	Pressure sensing mechanism	BMP390 capacitive MEMS sensor	6.1, 7
(b)	Mechanical transducer design	Optimized corrugated diaphragm	5.1, 6.1
(c)	Differential pressure measurement	Differential capacitive cell	6.1
(d)	Low-noise analog front-end	INA333 instrumentation amplifier	6.1, 7
(e)	Temperature compensation	On-chip real-time drift correction	5.1, 6.1
(f)	Long-period pressure equalization	Dual-vent / micro-leak vent assembly	5.1, 6.2
(g)	Environmental enclosure	Hermetically sealed IP65 housing	5.1, 6.2
(h)	Wind-noise reduction interface	Porous inlet + multi-sensor beamforming	6.2, 8
(i)	Calibration methodology	Auto-calibration via reference pressure source	6.2, 13.1

Annexure B: Glossary of Terms

Term	Definition
Infrasound	Acoustic pressure waves with frequency below approximately 20 Hz, inaudible to humans.
hPa (hectoPascal)	Unit of atmospheric pressure; 1 hPa = 100 Pa. Standard sea-level pressure is 1013.25 hPa.
MEMS	Micro-Electro-Mechanical Systems — miniaturized mechanical and electro-mechanical devices fabricated using semiconductor processing techniques.
Diaphragm	A thin, flexible membrane that deflects in response to pressure change; the core sensing element in a microbarometer.
Capacitive Readout	A sensing method that measures diaphragm displacement via the resulting change in electrical capacitance.
Optical (Fabry-Pérot) Readout	A sensing method that measures diaphragm displacement via changes in an optical interference pattern; higher potential sensitivity but currently research-stage.
ASIC	Application-Specific Integrated Circuit — a custom chip designed for one specific function.
Corrugated Diaphragm	A diaphragm with a wavy/ridged profile that increases mechanical compliance (sensitivity) while retaining structural strength.
Dual-Vent / Micro-Leak Vent	A controlled small air-leak path that equalizes slow atmospheric pressure changes while preserving response to fast (infrasound) events.
Beamforming	A signal-processing technique that combines data from multiple sensors to enhance a correlated signal and suppress uncorrelated noise (e.g., wind).
TinyML	Machine learning models optimized to run on small, low-power embedded devices such as microcontrollers.
LoRa	Long Range — a low-power, long-distance wireless communication technology.
GSM	Global System for Mobile Communications — standard cellular network technology.
CTBTO	Comprehensive Nuclear-Test-Ban Treaty Organization — operator of the global International Monitoring System (IMS), including infrasound stations.
Calibration	The process of comparing a sensor's output against a known reference to correct for measurement error.